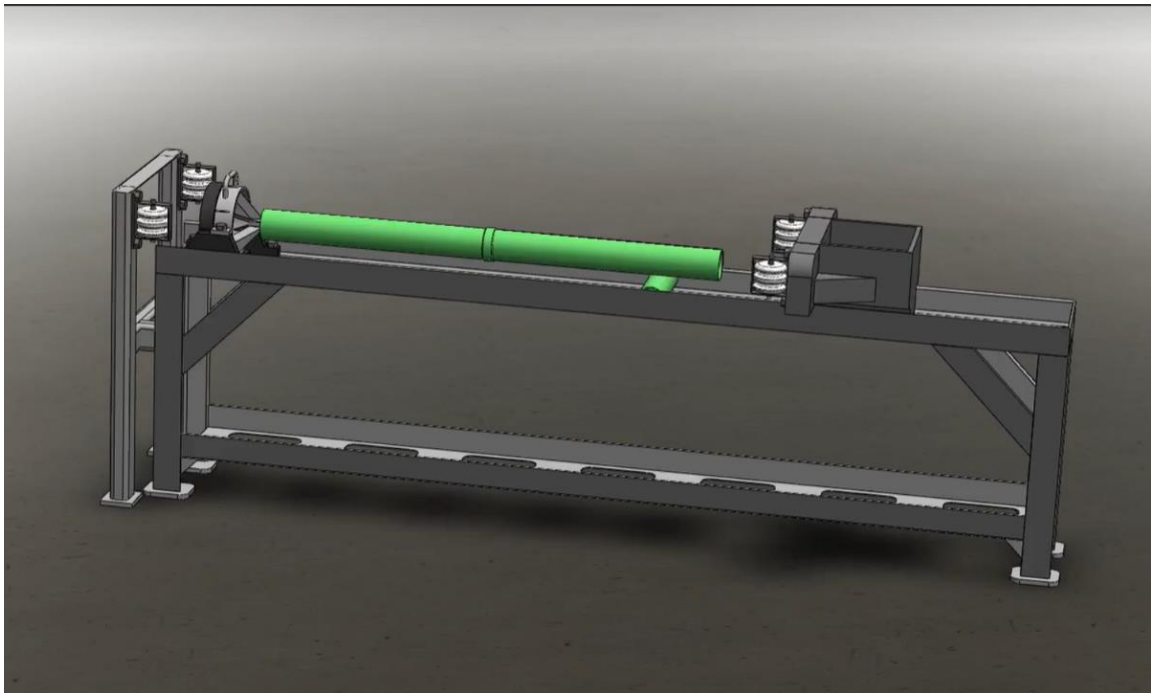
Bangladesh University of Engineering and Technology

A Report On

Convenient Bamboo Cutter



A Report On

Convenient Bamboo Cutter

Course No: IPE304

Course Title: Product Design-I Sessional

Date of Submission: 09/09/2023

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FORWARDING LETTER

September 9, 2023

Dr. Kais Bin Zaman

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Abu Taher

Department of IPE, BUET

Subject: A Report on "Convenient Bamboo Cutter".

Dear Teachers,

With due respect, we present you our report on "Convenient Bamboo Cutter". In our product, we have used different types of cutters to cut bamboo in different shapes. This machine is flexible, easy to use, easily affordable and time-saving. In this report we conducted a public survey, Quality Function Deployment, Functional Decomposition, Qualitative analysis of material, manufacturing process, Joining process selection. Finally, we did a cost and sensitivity analysis to find the total cost of our product.

We would like to express gratitude and acknowledgement to our teachers in helping us completing this report. We solemnly apologize for any involuntary mistakes, if any.

Sincerely,

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Acknowledgement

This project would not be completed without the help and participation of many people.

First of all, we are thankful to our teachers Dr. Kais Bin Zaman (Professor, Dept. of IPE, BUET), Dr. Ridwan Al Aziz (Assistant Professor, Dept. of IPE, BUET), Nafisa Anzum Sristi (Lecturer, Dept. of IPE, BUET) and Abu Taher (Lecturer, Dept. of IPE, BUET) for their cordial help and guidance in our product design. Their constant support helped us a lot to make this report better.

Finally, we thank all the people who participated in the surveying of our product.

Abstract

Bamboo, an eco-friendly and versatile material, has found numerous applications in various industries. In our country many rural people use bamboo for different purposes.

However, the process of cutting bamboo efficiently and in various shapes has been a labor intensive and time-consuming task. In response to this challenge, we introduce a “Convenient Bamboo Cutter”.

One of the standout features of this machine is its ability to cut bamboo into different shapes, meeting the diverse needs of various industries and village peoples. Whether it’s thin strips for handicrafts or sturdy sections for construction, the convenient bamboo cutting machine offers flexibility and precision.

Moreover, this innovative machine is designed with affordability in mind, ensuring that it is accessible to a wide range of users. Its user-friendly interface and straightforward operation make it suitable for both professionals and novices. Maintenance is hassle free and it significantly reduces the time required for bamboo processing, increasing overall productivity.

In this report, we first did a survey of our product on rural people who use bamboo regularly. Then we did a QFD to our product. After that we did a qualitative analysis on material selection, manufacturing process selection and joining process selection. After that using digital logic and performance index we selected the proper material, manufacturing process and joining process for every component of our product. After that cost and sensitivity analysis is done to find all costs related to our product.

Hopefully this report will encourage students to creative designing of a product.

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Chapter-1

Introduction

1.1 Product Design

Product design is a process that involves the creation and development of a product's appearance, functionality, and user experience. It consists of a broad range of disciplines and strategic activities, including planning, industrial design, engineering, ergonomics, aesthetics, manufacturing, all aimed at creating products that meet customer needs and desires.

Product design involves understanding the target audience, their requirements and preferences, and their problems to create solutions that affect the needs effectively. The primary goal is not only to apply visual effects and aesthetics but also to make the product practical, user-friendly and marketable. The process is typically comprised of a number of stages and involves a collaborative and cross-functional team initially conceptualizing and brainstorming product ideas and sketching concepts to explore different design possibilities. Other tools such as market research, user/customer research, and gathering idea about the target market are also crucial to identify the opportunities for innovative product design.

The ultimate objective of product design is to create products that meet the demands of people and is able to thrive in the competitive market. A balance between brainstorming ideas, creativity, functionality and market viability will lead to an effective product development and design.

1.2 Our Proposals

1.2.1 Product idea – 01

Coconut Cutting Machine

The manual Coconut Cutter with enhanced safety features is designed to simplify the process of opening coconuts while prioritizing user safety. The machine consists of a main frame, a moving coconut holder with clamping system, and a cutting blade with lever mechanism. The features are:

- **Main Frame:** built with a sturdy and durable frame, ensuring stability during operation. It is made with stainless steel, and it provides structural support and stability during the cutting process.
- **Moving Coconut Holder:** The machine consists of a linearly moving coconut holder with clamping mechanism. To cut in different positions of the coconut, the holder moves the coconut accordingly.
- **Secure Clamping:** The Coconut Holder features adjustable clamps that securely hold the coconut in place, ensuring stability during the cutting process.
- **Lever Operation:** The machine utilizes a user-friendly lever mechanism. By pulling the lever, the cutting blade swiftly and effortlessly slices through the coconut shell.
- **Non-slip Rubber Feet:** The machine is equipped with non-slip rubber feet, providing a stable grip on any surface and minimizing the risk of unintended movement during use.

The machine's design and functionality enable quick and efficient coconut cutting, saving time and effort in opening coconuts. The machine should be able to handle coconuts of various sizes, allowing users to cut both small and large coconuts with ease.

1.2.2 Product Idea – 02

Stairs Cleaning Machine

The objective of this design idea is to come up with a device that can conveniently clean stairs (both horizontal and vertical part simultaneously). Moreover, it may be used to clean any kind of floors as well. This stair-cleaning machine will be used in various settings such as commercial buildings, schools, hospitals etc. In this product, we want to highlight some features, one of them being a compact and maneuverable design that allows them to navigate and clean stairs easily. It will have rotating brushes, suction systems for dust collection, power supply system and sometimes water tanks to carry out the cleaning process.

The features are as follows:

- **Compact Design**, allowing them to easily be handled and maneuvered on stairs and tight spaces.
- **Rotating Brushes:** To scrub the stair surfaces, loosening dirt and debris for efficient cleaning.
- **Suction System:** To collect the loosened dirt and debris from the stairs, leaving them clean.
- **Built-in Water Tanks**, that can be used to apply water or cleaning solution onto the stairs for deep cleaning.
- **Adjustable Handle and Attachments:** various attachments and accessories may be added for specialized cleaning requirements.
- **Safety Features**, such as non-slip wheels or tracks to ensure stability during operation and prevent accidents.

1.2.3 Product Idea – 03

Wheel Controlled Pesticide Sprayer

The basic pesticide sprayer which is available in the current market needs to be carried on the back of the farmers and to be sprayed manually by hands. Carrying these heavy weights makes it a very long tiring and physically challenging process. Farmers even go through back pain in their later life. The wheel-controlled pesticide sprayer is an economically inexpensive and manually controllable sprayer.

The main objective of our design is to manufacture mechanically operate-able wheel-controlled pesticide sprayer which we desire to have the following features:

- **Mechanical Automation:** using cam mechanism, we can operate the handle of the sprayer mechanically through the rotating shaft of the wheel of the cart.
- **Multiple Nozzle**: we can add multiple nozzle which will spray at 180-degree angle covering a wide range of area.

- **Safety:** the farmer will be able to keep himself at a safe distance using the cart. We can also add a transparent shield fulfilling our safety issue.
- **Movability:** the farmer can place the heavy containers on the cart and move it easily using the wheel mechanism.

1.2.4 Product Idea – 04

Convenient Stair-Climbing Trolley

Stair-climbing trolleys provide a practical solution for individuals who often need to transport or deliver heavy objects within a building or any location where stairs might be present. With unique wheel systems, these trolleys can greatly reduce the physical effort needed for carrying objects up and down stairs, thus making physically demanding tasks easier.

The objective of our design idea is to manufacture such a stair-climbing trolley for carrying bulky loads, with more concentration on its ease of use, versatility, storage and safety features.

The following features were planned to be incorporated:

- **A removable platform**, which will allow the trolley to be converted to a flatbed cart when not climbing stairs.
- **Foldable/extendable platforms**, which will allow the accommodation of larger or irregularly shaped objects.
- **Built-in Straps or Hooks** for secure fastening of the load during transportation.
- **A Locking Mechanism**, to secure the trolley in place and avoid accidental movements or collapses.
- Other Safety Features, such as **a braking mechanism** for better control amidst stairs, and **non-slip handle surfaces** for firm gripping.

1.2.5 Product Idea – 05

One-Wheel Seed Planter

The One-Wheel Seed Planter is a manually operated tool designed for small-scale agricultural fields. It consists of a large wheel connected to a handle, a seed hopper, a seed metering mechanism, a furrow opener, and a furrow closing mechanism.

- The user pushes the planter along the field, and the **rotation of the wheel activates the seed metering mechanism.**
- The seed metering mechanism controls the flow of seeds from the hopper, **ensuring consistent and controlled dispensing**.
- **A furrow opener** in front of the wheel creates a furrow in the soil at the desired depth.
- The dispensed seeds fall into the furrow as the planter moves forward, providing **regular seed placement along the furrow.**
- A mechanism behind the furrow opener closes or covers the furrow with soil, protecting the seeds.
- The planting depth and spacing can be adjusted by modifying the furrow opener and seed metering mechanism setting.

  Overall, the One-Wheel Seed Planter offers a straightforward and efficient solution for sowing seeds in small-scale agricultural fields, providing control over planting depth and spacing while reducing manual effort.

1.2.6 Product idea – 06

Adjustable Umbrella Holder Backpack

An umbrella holder backpack is a specialized product that is designed to attach your umbrella as a backpack, keeping your hands free and your belongings dry. It offers protection from sunlight and rain without any hassle of having to hold the umbrella. The umbrella backpack holder consists

of various components, including an umbrella, a holder, shafts, a locking mechanism, bevel gears, a baseplate, and a handle.

The elements and components of our design idea is briefly outlined below:

1. **Holder:** Holds the umbrella in place and allows to adjust the angle of the umbrella to shield us from the sun or rain at the desired angle.
2. **Shaft:** Rotational motion from the handle is transferred to the holder through shafts and bevel gears. The shafts are supported by **bearings** to facilitate rotations.
3. **Locking mechanism**: Involves a **clutch system** incorporated to the handle, which, when drawn, will **release a brake shoe, which is normally in contact to a shaft**, and thus allowing the handle to be adjusted according to need. When the clutch is released, the brake shoe will contact against the shaft, resisting any rotation and thus positioning the umbrella securely in the desired adjusted angle.
4. **Bevel gear mechanism:** Bevel gears will be used to transfer shaft rotations perpendicularly.
5. **Baseplate**: The baseplate will accommodate the components, and provide stability, attachment.
6. **Handle**: Ensuring a secure and firm hold, enhancing control over the holder and the attached umbrella.
7. **Cost-effective:** The components should not be very expensive. So, the manufacturing cost should also be low.

1.2.7 Product Idea – 07

Convenient Bamboo Cutter

The Convenient Bamboo Cutting Machine is an innovative design that streamlines the bamboo cutting process while prioritizing safety, efficiency, and affordability. This machine harnesses the power of a **pulley mechanism to generate a substantial amount of force** required to cut multiple bamboos simultaneously, enhancing productivity and reducing production time. With its simple

yet effective design, it **eliminates the need for motors or electrical devices**, making it a cost-effective solution for bamboo processing.

Key Features:

- **Sturdy Main Frame**: The machine boasts a robust main frame that serves as the foundation, providing stability and support for all its components.
- **Adjustable Cutting Blade:** Positioned at the front of the machine, the cutting blade remains stationary while operation, ensuring precise and consistent cuts for every bamboo. These cutting blades are versatile and easily adjustable to the necessary cuts required.
- **Plunger Mechanism:** A plunger mechanism, smoothly sliding along the main frame, is responsible for driving the bamboos forward, delivering the required kinetic energy to facilitate clean cuts.
- **Pulley System:** The pulley system incorporated into the machine enhances its cutting capacity, allowing multiple bamboos to be processed simultaneously, significantly increasing productivity.
- **Safety Measures:** By automating the cutting process, the machine minimizes the risks associated with traditional manual methods, ensuring a safer working environment for operators.
- **Cost-Effective Design**: With no reliance on motors or electrical devices, the Bamboo Cutting Machine offers a cost-effective solution, reducing both initial investment and operational expenses.
- **Enhanced Efficiency:** The machine's innovative design streamlines the bamboo cutting process, improving overall efficiency and output while maintaining consistent quality.

By leveraging the power of pulleys, the Bamboo Cutting Machine revolutionizes bamboo processing, combining safety, efficiency, and affordability into a single solution. With its ability to handle multiple bamboo at once, it drastically reduces production time, making it an invaluable asset for bamboo manufacturers seeking to optimize their operations.

1.2.8 Product Idea – 08

Height Adjustable Table

The Height Adjustable Table is a versatile and ergonomic piece of furniture designed to meet the needs of individuals who require flexibility in their working or living spaces. It allows users to easily adjust the height of the table to their desired level, promoting better posture, comfort, and productivity.

- **Rotary Wheel for Height Adjustment:** The table incorporates a rotary wheel that serves as the control mechanism for adjusting the table's height. When the user rotates the wheel, it engages with the gear mechanism, initiating the lifting or lowering process.
- **Gear Mechanism for Height Adjustment:** The Table features a gear mechanism that enables the table to be lifted or lowered according to the user's needs. This mechanism utilizes interlocking gears to transmit rotational motion into vertical movement.
- **Mobility Wheels**: The Height Adjustable Table is equipped with wheels to facilitate easy movement and repositioning. These wheels are typically attached to the base of the table, allowing users to effortlessly roll the table from one location to another.
- **Locking Wheel in Stationary Use:** To keep the table stationary when needed, the mobility wheels are often designed with a locking mechanism. When engaged, the locking wheel securely immobilizes the table, providing stability during use, especially when performing tasks that require minimal or no movement.

In summary, the Height Adjustable Table with a gear mechanism for lifting and wheels for mobility combines functionality, versatility, and user-friendly design. The gear mechanism allows for precise height adjustments, while the mobility wheels and locking mechanism provide effortless movement and stability as needed. This table is an excellent choice for individuals seeking a customizable and mobile solution for their work or living spaces.

1.3 Selected Proposal

After consulting with our course teachers, our seventh proposal **“Convenient Bamboo Cutter”** was selected as our project.

Causes for selection: Bamboo is a strong and fibrous material, making it difficult to cut manually. Its dense structure requires significant force and effort to make clean and precise cuts. This can be physically demanding and time-consuming, also leading to an injurious process. Bamboo is used in agricultural practices, such as stakes for supporting plants (betel leaf), trellises and fencing. It has also a wide range of applications in fishing. The pulley-operated bamboo cutting machine can greatly reduce the effort and time needed to cut bamboo and increase efficiency, at relatively lower costs than fully automatic bamboo cutting machines.

Chapter-2

Understanding Customer Needs Through Survey

2.1 Introduction

In the world of bamboo processing, innovation continues to push boundaries, revolutionizing the way artisans and manufacturers work with this remarkable resource. Among these groundbreaking advancements is the manual bamboo cutting machine, a game-changer that utilizes a pulley mechanism and offers unparalleled customization options.

The manual bamboo cutting machine with a pulley mechanism brings a new level of precision and efficiency to the process of cutting bamboo. By leveraging the power of pulleys, this machine ensures smooth and controlled movements, allowing operators to effortlessly guide the cutting process. The pulley system minimizes effort and maximizes output, making it an ideal choice for both small-scale workshops and larger production facilities.

After our research, we have identified our main customers. They are:

1. Artisans and Craftsmen
2. Bamboo Furniture Manufacturers
3. Construction Companies
4. Bamboo Handicraft Producers
5. Agricultural and Farming Communities

We carried out a survey for developing customer requirements for our Bamboo Cutting Machine among some of the above customers.

2.2 Areas and Locations of Survey:

1. Kotchandpur Bazar, Jhenaidah, Bangladesh
2. Ibrahimpur Bazar, Nabinagar, Brahmanbaria
3. Harong, Chandina, Cumilla

4. Mollapara Bazar, Agailjhara, Barishal

2.3 Survey Result:

The survey was done among 32 people. The results of the survey are shown below with the help of pie chart along with percentages:

1. Have you used a bamboo cutting machine before?
   a) Yes
   b) No

Table 2.1 Data collection table for question 1

Option	Response Count	Response Percent (%)
Yes	4	12
No	28	88

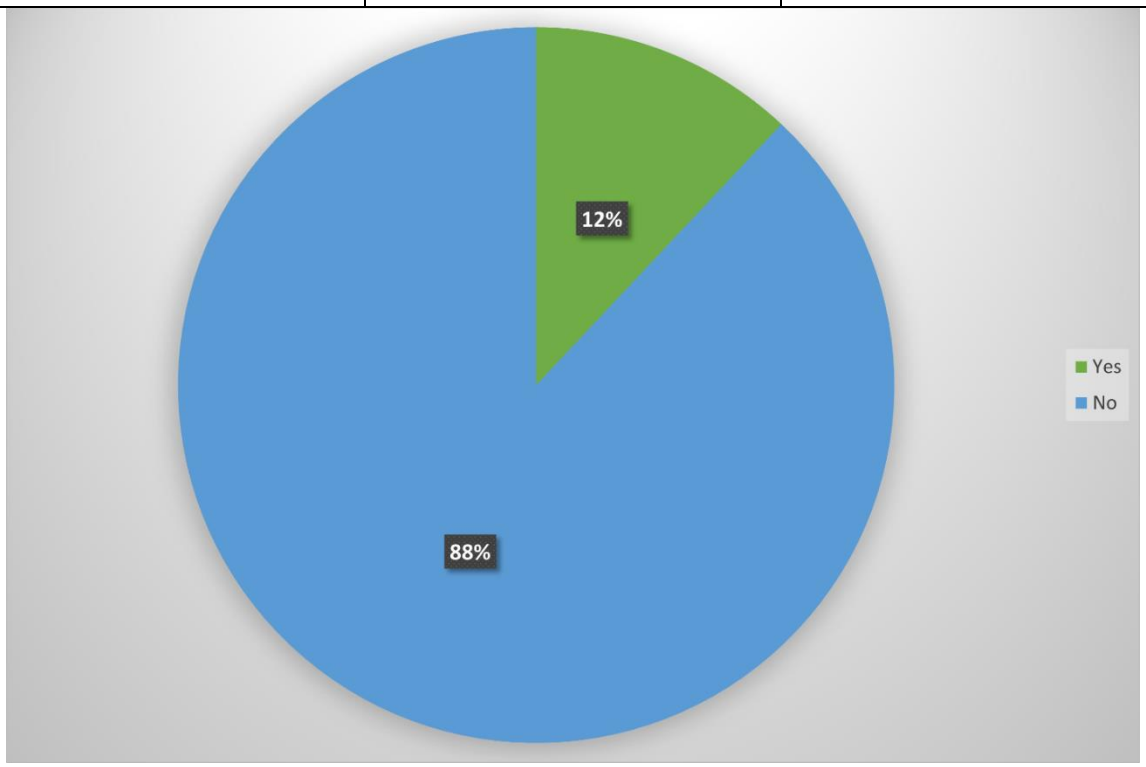


Figure 2.1 Reviewer opinion percentage for question 1

2. How often do you work with bamboo or bamboo related products?
    a) Once a year
    b) Once every 6 months
    c) Depends on seasonal use.
    d) Frequently (2-3 times a week)
    e) Daily

Table 2.2 Data collection table for question 2

Option	Response Count	Response Percent (%)
Once a year	2	6
Once every 6 months	6	19
Depends on seasonal use	12	38
Frequently (2-3 times a week)	9	28
Daily	3	9

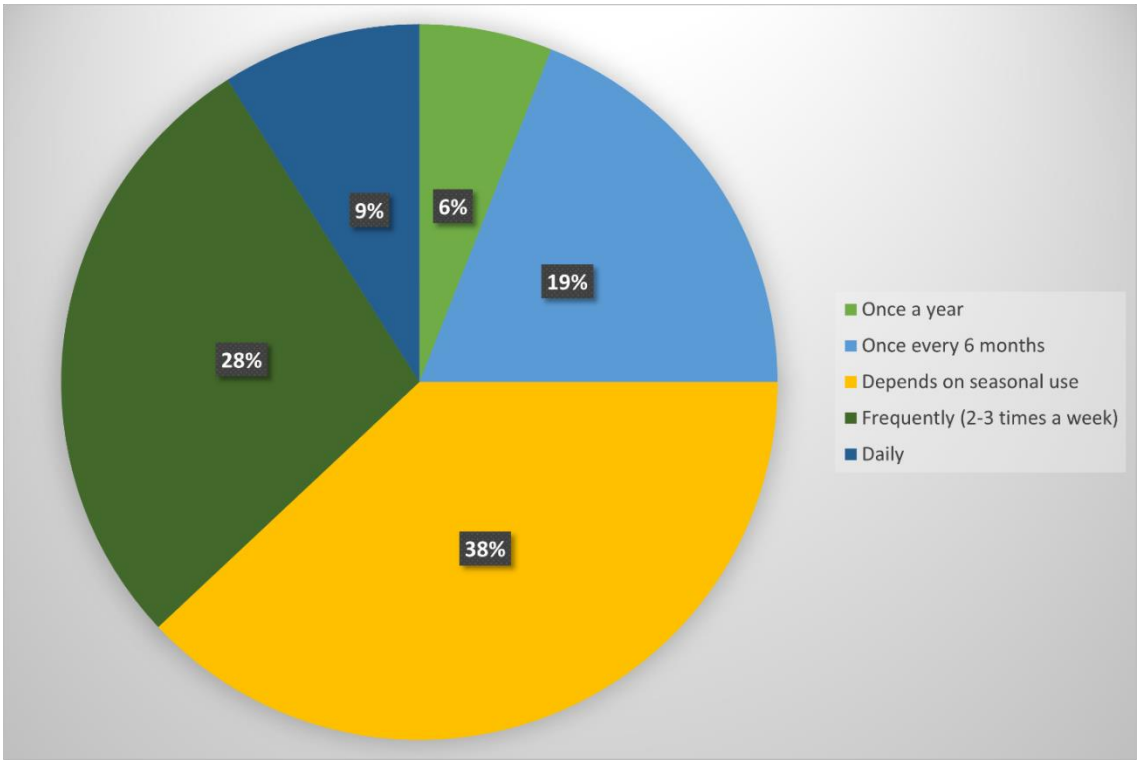


Figure 2.2 Reviewer opinion percentage for question 2

3. How much time do you need to cut one average size bamboo (8-10 feet) into 8 pieces?
    a) Less than 5 minutes
    b) Less than 10 minutes
    c) Less than 15 minutes
    d) More than 15 minutes

Table 2.3 Data collection table for question 3

Option	Response Count	Response Percent (%)
Less than 5 minutes	5	16
Less than 10 minutes	14	44
Less than 15 minutes	9	28
More than 15 minutes	4	12

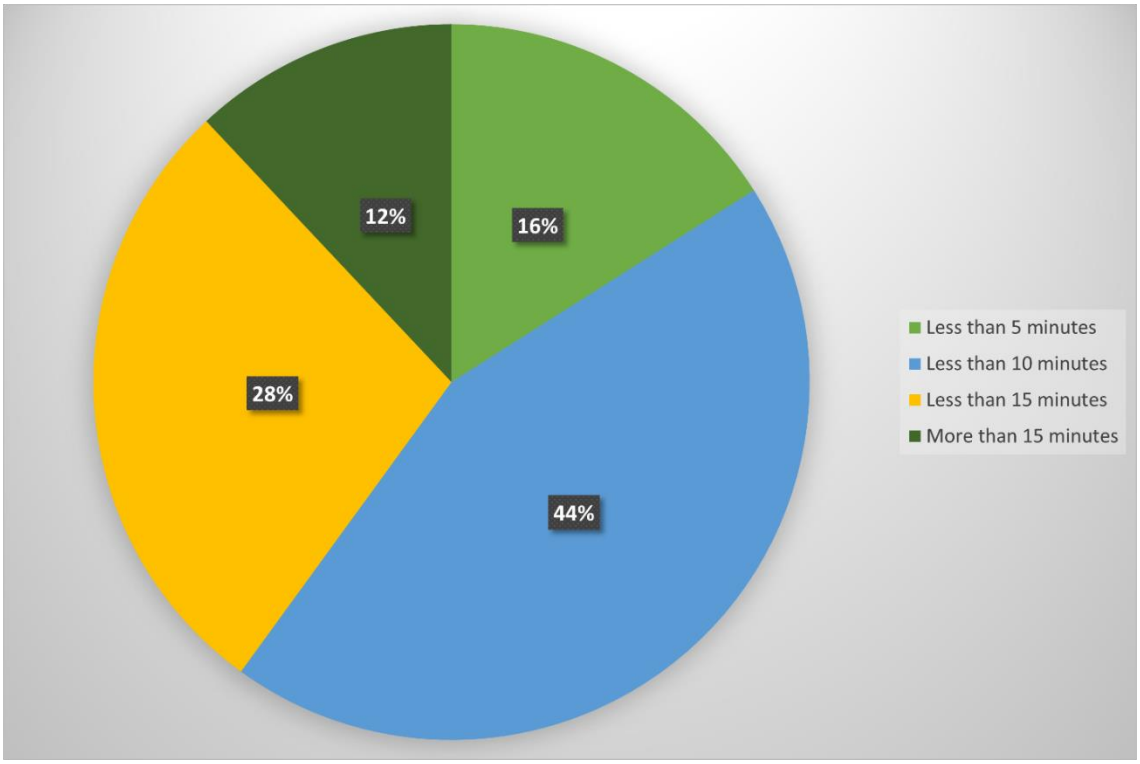


Figure 2.3 Reviewer opinion percentage for question 3

4. What is the primary purpose for which you use bamboo?
    a) Construction
    b) For making fish traps
    c) Fencing
    d) Furniture
    e) Crafts and artistic projects

Table 2.4 Data collection table for question 4

Option	Response Count	Response Percent (%)
Construction	6	19
For making fish traps	7	22
Fencing	11	34
Furniture	3	9
Crafts and artistic projects	5	16

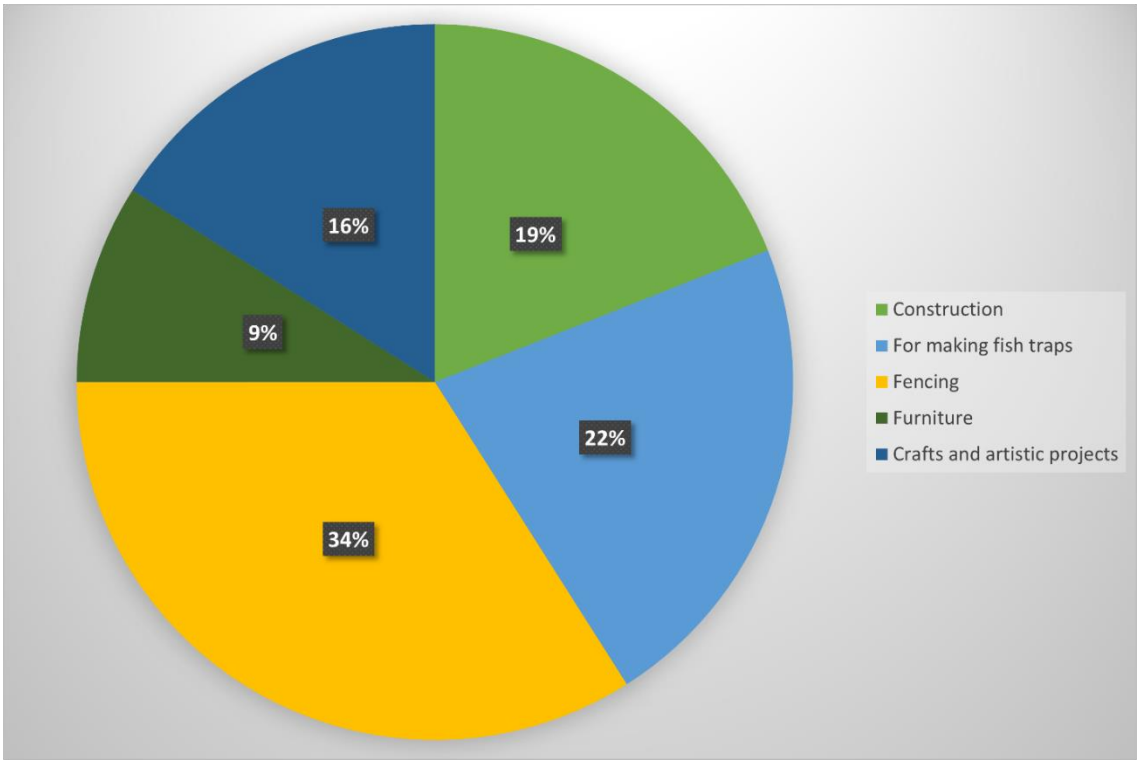


Figure 2.4 Reviewer opinion percentage for question 4

5. From the list below what problem you have faced the most while cutting bamboo
    a) Thorns and splinters related issues.
    b) Physical pain
    c) Inadequate tools
    d) Time consuming process

Table 2.5 Data collection table for question 5

Option	Response Count	Response Percent (%)
Thorns and splinters related issues	7	22
Physical pain	13	41
Inadequate tools	4	12
Time consuming process	8	25

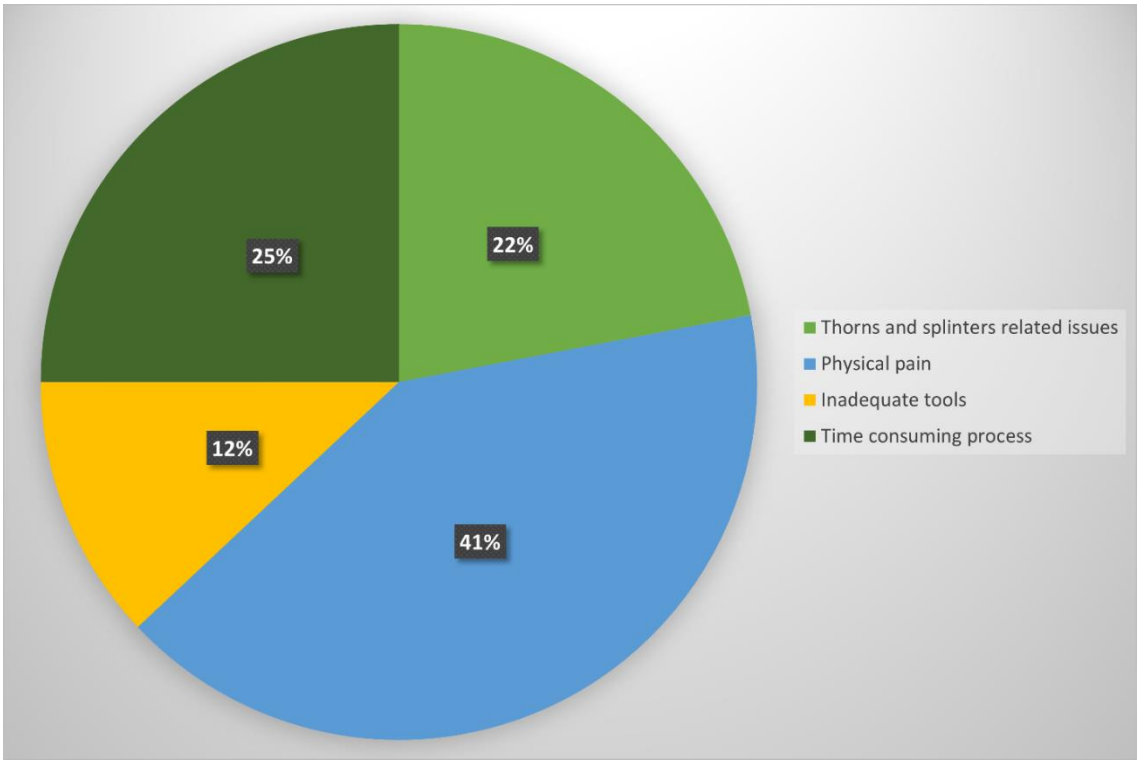


Figure 2.5 Reviewer opinion percentage for question 5

6. Are you aware of the benefits of using a bamboo cutting machine over traditional manual methods?
    a) Yes
    b) No

Table 2.6 Data collection table for question 6

Option	Response Count	Response Percent (%)
Yes	10	31
No	22	69

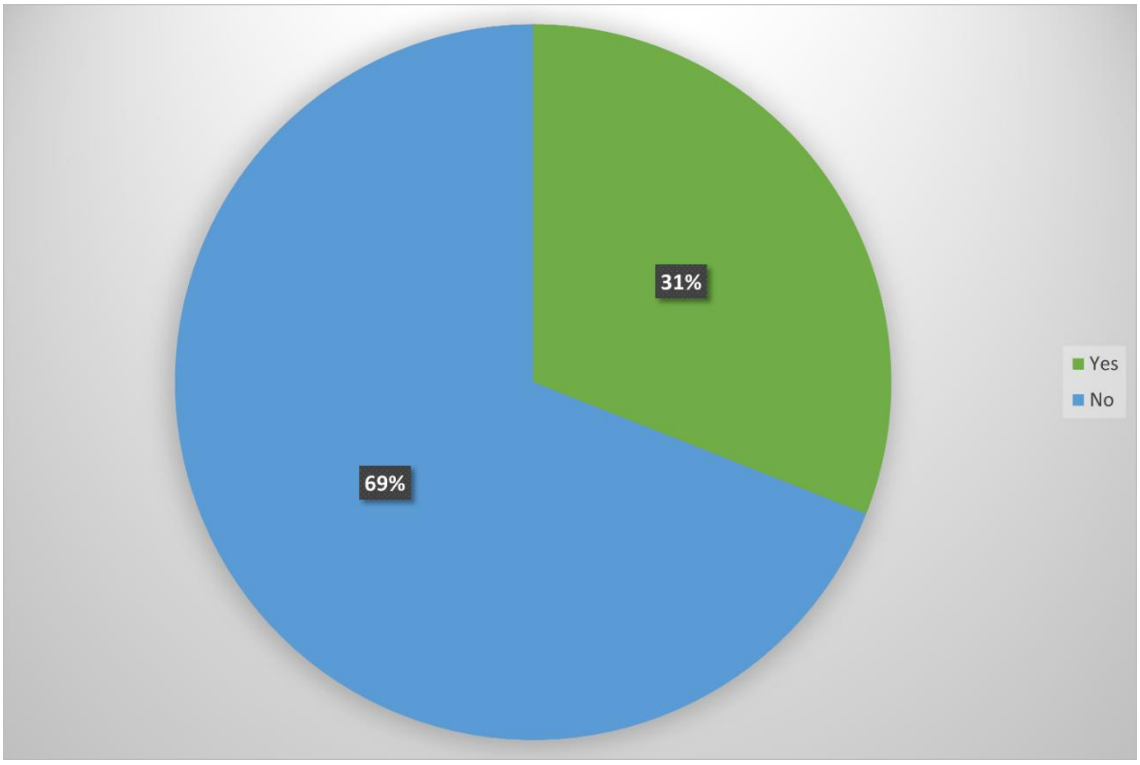


Figure 2.6 Reviewer opinion percentage for question 6

7. Are you interested in automizing this bamboo cutting process using a machine?
   a) Yes
   b) No

Table 2.7 Data collection table for question 7

Option	Response Count	Response Percent (%)
Yes	24	75
No	8	25

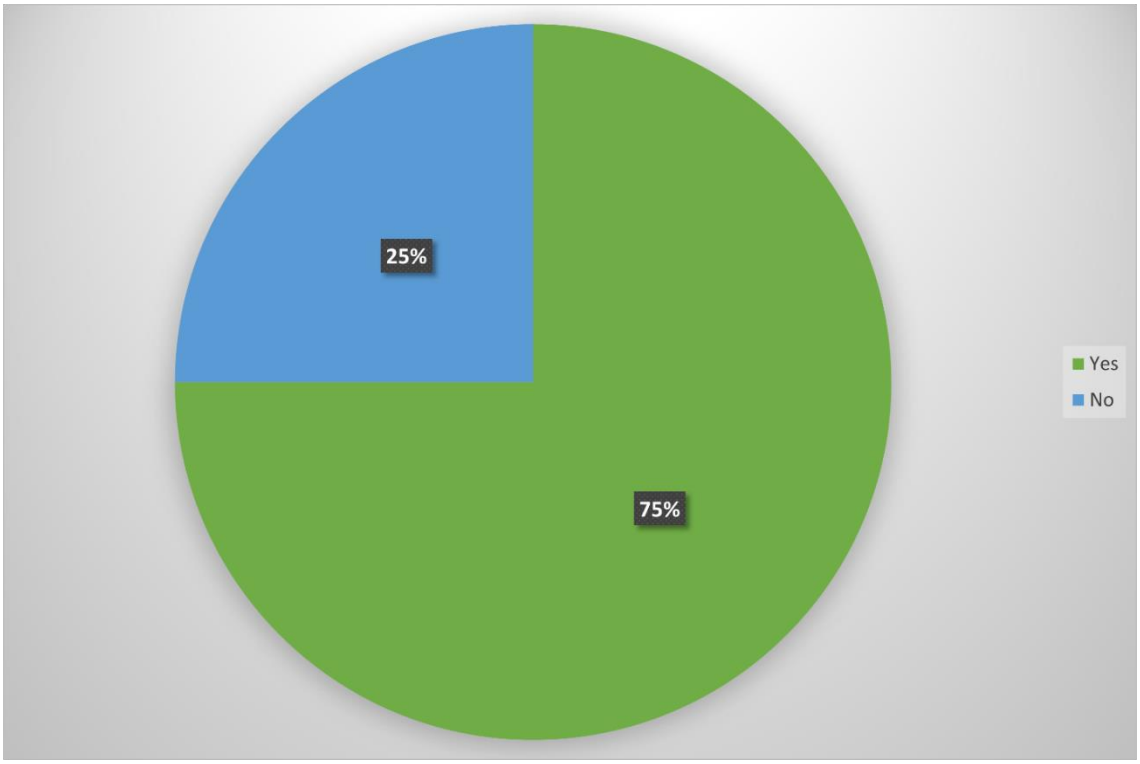


Figure 2.7 Reviewer opinion percentage for question 7

8. Which of the following method do you prefer?
    a) Traditional (Manual)
    b) Electrically operable automatic machine with motor
    c) Mechanically designed semi-automatic

Table 2.8 Data collection table for question 8

Option	Response Count	Response Percent (%)
Traditional (Manual)	8	25
Electrically operable automatic machine with motor	13	41
Mechanically designed semi-automatic	11	34

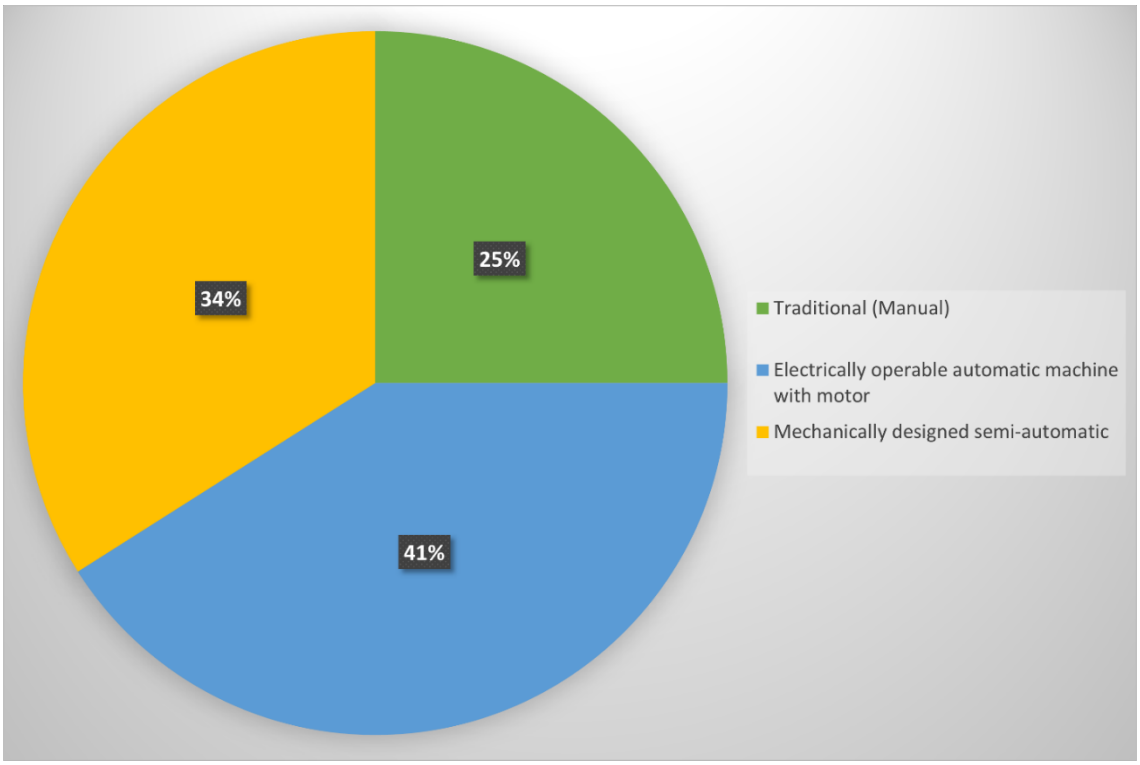


Figure 2.8 Reviewer opinion percentage for question 8

9. How much would you like to spend over a bamboo cutting machine?
    a) 5,000-10,000tk
    b) 10,000-15,000tk
    c) 15,000-20,000tk
    d) 20,000-25,000tk

Table 2.9 Data collection table for question 9

Option	Response Count	Response Percent (%)
5,000-10,000tk	20	62
10,000-15,000tk	7	22
15,000-20,000tk	4	13
20,000-25,000tk	1	3

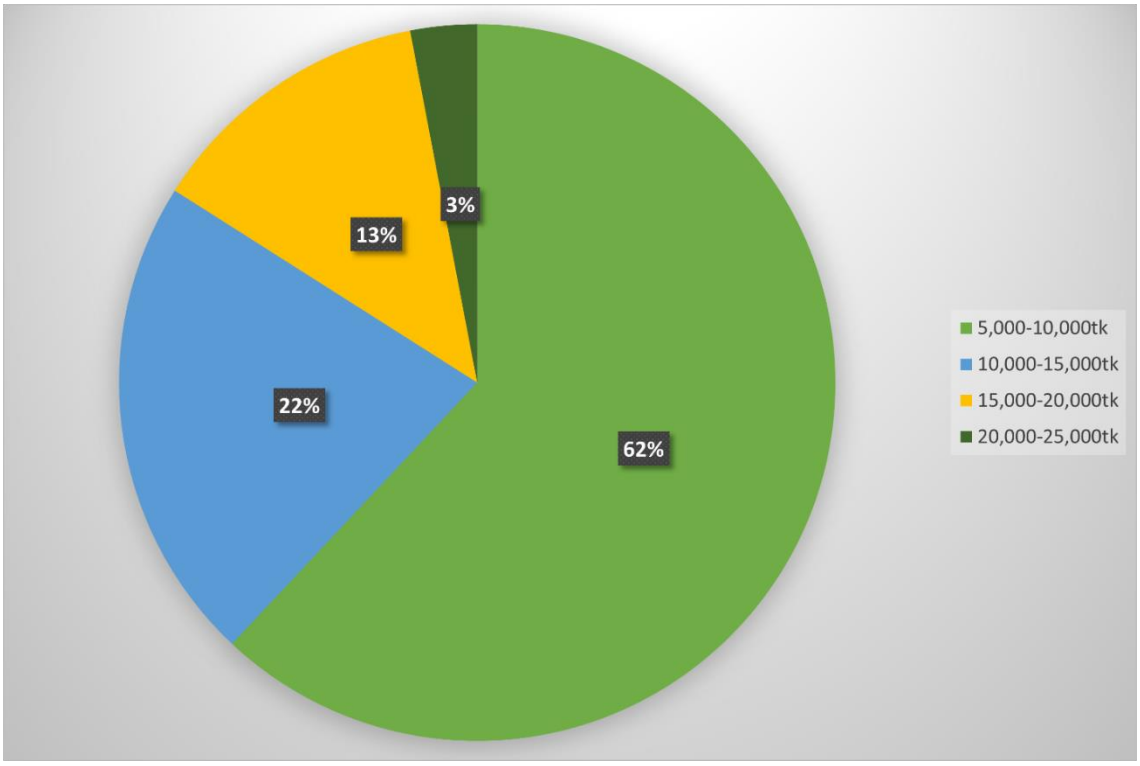


Figure 2.9 Reviewer opinion percentage for question 9

10. Are you interested in a machine which can handle different sizes and thicknesses of bamboo?
    a) Yes
    b) No

Table 2.10 Data collection table for question 10

Option	Response Count	Response Percent (%)
Yes	27	84
No	5	16

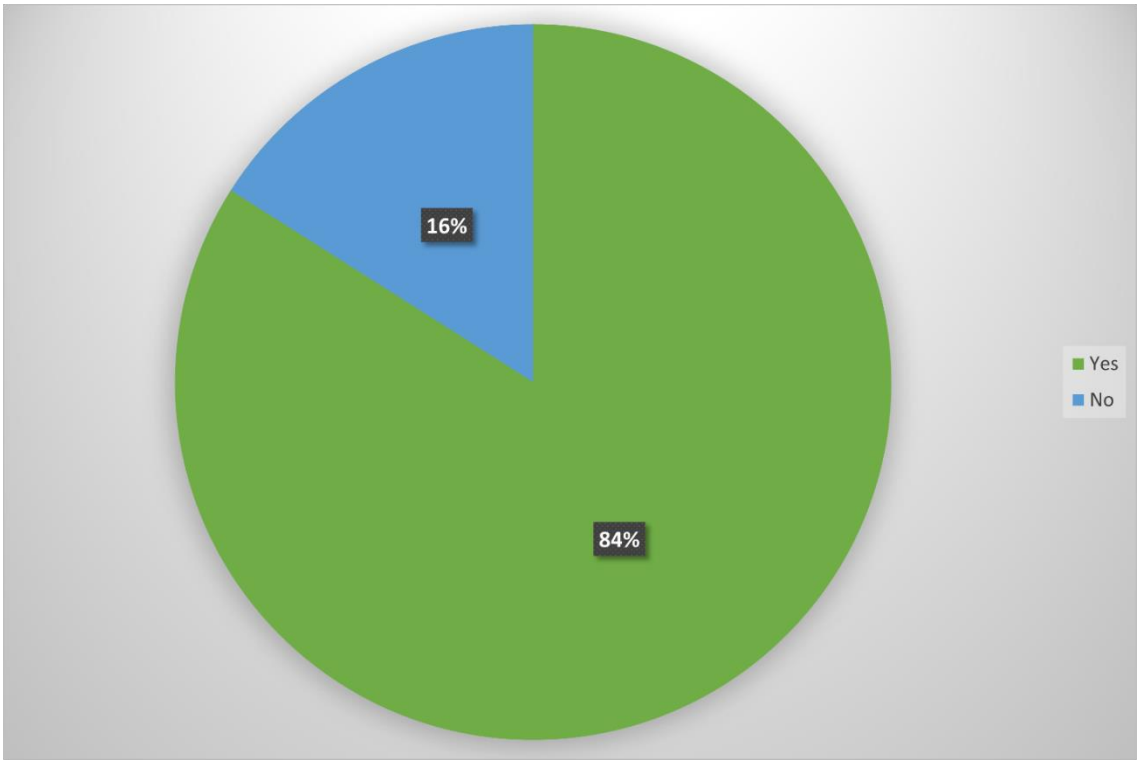


Figure 2.10 Reviewer opinion percentage for question 10

11. How important is the safety feature of a bamboo cutting machine to you?
    a) Very much
    b) At moderate level
    c) Not that much

Table 2.11 Data collection table for question 11

Option	Response Count	Response Percent (%)
Very much	22	68.75
At moderate level	8	25
Not that much	2	6.25

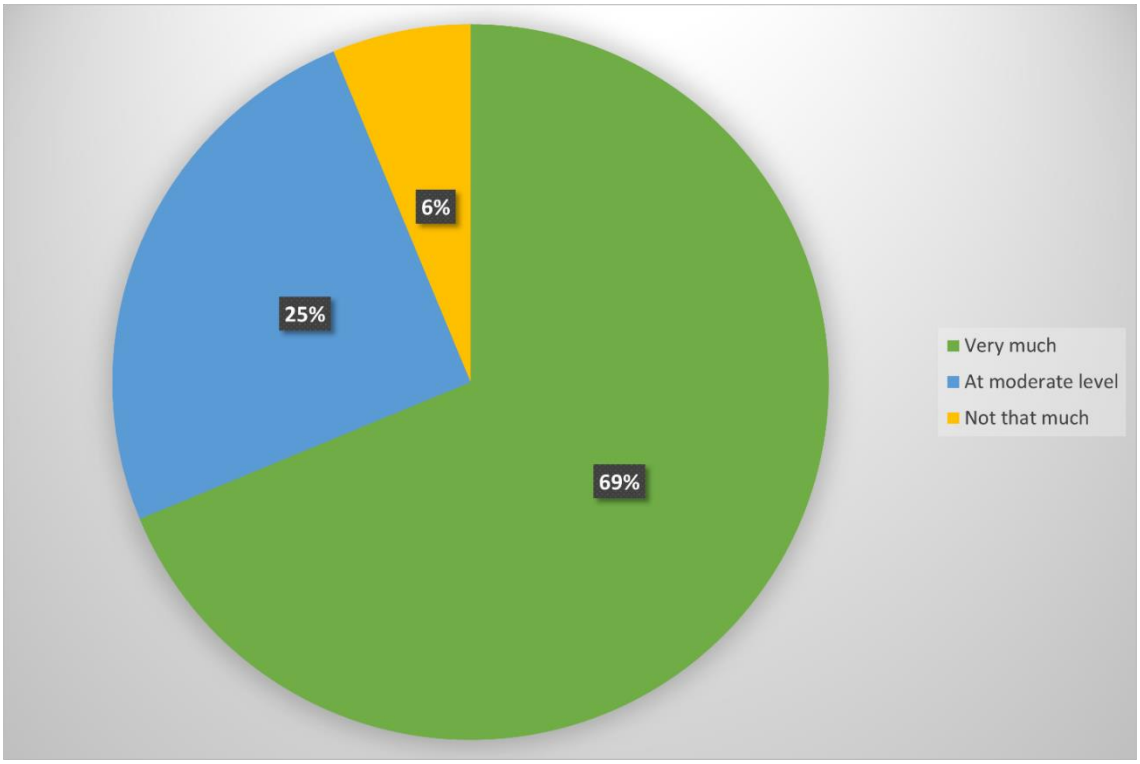


Figure 2.11 Reviewer opinion percentage for question 11

12. How does using a bamboo cutting machine impact your productivity or efficiency compared to a manual cutting method?
    a) Very much
    b) At moderate level
    c) Little bit
    d) Not that much

Table 2.12 Data collection table for question 12

Option	Response Count	Response Percent (%)
Very much	17	53.125
At moderate level	8	25
Little bit	4	12.5
Not that much	3	9.375

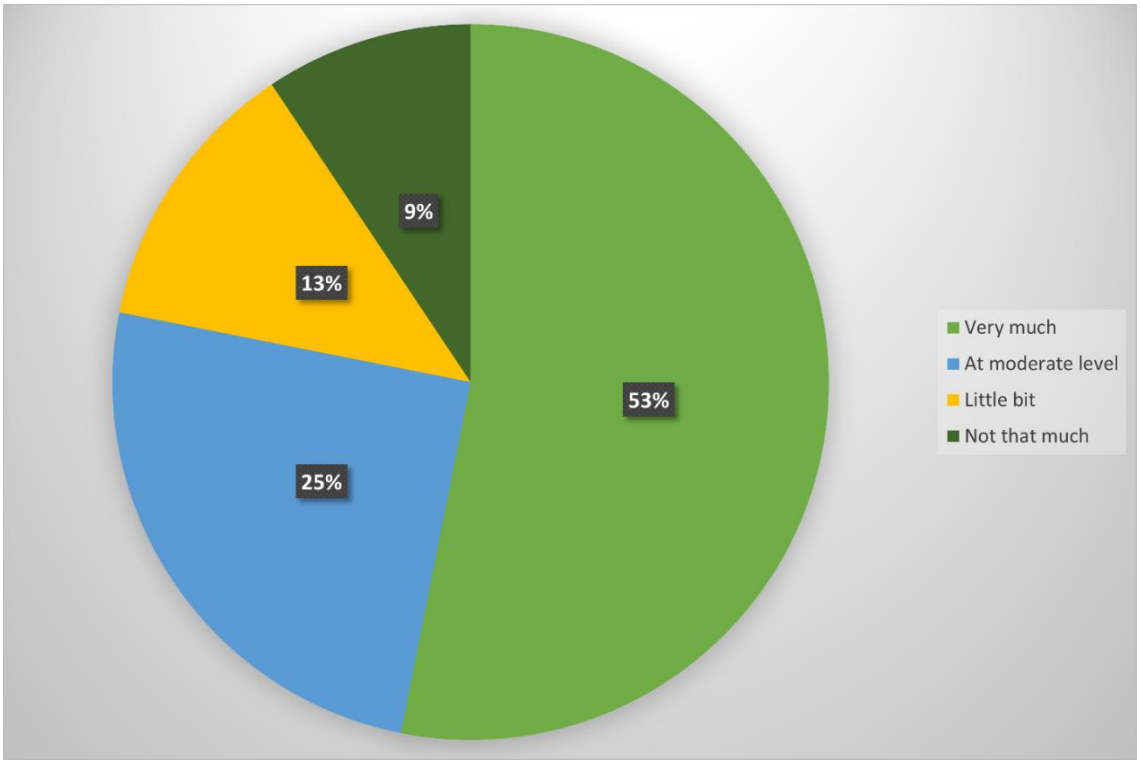


Figure 2.12 Reviewer opinion percentage for question 12

13. What improvement or additional features you would like to see in a bamboo cutting machine?
    a) Cost effectiveness
    b) Safety features
    c) Variety of functions
    d) Ease of use

Table 2.13 Data collection table for question 13

Option	Response Count	Response Percent (%)
Cost effectiveness	13	40.63
Safety features	10	31.25
Variety of functions	2	6.25
Ease of use	7	21.88

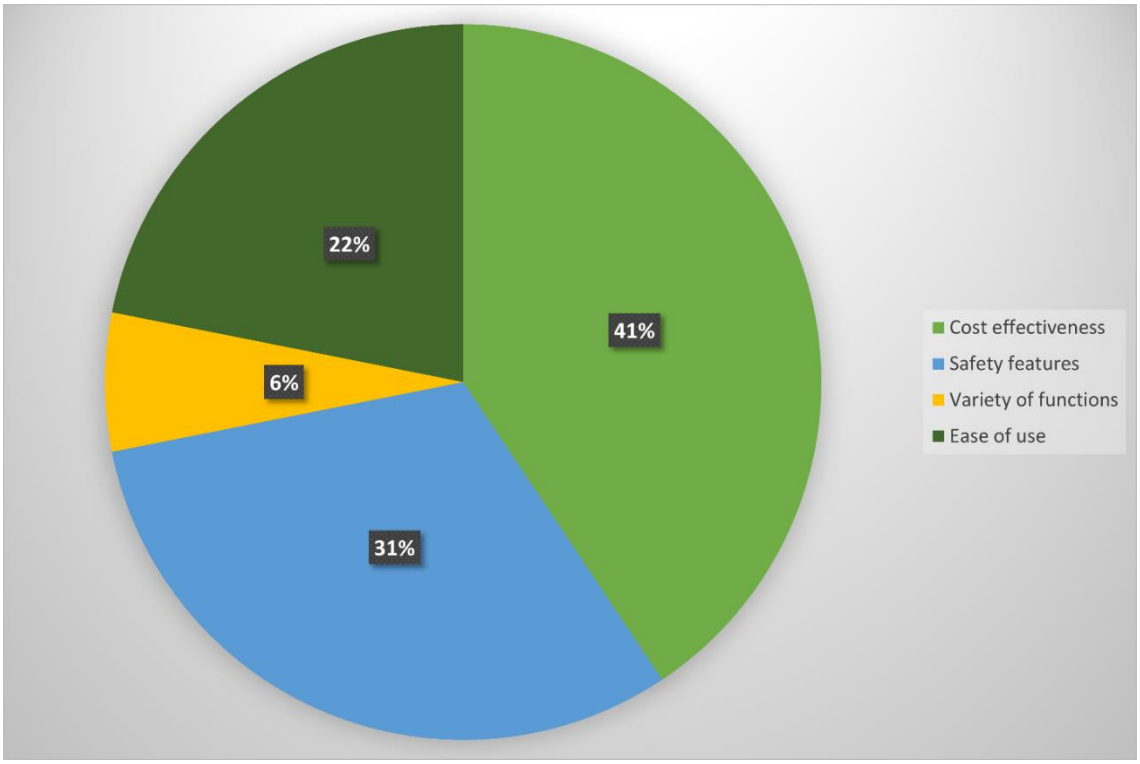


Figure 2.13 Reviewer opinion percentage for question 13

14. Motor-driven automatic bamboo cutters are available in the market. How likely would you prefer buying one of those, given the difficulties of manual bamboo cutting?
    a) I would buy one.
    b) I would buy one if I could afford it.
    c) I am only interested in traditional methods.

Table 2.14 Data collection table for question 14

Option	Response Count	Response Percent (%)
I would buy one.	9	28.125
I would buy one if I could afford it.	15	46.875
I am only interested in traditional methods.	8	25

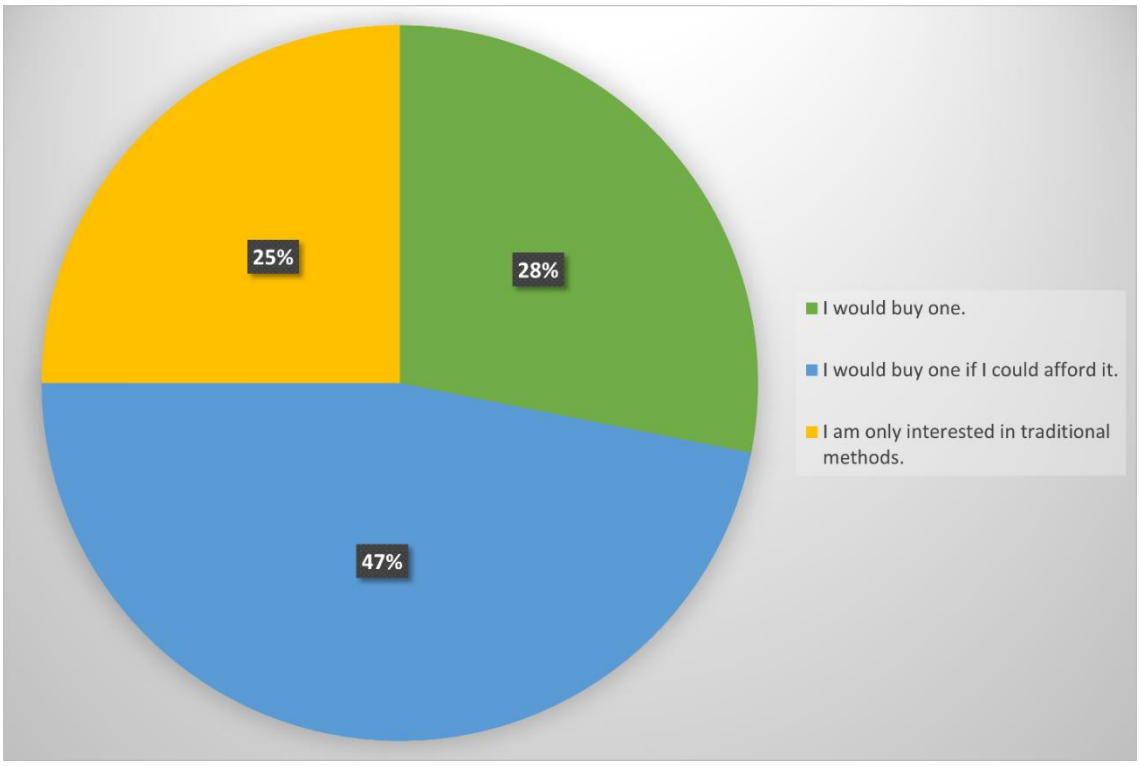


Figure 2.14 Reviewer opinion percentage for question 14

15. If you could cut bamboo more efficiently and with less effort by means of a relatively cheaper, pulley-operated bamboo cutter, would you prefer to purchase it?
    a) Yes
    b) No
    c) Cannot say for sure.

Table 2.15 Data collection table for question 15

Option	Response Count	Response Percent (%)
Yes	19	59.375
No	7	21.875
Cannot say for sure.	6	18.75

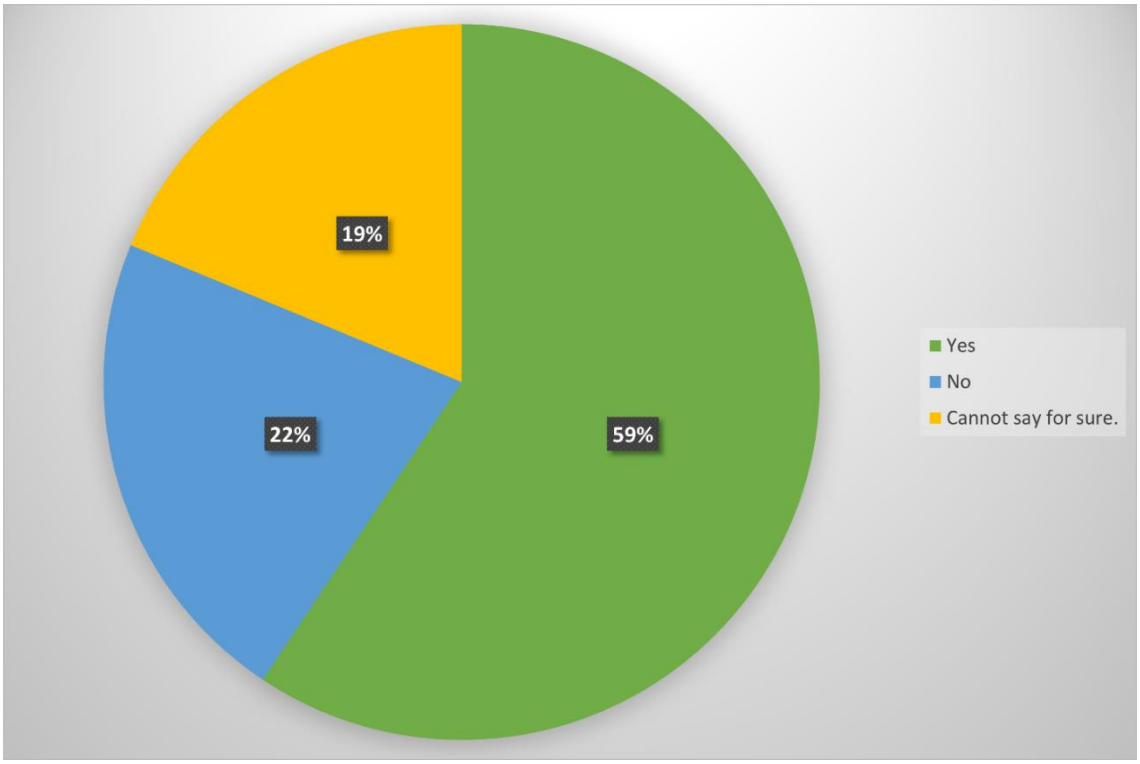


Figure 2.15 Reviewer opinion percentage for question 15

16. On a scale 1 to 10, how likely are you to recommend a bamboo cutting machine to others?
    a) Scale 1
    b) Scale 2
    c) Scale 3
    d) Scale 4
    e) Scale 5

Table 2.16 Data collection table for question 16

Option	Response Count	Response Percent (%)
Scale 1	1	3
Scale 2	1	3
Scale 3	3	10
Scale 4	11	34
Scale 5	16	50

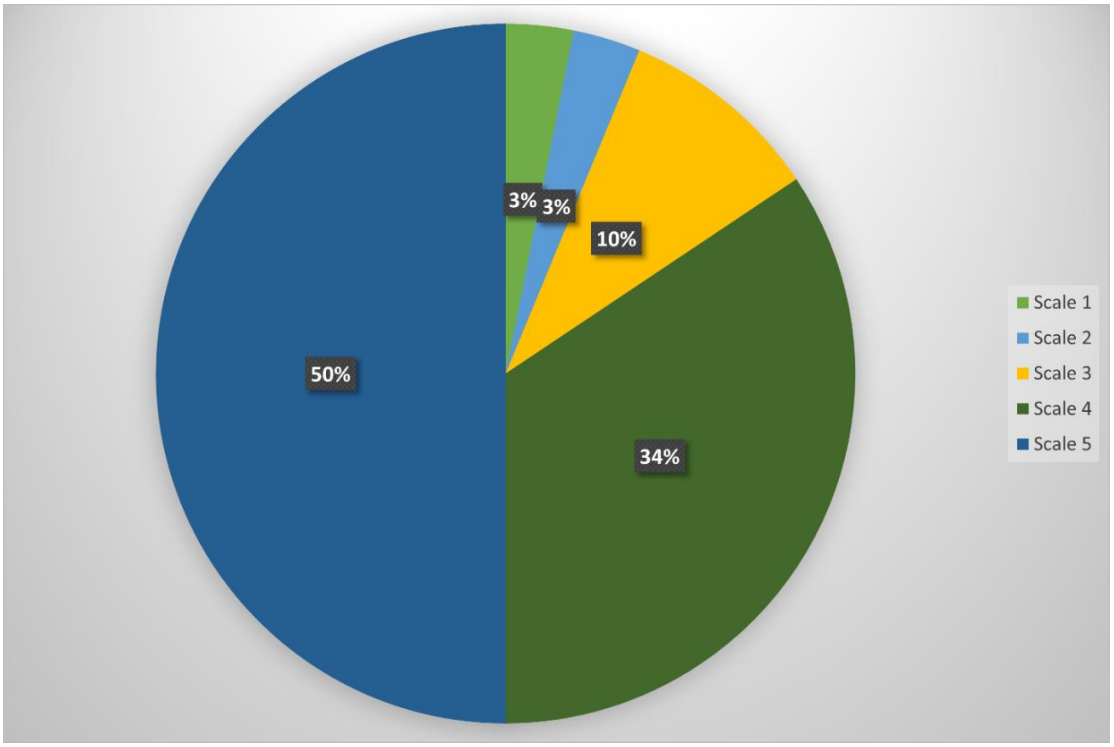


Figure 2.16 Reviewer opinion percentage for question 16

2.4 Customer Requirement Evaluation

Customer Requirement	**Relative Weight/ Importance (Scale of 10)**
Easy to Move	07
Safety Features	09
Time Saving	09
Precision and Accuracy	07
Variety of Functions	08
Low Cost	10
Easy To Maintain/Repair	08
Durability and Reliability	07
Ramming Force	09

2.5 Conclusion

In conclusion, the bamboo cutting machine has proven to be a highly efficient and reliable solution for bamboo processing, offering significant time and labor savings. Its precise cutting capabilities, adjustable settings, and user-friendly interface have garnered positive feedback from users, highlighting its effectiveness in meeting the demands of various industries. With its ability to streamline production processes and enhance overall productivity, the bamboo cutting machine is undoubtedly an asset in the bamboo industry.

Chapter-3

Incorporating the Voice of Customer in Product Design with Quality Function Deployment (QFD)

3.1 Introduction

Quality Function Deployment (QFD) is a product development and design methodology that originated in Japan. This tool is a customer-driven, systematic procedure which is used to translate customer needs and demands into engineering functionalities and requirements. As we know, quality is a determinant of the customer, not the producer. The primary goal of QFD is to ensure that the final product or service aligns with customer requirements, and possibly exceeds the expectations of the customer.

The "Voice of the Customer" (VOC) involves describing the stated and unstated (but implied) customer needs or requirements. The core concept of QFD revolves around the "Voice of the Customer". Capturing the VOC involves gathering and analyzing customer needs in a variety of ways:

- Discussion through interviews
- Survey data
- Focus groups
- Opinions from experts
- Educated Guess
- Other means of market research

The next step after gathering data about customer requirements involves summarizing the data in the "House of Quality" (HOQ), which is a unique matrix-like representation resembling a house. For our project, we have collected relevant information through interviews and survey questionnaires with our target customers, a few educated guesses where specific data was absent, and some internet research. Ultimately, by applying the Quality Function Deployment, we plan to enhance our product and its development process such that it addresses the customer needs satisfactorily.

3.2 Customer Requirements and Engineering Requirements Relationship

The relationships between customer requirements and engineering requirements have been tabulated below.

Table 3.1: Relationship Explanation

Customer Requirements	Engineering Requirements	Relationship	Explanation
Easy to Move	Design of Components	Strong	Compared to automatic cutter machines, our manually controllable product is easier to move due to its better design components.
Safety Features	Pulley System	Medium	Traditional cutting process involves various safety concerns. Our pulley system is designed considering all these safety issues.
	Cutter Sharpness	Low	Cutter sharpness ensures safety by reducing required force and vibration while providing a smooth finish.
	Manufacturing Cost	Medium	Manufacturing cost increases a bit with additional safety features.
	Rope Strength	Strong	If rope is not strong enough to endure the applied force, it will tear, and accidents may happen.
	Design of Components	Strong	Better design of components ensures better safety.

Time Saving	Pulley System	Strong	Pulley system empowered the whole process to be more efficient and quicker.
	Cutter Sharpness	Strong	A sharp cutter helps bamboos to pass through it quickly and smoothly.
	Design of Components	Low	Performance of the components can be enhanced by designing them properly.
Precision & Accuracy	Pulley System	Strong	A well calibrated pulley system ensures minimal vibrations, resulting in precise and accurate cuts for smoother operation.
	Cutter Sharpness	Strong	A finer cutter sharpness leads to reduced force requirements, minimized vibrations and smoother bamboo cuts.
	Rope Strength	Medium	The rope strength directly impacts both precision and accuracy as insufficient rope's strength may break during operation leading to inaccuracies and inconsistencies in the cutting process.
	Design of Components	Low	The precision and accuracy of a bamboo cutting machine are influenced by the design of its components as their proper engineering and alignment ensures consistent and reliable cutting performance.
Variety of Functions	Pulley System	Low	As the variety of functions increases, requiring it to cut multiple bamboos, the pulley system must be designed to handle higher forces to ensure efficient and effective cutting.

	Manufacturing Cost	Strong	As the variety of functions increases, the manufacturing cost will also increase due to the additional complexity and customization required to produce and integrate multiple types of cutters.
	Operational Cost	Low	The operational cost tends to slightly increase due to the higher maintenance and complexity associated with the multiple cutter options.
	Design of Components	Medium	The variety of functions are correlated with the design of components, as providing different shapes of cutters enables the machine to cut bamboo of various sizes effectively and efficiently.
Low Cost	Pulley System	Low	The relationship between low cost and pulley system is that the pulley system serves as a cost-effective alternative to a motorized bamboo cutting machine which offers an economical solution for cutting bamboo efficiently.
	Cutter Sharpness	Medium	To provide perfect cutter sharpness overall cost will increase.
	Strength of Main Body	Strong	The cost will increase due to the requirement for expensive materials to achieve the desired strength level of the main body.
	Cutter Material	Strong	Investing in more expensive, high-quality materials is necessary to ensure proper strength and functionality which has inverse relationship with “low cost”.

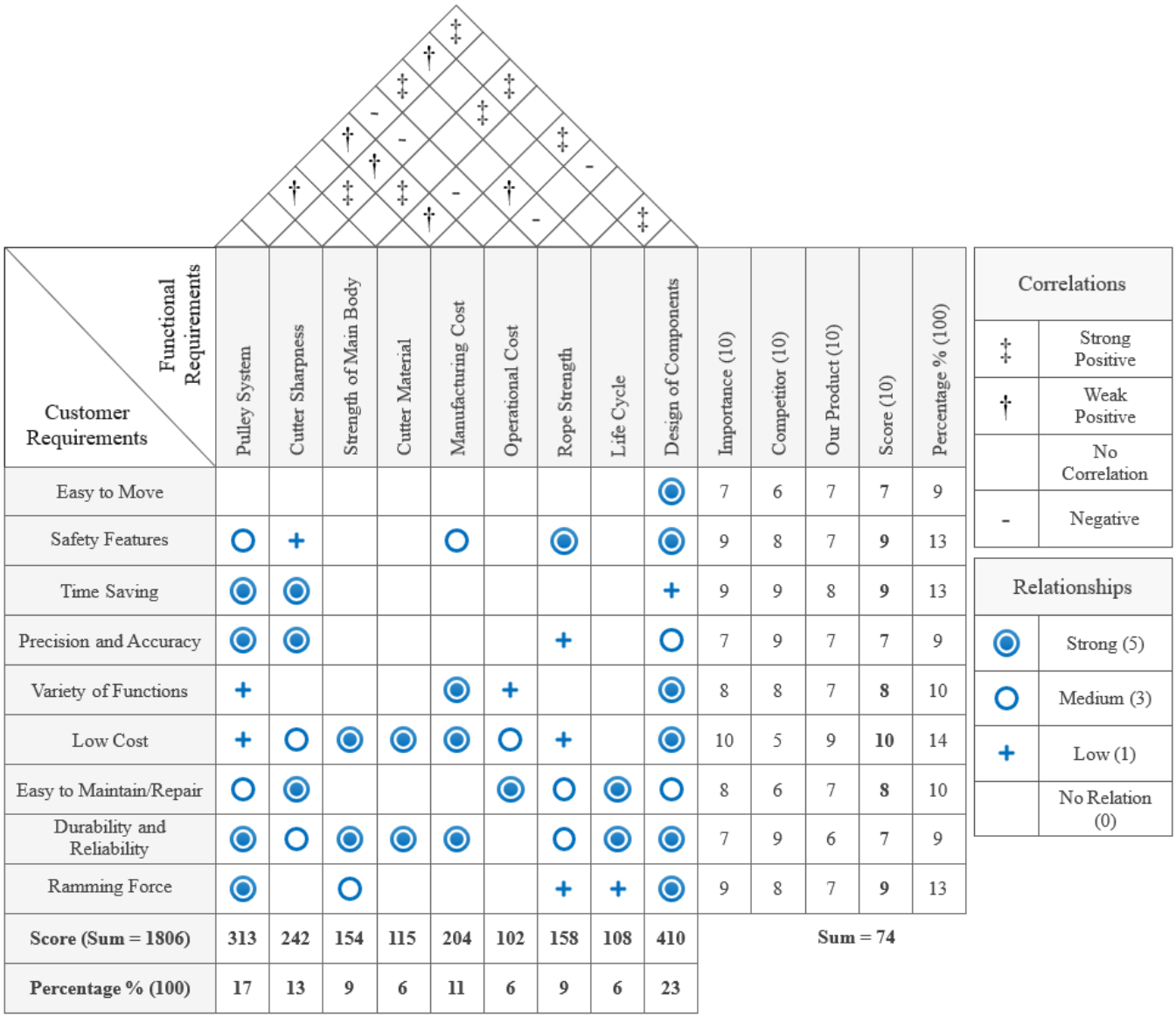
	Manufacturing Cost	Strong	The manufacturing cost will be low for our manually operated bamboo cutting machine compared to motorized automatic bamboo cutting machine.
	Operational Cost	Medium	The use of different shaped cutters to cut bamboo could lead to higher operational costs due to increased maintenance expenses associated with maintaining and replacing various cutters.
	Rope Strength	Low	A less expensive rope may compromise the strength required for efficient bamboo cutting so we need more expensive, high quality, higher strength rope which will increase cost.
	Design of Components	Strong	A well-engineered and simplified design of components for the bamboo cutting machine, utilizing an efficient pulley system and carefully selected cutter shapes, will lead to a lower overall cost for the machine.
Easy to Maintain/ Repair	Pulley System	Medium	Complex pulley system will make the maintenance operation difficult.
	Cutter Sharpness	Strong	A cutter with worn-down edge will lead to frequent repairing. On the contrary, a sharp cutter will lessen often repairing.
	Operational Cost	Strong	Maintenance or repairing of the machine will increase the overall operational cost.
	Rope Strength	Medium	Rope strength should be higher so that it doesn’t need to repair regularly.

	Life Cycle	Strong	If the maintenance management is good, the life cycle of the machine will increase. A good maintenance management can be ensured with a simple machine design.
	Design of Components	Medium	The complex design of components will make it laborious to repair.
Durability & Reliability	Pulley System	Strong	A high-quality pulley system can make the machine more reliable while lessening repair.
	Cutter Sharpness	Medium	The better the cutter sharpness, the higher possibility of being the machine durable. A less sharp cutter will require to fix often which will reduce durability and reliability.
	Strength of Main Body	Strong	The main frame must be strong enough to provide a more durable machine.
	Cutter Materials	Strong	If the cutter material is not so good, it can break with a low applied force which will result a low durability and the machine will also be less reliable.
	Manufacturing Cost	Strong	To make the machine more durable and reliable, the overall manufacturing cost will be higher.
	Rope Strength	Medium	Rope with less strength will not be able to provide a good durable and reliable machine.

	Life Cycle	Strong	A durable machine will provide a better machine life cycle compared to a less durable machine.
	Design of Components	Strong	The better the design of the components, the more durable and reliable the machine is.
Ramming Force	Pulley System	Strong	The numbers of pulleys will strongly affect the ramming force. The higher the number of pulleys, the higher the ramming force will be.
	Strength of Main Body	Medium	A machine with higher main frame strength can handle a higher ramming force.
	Rope Strength	Low	A low strength rope sometimes will not be able to provide a good ramming force.
	Life Cycle	Low	If the ramming force is excessive, the machine will encounter vibrations and shaking which may increase repair operation and hence reduce the life cycle.
	Design of Components	Strong	Components with better strength will be able to endure higher ramming force. Also, proper design of bearings and slider will significantly increase the ramming force.

3.3 House of Quality

Quality Function Deployment's House of Quality is given below:



Customer Requirements \ Functional Requirements	Pulley System	Cutter Sharpness	Strength of Main Body	Cutter Material	Manufacturing Cost	Operational Cost	Rope Strength	Life Cycle	Design of Components	Importance (10)	Competitor (10)	Our Product (10)	Score (10)	Percentage % (100)
Easy to Move									◉	7	6	7	**7**	9
Safety Features	○	+			○		◉		◉	9	8	7	**9**	13
Time Saving	◉	◉							+	9	9	8	**9**	13
Precision and Accuracy	◉	◉					+		○	7	9	7	**7**	9
Variety of Functions	+				◉	+			◉	8	8	7	**8**	10
Low Cost	+	○	◉	◉	◉	○	+		◉	10	5	9	**10**	14
Easy to Maintain/Repair	○	◉				◉	○	◉	○	8	6	7	**8**	10
Durability and Reliability	◉	○	◉	◉	◉		○	◉	◉	7	9	6	**7**	9
Ramming Force	◉		○				+	+	◉	9	8	7	**9**	13
Score (Sum = 1806)	**313**	**242**	**154**	**115**	**204**	**102**	**158**	**108**	**410**			**Sum = 74**		
Percentage % (100)	**17**	**13**	**9**	**6**	**11**	**6**	**9**	**6**	**23**					

Correlations	
‡	Strong Positive
†	Weak Positive
	No Correlation
-	Negative

Relationships	
◉	Strong (5)
○	Medium (3)
+	Low (1)
	No Relation (0)

Figure 3.1: House of Quality

3.4 Importance Rating Table

The functional requirements have been arranged in the table below according to their importance rating (percentage of total score) in the House of Quality:

Table 3.2: Importance Rating

Rank	Functional Requirement	Importance Rating (%)
1	Design of Components	23
2	Pulley System	17
3	Cutter Sharpness	13
4	Manufacturing Cost	11
5	Rope Strength	9
6	Strength of Main Body	9
7	Cutter Material	6
8	Life Cycle	6
9	Operational Cost	6

In conclusion, QFD is a tool of utmost importance that helps convert customer requirements and demands into actual design parameters to be used in the production process. It also helps prevent the wastage of time and resources on features that may not be adding any value to the product. By the systematic linking of customer needs to technical specifications, QFD bridges the gap between customer expectations and the final product, thus leading to satisfaction of the customers and success in business.

Chapter-4

Functional Decomposition

4.1 Introduction

Functional Decomposition is a systematic method of breaking down a complicated system or product into smaller functions or components, which permits for a more manageable and understandable system. Customers want to use a product efficiently and with ease. To them, the working procedure of the components are not of importance; rather it is about how conveniently they are able to maneuver or operate the individual functions of the product to get the job done. The black box method is a great tool to demonstrate this. The process of Functional Decomposition also involves preparing a functional hierarchy chart which mainly distinguishes among the various components from the surface of the product to its core, and a cluster function structure which clusters related functions together and helps reduce the complexity of the overall product or system.

For successful functional decomposition, there are mainly four basic steps to be followed:

Step 1: Identify the most general function to be accomplished

Almost every design problem has one (or two) “major” function(s). This function (or functions) needs to be reduced to a simple clause and placed in a *black box*. The inputs to this box consist of the energy, materials and information that flow into the system. The outputs form the box are what flow out of the system.

Step 2: Determine the closest Sub-functions

This step involves identifying the sub-functions that are nearest to the major function(s).

Step 3: Order the Sub-functions

Rearrange the functions identified in Step 2. Then the goal is to accomplish the general function in Step 1.

Step 4: Refine the Sub-functions

The objective of this step is to analyze and decompose the sub-functions as finely as possible; we check if the sub-functions identified in the previous steps can be further divided into more sub-functions.

4.2 Black Box Model of Functional Decomposition

A Black Box Model is an abstracted entity that represents a system by focusing solely on the input-output relationships, without explicitly showing any internal processes and details. The product or system is viewed as an opaque box with inputs entering from one side and outputs emerging from the other side. The internal mechanisms or operations of the system or product are invisible and irrelevant for understanding its overall function.

The Black Box Model for Convenient Bamboo Cutter is shown below:

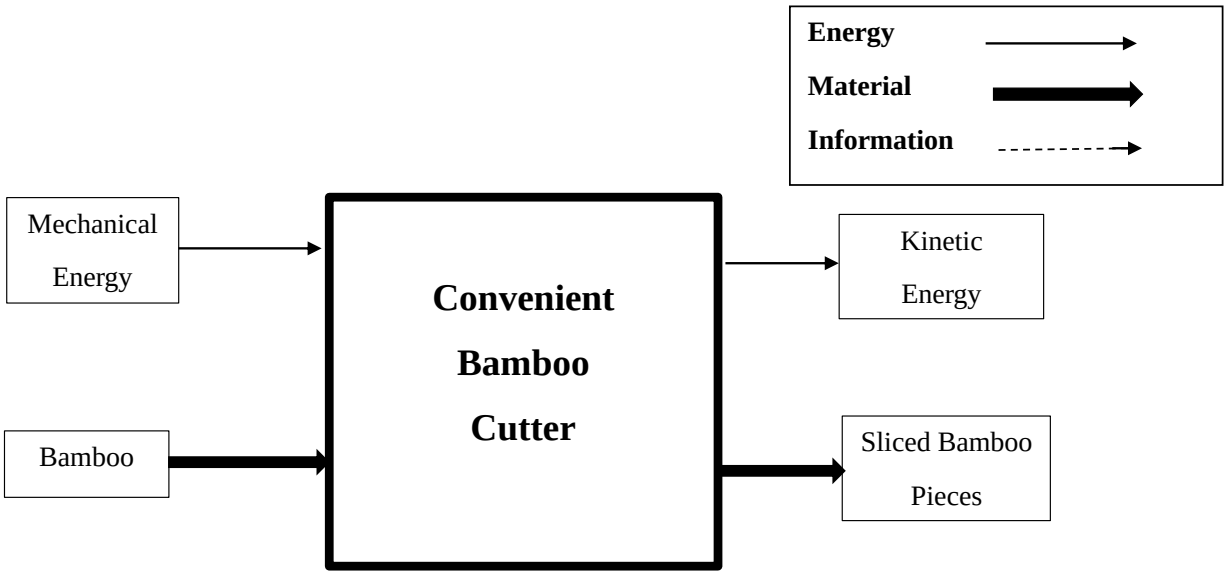


Figure 4.1: Black Box Model of Convenient Bamboo Cutter

There are generally three types of flows that are shown in the black-box model. These are:

- Energy flow
- Material flow
- Information flow.

Our product has no Information flow. The other two types of flow are discussed below:

1. **Energy flow**: Our product will receive mechanical energy from the operator which is converted into kinetic energy and it will be used to compact the bamboo for slicing.

2. **Material flow**: The frame will feature a bamboo supporting slot that allows the bamboo to connect with the cutter. Subsequently, the rammer will apply kinetic energy to strike the bamboo, resulting in the slicing of the bamboo. Our product will take bamboo as input and give sliced bamboo as output.

4.3 Component Hierarchy

Component hierarchy is an effective method which involves decomposing the major function(s) of a product or system into sub-functions. This methodology helps distinguish the core parts of the product and breaks these down into smaller, more manageable parts, and then arranges them in a hierarchical manner based on their relationships and dependencies. The top level of the hierarchy chart is the main assembly (the product itself), followed by sub-assemblies and then components (and so on, if applicable). When all sub-assemblies are satisfied, the prime function is satisfied. This can be repeated iteratively down several levels, developing a function tree.

Component Hierarchy chart for our Convenient Bamboo Cutter is shown in the following page:

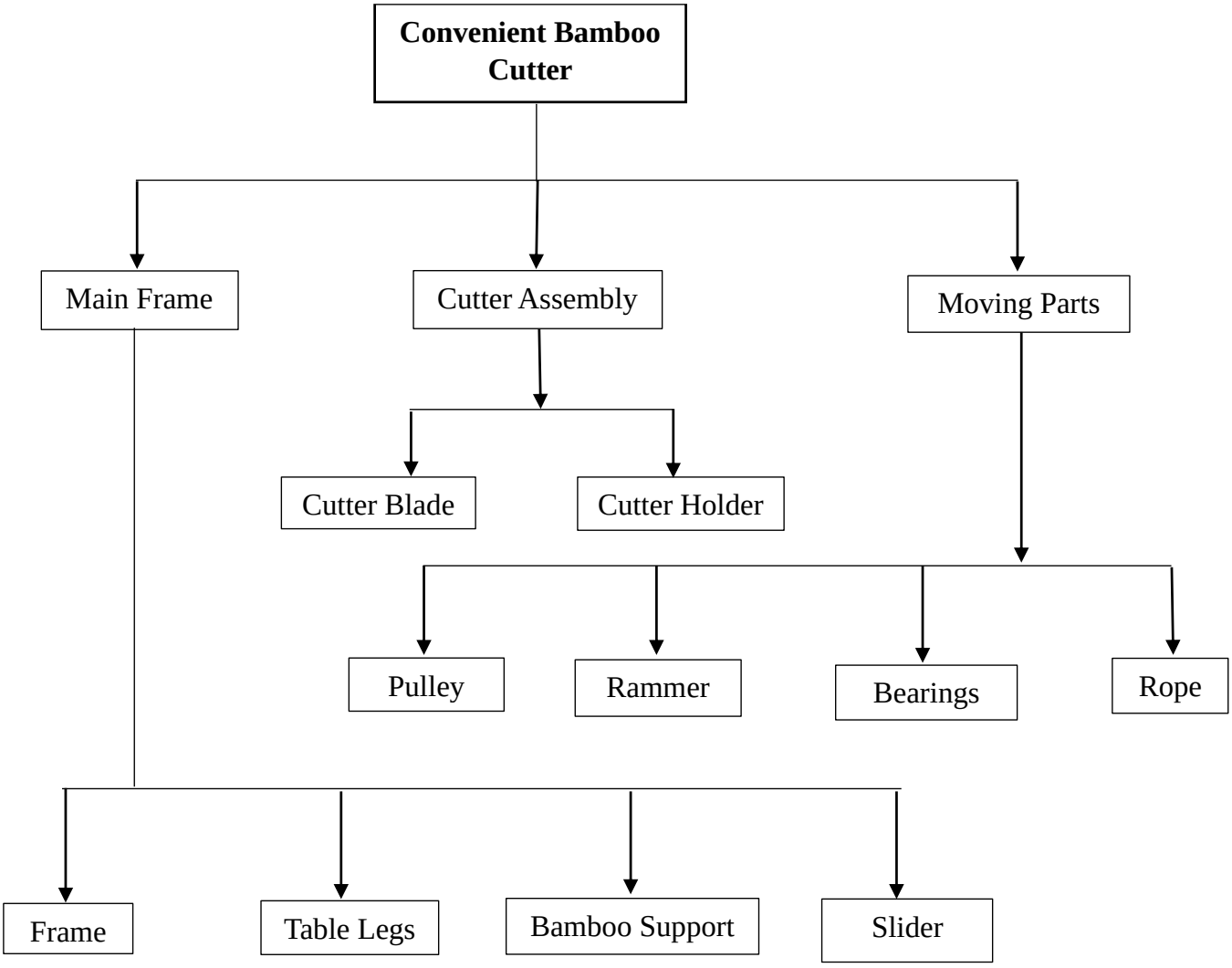


Figure 4.2: Component Hierarchy Chart for Convenient Bamboo Cutter

4.4 Cluster Function Structure of Convenient Bamboo Cutter

The Cluster Function Chart illustrates the functions performed by different assemblies or components of the product. After identifying the different functions, they are grouped or clustered based on their similarities or relationships. Mainly the energy transfers from one component to another is illustrated in this structure.

The Cluster Function Structure for Convenient Bamboo Cutter is shown below:

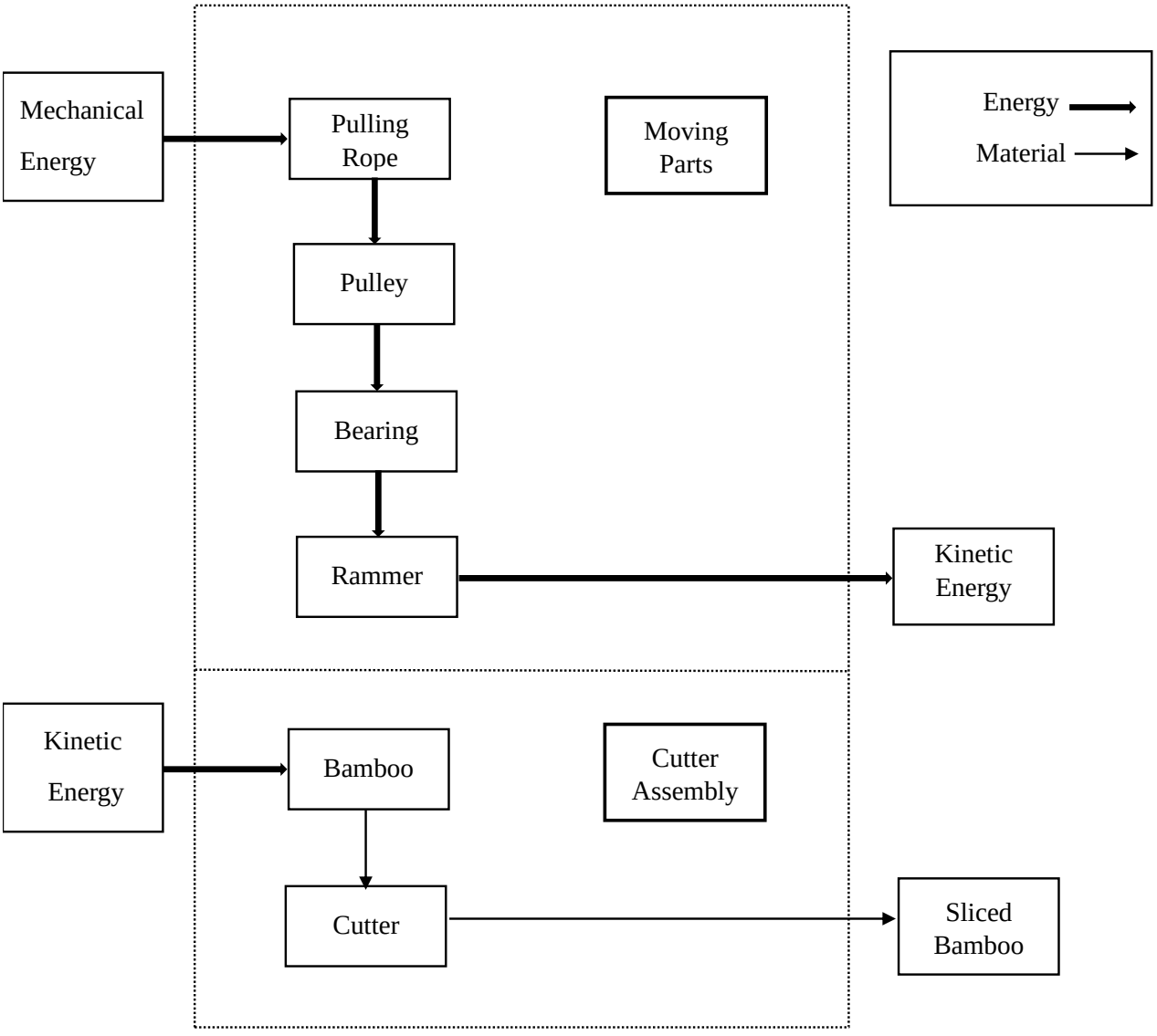


Figure 4.3: Cluster Function Structure for Convenient Bamboo Cutter

4.5 Functional Analysis System Technique (FAST) Diagram

Functional Analysis System Technique (FAST) is a technique which is used to diagrammatically represent the logical relationships between the functions of a product, system or project based on the questions “How”, “Why” and “When”.

The FAST diagram for Convenient Bamboo Cutter is shown below:

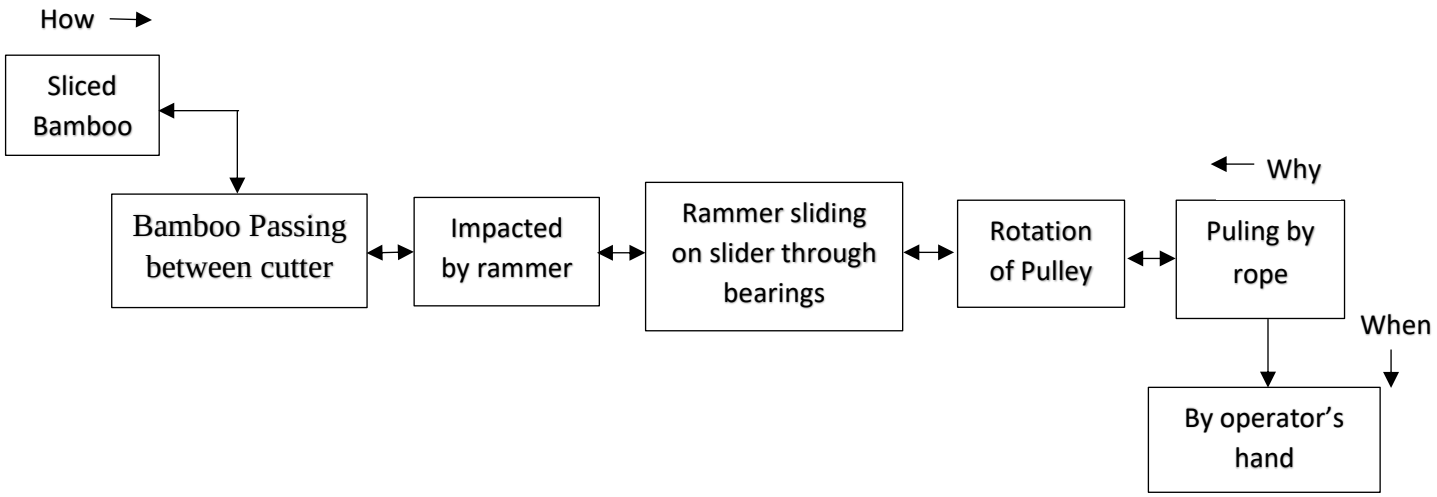


Fig 4.4: FAST Diagram for Convenient Bamboo Cutter

The methods of functional decomposition used in this chapter are important for reducing the overall complexity of our product and help in the streamlined designing of the product. The Black Box Diagram shows three different kinds of inputs and outputs without any details of the internal processes; the Component Hierarchy Chart and the Cluster Function Structure illustrates the different parts and sub-parts and the transfer of energy within them; and the FAST diagram logically relates the various functions. Ultimately, the goal of functional decomposition is to help set up the conceptual design of the product.

Chapter-5

Qualitative Analysis of Material, Manufacturing Process & Joining Process Selection

As the conceptual development of Convenient Bamboo Cutter is progressing, it is important that we assign specific materials to the various components of our product. Along with basic knowledge of material properties and manufacturing methods, we will qualitatively analyze and select the materials, manufacturing processes and joining processes, along with finishing and coloring processes, to be used for building our product.

5.1 Selection of Materials for Different Sections

5.1.1 Main Body:

The qualitative analysis of material selection for the parts of main body is given below:

Table 5.1: Qualitative analysis of material selection for the parts of main body

Section	Parts	Material selection	Option for other materials	Reasons for selection
Main Body	Frame	Mild Steel	Aluminum Alloy, Stainless Steel	Durability, Long- lasting, Inexpensive
	Bamboo Support	Bamboo	Steel, Wood	Easily available, Inexpensive and light weight
	Bamboo Storage	Wood	Steel, Bamboo	Sustainable, Inexpensive, Light Weight and availability

	Supporting Feet	Natural and Synthetic Rubber	Nylon, Leather, Steel	Sustainable, Inexpensive, Light Weight and availability
	Table Legs	Mild Steel	Wood, Aluminum, Stainless Steel	Sustainable, Inexpensive, Light Weight and availability

5.1.2 Cutter Assembly:

The qualitative analysis of material selection for the parts of cutter assembly is given below:

Table 5.2: Qualitative analysis of material selection for the parts of cutter assembly

Section	**Parts**	**Material Selection**	**Option for other materials**	**Reasons for selection**
Cutter Assembly	Cutter Blade	High Speed Steel	Carbide, Mild Steel, Stainless Steel	Wear resistance, Impact resistance and high strength
	Cutter Holder	Cast iron	Stainless Steel, Mild Steel, Aluminum alloy	Inexpensive, Optimum hardness and strength

5.1.3 Moving Parts:

The qualitative analysis of material selection for the moving parts is given below:

Table 5.3: Qualitative analysis of material selection for the moving parts

Section	Parts	Material Selection	Option for other materials	Reasons for selection
Moving Parts	Pulley	Cast Iron	Aluminum, Stainless Steel, Nylon	Resistance to abrasion, High hardness and toughness
	Rope	Nylon	Jute Fiber, Cotton, Polypropylene, Wire rope	High strength, Resistance to wear and tear
	Rammer	Steel (outer part) Concrete (inner part)	Cast Iron, Wood	Reduces the amount of steel needed while maintaining sufficient impact force, Inexpensive, Durable
	Bearings	Carbon Steel	Chrome Steel, Stainless Steel, Ceramics	Wear and friction resistance, High strength, Durability

5.2 Qualitative Analysis of Manufacturing Process Selection

5.2.1 Main Body

The qualitative analysis of manufacturing process selection for the components of the main body is given below:

Table 5.4: Qualitative analysis of manufacturing process selection for the components of the main body

Section	Components	Possible Manufacturing Processes	Make or Buy	Manufacturing Process Selection	Reasons Behind Selection
Main Body	Frame	Welding, Casting, Forging	Make	Welding	Cost effective, Available method
	Bamboo Storage	Welding, Casting, Machining	Make	Welding and Machining	High strength, Easy to perform
	Supporting Feet	Welding, Casting, Machining	Make	Welding and Machining	Cheaper to perform
	Table Legs	Welding, Casting	Make	Welding	Cost effective, Available method

5.2.2 Cutter Assembly

The qualitative analysis of manufacturing process selection for the components of the cutter assembly is given below:

Table 5.5: Qualitative analysis of manufacturing process selection for the components of the cutter assembly

Section	Components	Possible Manufacturing Processes	Make or Buy	Manufacturing Process Selection	Reasons Behind Selection
Cutter Assembly	Cutter Blade	Forging, Machining, Grinding, Laser cutting, EDM	Make	Forging, Machining, Grinding	Easy to perform, Cost effective
	Cutter Holder	Welding, Casting, Machining	Make	Casting, Machining	Less cost, Easy to perform

5.2.3 Moving Parts

The qualitative analysis of manufacturing process selection for the components of the moving parts is given below:

Table 5.6: Qualitative analysis of manufacturing process selection for the components of the moving parts

Section	Components	Possible Manufacturing Processes	Make or Buy	Manufacturing Process Selection	Reasons Behind Selection
	Pulley	Casting	Buy	Outsource	Cost-effective to

Moving Parts					buy it than to make it
	Rope	Laying	Buy	Outsource	Cost-effective to buy it than to make it
	Rammer	Welding, Casting, Machining	Make	Casting, Machining	Easy to perform
	Bearings	Forging, Casting	Buy	Outsource	Cost-effective to buy it than to make it

5.3 Qualitative Analysis of Parts Joining Method Selection

5.3.1 Main Frame

The qualitative analysis of joining process selection for the components of the main frame is given below:

Table 5.7: Qualitative analysis of joining process selection for the components of the main frame

Joining Parts	**Type of Joint**	**Joining Method Options**	**Joint Selected**	**Reason Behind Selection**
Frame Top	Permanent	Arc Welding, TIG, MIG	Arc Welding	Cheap, Provides strong joint

Bamboo Storage And Frame	Permanent	Arc Welding, TIG, MIG	TIG Welding	Better joining strength, Precision in joining
Table Legs and Frame	Permanent	Spot Welding, Arc Welding, TIG, MIG	Arc Welding	Cheap, Easy to perform
Supporting Feet and Table Legs	Temporary	Fastening, Hot Glue, Epoxy Adhesive	Epoxy Adhesive	Cheap, Provides sufficient strength to the joint

5.3.2 Cutter Assembly

The qualitative analysis of joining process selection for the components of the cutter assembly is given below:

Table 5.8: Qualitative analysis of joining process selection for the components of the cutter assembly

Joining Parts	**Type of Joint**	**Joining Method Options**	**Joint Selected**	**Reason Behind Selection**
Cutter Blade	Permanent	Spot Welding, Arc Welding, TIG, MIG	TIG Welding	Precision in joining

Cutter Blade and Cutter Holder	Temporary	Bolted, Nut-Bolt	Nut-Bolt	Provides sufficient strength to the joint
Cutter Holder and Frame	Temporary	Bolted, Nut-Bolt	Nut-Bolt	Provides sufficient strength to the joint

5.3.3 Moving Parts

The qualitative analysis of joining process selection for the components of the moving parts is given below:

Table 5.9: Qualitative analysis of joining process selection for the components of the moving parts

Joining Parts	**Type of Joint**	**Joining Method Options**	**Joint Selected**	**Reason Behind Selection**
Rammer and Bearings	Temporary	Bolted, Nut-Bolt, Rivet	Rivet	Higher strength than other joints
Rammer and Frame	Temporary	Slide	Slide	Easy to move, Less friction
Pulley and Rammer	Permanent	TIG, MIG, Arc Welding	TIG Welding	Better joining strength

5.4 Qualitative Analysis of Parts Finishing Method Selection

The Qualitative Analysis of Parts Finishing Method Selection is shown below for the parts to be finished:

Table 5.10: Qualitative Analysis of Parts Finishing Method Selection

Parts to Finish	**Finishing Process Option**	**Process Selected**	**Reason Behind Selection**
Frame	Grinding, Polishing, Annealing	Grinding followed by Polishing	Reduce surface roughness, Remove bumps and flashes
Wooden Bamboo Storage	Polishing	Polishing	For appearance, Inexpensive
Cutter Holder	Anodizing, Grinding, Polishing	Grinding followed by Polishing	Reduce surface roughness, For appearance
Cutter Blade	Annealing, Coating, Grinding, Polishing, Honing	Annealing, Grinding and Polishing	To enhance strength, to retain sharpness,
Rammer	Grinding, Polishing	Grinding	Appearance, to eliminate any sharp corners

5.5 Qualitative Analysis of Coloring Method Selection

The Qualitative Analysis of Parts Coloring Method Selection for the aesthetics of our product is shown below for the parts to be finished:

Table 5.11: Qualitative Selection of Coloring Method

Parts to Color	**Coloring Process Option**	**Process Selected**	**Reason Behind Selection**
Frame	Powder Coating, Spraying, Brush Painting, Enameling	Spraying	Appearance, Durable Paint Layer
Cutter Holder	Brush Painting, Spraying, Powder Coating	Brush Painting	Inexpensive, For Appearance
Rammer	Brush Painting, Spraying, Powder Coating	Brush Painting	Inexpensive, For Appearance

The qualitative selection of materials requires no numeric evaluations and is useful for assigning specific materials to the different parts and components of the product in such a way that the parts are able to perform their required functions while having sufficient mechanical strength and other mechanical properties. Sometimes, the mechanical strength is extremely important for a part while some parts do not require very high mechanical strength. While selecting the methods for manufacturing, joining, finishing and coloring, availability and cost-effectiveness have been taken into consideration.

Chapter-6

Material, Manufacturing Process and Joining Process Selection Using Weighted Average Method

6.1 Introduction

In the previous chapter, a qualitative analysis of material, manufacturing process and joining method selection has been carried out. But it is nothing but implementation of intuitive knowledge. The goal is to ensure that the chosen material or process align with the intended purpose of the project, offering optimal performance, longevity and efficiency. Quantitative analysis serves this purpose profoundly. Quantitative analysis of a material or process plays a pivotal role in diverse industries, ranging from manufacturing and construction to product design. This process involves identifying and choosing the most suitable material or process for a particular application.

"Digital Logic Method" has been used for selecting materials and processes based on their performance indices.

6.2 List of Parts for Each Sub-assembly

6.2.1 Main Body

- Frame
- Bamboo Support
- Bamboo Storage
- Supporting Feet
- Table Legs

6.2.2 Cutter Assembly

- Cutter Blade
- Cutter Holder

6.2.3 Moving Parts

- Pulley
- Rope
- Rammer
- Bearings

6.3 Equations

a) Relative Emphasis Coefficient, $\alpha = \frac{No.\ of\ Postive\ Decisions\ acquired\ by\ a\ criteria}{Total\ no.\ of\ positive\ decsions}$

b) For properties to be maximized,

Scaled Property, $\beta = \frac{Numerical\ Value\ of\ the\ Property}{Maximum\ value\ in\ the\ list} \times 100\%$

c) For properties to be minimized,

Scaled Property, $\beta = \frac{Minimum\ value\ in\ the\ list}{Numerical\ Value\ of\ the\ property} \times 100\%$

d) Weighted Score = Relative Emphasis Coefficient (α) × Scaled Property (β)

e) Performance Index, $\gamma = \Sigma$ Weighted Score ($\alpha \times \beta$)

6.4 Likert scale

A Likert scale is a tool used in surveys and research to measure attitudes, opinion or perceptions of individuals towards a particular subject. Named after its creator Rensis Likert, this scale involves presenting respondents with a series of statements and asking them to indicate their level of agreement or disagreement with each statements using a predetermined set of response options.

In this chapter, we are going to use the Likert scale for those criteria that can't be assigned with numerical values. The scale is shown in the following page:

Table 6.1: Likert scale

Very Good	5
Good	4
Moderate	3
Poor	2
Very Poor	1

6.5 Quantitative Analysis of Material Selection of Convenient Bamboo Cutter

6.5.1 Material selection for Frame and Table Legs

Table 6.2: List of selection criteria with numerical values

Selection Criteria	Mild Steel	Aluminum Alloy	Stainless Steel
Strength (MPa)	550 [1]	310 [2]	621 [3]
Density (kg/m^3)	7680	2810	8000
Cost (BDT/ per kg)	80	350	317

Table 6.3: List of Selection criteria with Likert scale

Selection Criteria	**Mild Steel**	**Aluminum Alloy**	**Stainless Steel**
Corrosion Resistance	2	4	5
Durability	3	4	5
Availability	5	3	3

Table 6.4: Determination of relative importance of material selection criteria for Bamboo support using Digital Logic Method

Selection Criteria	Numbers of positive decision, n(n-1)/2=6(6-1)/2=15															Positive Decisions	Relative Emphasis Coefficient, α
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15		
Strength	1	0	1	0	0											2	0.133
Density	0					0	1	0	0							1	0.067
Cost		1				1				1	0	1				4	0.267
Corrosion Resistance			0				0			0			1	0		1	0.067
Durability				1				1			1		0		0	3	0.2
Availability					1				1			0		1	1	4	0.266
Total numbers of positive decisions																15	Σα=1

Table 6.5: Calculation of the performance index for material selection of Frame and Table Legs

Selection Criteria	Relative Emphasis Coefficient α	Mild Steel		Aluminum Alloy		Stainless Steel	
		Scaled property β	Weighted score αβ	Scaled property β	Weighted score αβ	Scaled property β	Weighted score αβ
Strength	0.133	88.57	11.78	49.92	6.64	100	13.3
Density	0.067	36.59	2.45	100	6.7	35.13	2.35
Cost	0.267	100	26.7	22.86	6.1	25.24	6.74
Corrosion Resistance	0.067	40	2.68	80	5.36	100	6.7
Durability	0.2	60	12	80	16	100	20
Availability	0.266	100	26.6	60	15.96	60	15.96
Material Performance Index			**82.21**		56.76		65.05

Table 6.5 shows the weighted score for different selection criteria for three different materials and the performance index for each of the materials. Among the three materials, Mild Steel has the maximum performance index. Therefore, for Frame and Table Legs **Mild Steel** should be selected as material which we also selected in the previous chapter.

6.5.2 Material Selection for Bamboo Support

Table 6.6: List of selection criteria with numerical values

Selection Criteria	Bamboo	Mild Steel	Wood
Strength (MPa)	265 [4]	550	110 [5]
Density (kg/m^3)	360 [6]	7680	500 [7]
Cost (BDT/kg)	100 [8]	80	50 [9]

Table 6.7: List of selection criteria with Likert scale

Selection Criteria	Bamboo	Mild Steel	Wood
Durability	4	5	3
Availability	5	4	5

Table 6.8: Determination of relative importance of material selection criteria for Bamboo support using Digital Logic Method

Selection Criteria	Numbers of positive decision, n(n-1)/2=5(5-1)/2=10										Positive Decisions	Relative Emphasis Coefficient, α
	1	2	3	4	5	6	7	8	9	10		
Strength	0	1	0	0							1	0.1
Density	1				0	1	0				2	0.2
Cost		0			1			1	1		3	0.3
Durability			1			0		0		0	1	0.1
Availability				1			1		0	1	3	0.3
Total numbers of positive decisions											10	Σα=1

Table 6.9: Calculation of the performance index for material selection of Bamboo Support

Selection Criteria	Relative Emphasis Coefficient α	Bamboo		Mild Steel		Wood	
		Scaled property β	Weighted score αβ	Scaled property β	Weighted score αβ	Scaled property β	Weighted score αβ
Strength	0.1	48.18	4.82	100	10	20	2
Density	0.2	100	20	4.69	0.94	6.51	1.302
Cost	0.3	50	15	62.5	18.75	100	30
Durability	0.1	80	8	100	10	75	7.5
Availability	0.3	100	30	80	24	100	30
Material Performance Index			**77.82**		63.69		70.80

Table 6.9 shows the weighted score for different selection criteria for three different materials and the performance index for each of the materials. Among the three materials, Bamboo has the maximum performance index. Therefore, for Bamboo Support, **Bamboo** should be selected as material which we also selected in the previous chapter.

6.5.3 Bamboo Storage

Table 6.10: List of selection criteria with numerical values

Selection Criteria	**Bamboo**	**Mild Steel**	**Wood**
Strength (MPa)	265	550	110
Density (kg/m^3)	360	7680	500
Cost (BDT/kg)	100	80	50

Table 6.11: List of selection criteria with Likert scale

Selection Criteria	**Bamboo**	**Mild Steel**	**Wood**
Durability	4	5	3
Availability	5	4	5

Table 6.12: Determination of relative importance of material selection criteria for Bamboo Storage using Digital Logic Method

Selection Criteria	Numbers of positive decision, n(n-1)/2=5(5-1)/2=10										Positive Decisions	Relative Emphasis Coefficient, α
	1	2	3	4	5	6	7	8	9	10		
Strength	1	0	0	0							1	0.1
Density	0				0	1	0				1	0.1
Cost		1			1			1	1		4	0.4
Durability			1			0		0		0	1	0.1
Availability				1			1		0	1	3	0.3
Total numbers of positive decisions											10	Σα=1

Table 6.13: Calculation of the performance index for material selection of Bamboo Support

Selection Criteria	Relative Emphasis Coefficient α	Bamboo		Mild Steel		Wood	
		Scaled property β	Weighted score αβ	Scaled property β	Weighted score αβ	Scaled property β	Weighted score αβ
Strength	0.1	48.18	4.82	100	10	20	2
Density	0.1	100	10	4.69	0.47	6.51	0.65
Cost	0.4	50	20	62.5	25	100	40
Durability	0.1	80	8	100	10	75	7.5
Availability	0.3	100	30	80	24	100	30
Material Performance Index			72.82		69.47		**80.15**

Table 6.13 shows the weighted score for different selection criteria for three different materials and the performance index for each of the materials. Among the three materials, Wood has the maximum performance index. Therefore, for Bamboo Storage, **Wood** should be selected as material which we also selected in the previous chapter.

6.5.4 Supporting Feet

Table 6.14: List of selection criteria with numerical values

Selection Criteria	**Natural and Synthetic Rubber**	**Nylon 6**	**Leather**
Cost (BDT/kg)	200 [10]	400	350 [11]
Shear Modulus (MPa)	0.9	40	25

Table 6.15: List of selection criteria with Likert scale

Selection Criteria	Natural and Synthetic Rubber	Nylon	Leather
Grip	5	1	4
Wear Resistance	3	5	4
Availability	5	3	3

Table 6.16: Determination of relative importance of material selection criteria for Supporting Feet using Digital Logic Method

Selection Criteria	Numbers of positive decision, n(n-1)/2=5(5-1)/2=10										Positive Decisions	Relative Emphasis Coefficient, α
	1	2	3	4	5	6	7	8	9	10		
Cost	0	0	0	1							1	0.1
Shear Modulus	1				0	0	0				1	0.1
Grip		1			1			1	1		4	0.4
Wear Resistance			1			1		0		0	2	0.2
Availability				0			1		0	1	2	0.2
Total numbers of positive decisions											10	Σα=1

Table 6.17: Calculation of the performance index for material selection of Supporting Feet

Selection Criteria	Relative Emphasis Coefficient α	Natural and Synthetic Rubber		Nylon 6		Leather	
		Scaled property β	Weighted score αβ	Scaled property β	Weighted score αβ	Scaled property β	Weighted score αβ
Cost	0.1	100	10	50	5	57.14	5.71
Shear Modulus	0.1	2.25	0.225	100	10	62.5	6.25
Grip	0.4	100	40	20	8	80	32
Wear Resistance	0.2	60	12	100	20	80	16
Availability	0.2	100	20	60	12	60	12
Material Performance Index			**82.23**		55		71.96

Table 6.17 shows the weighted score for different selection criteria for three different materials and the performance index for each of the materials. Among the three materials, Natural and Synthetic Rubber has the maximum performance index. Therefore, for Supporting Feet, **Natural and Synthetic Rubber** should be selected as material which we also selected in the previous chapter.

6.5.5 Cutter Blade

Table 6.18: List of selection criteria with numerical values

Selection Criteria	High Speed Steel	Mild Steel	Carbide (Tungsten)
Strength (MPa)	1200 [12]	550	2600 [13]
Hardness (Mohs Scale)	8	4	9.5
Cost (BDT/kg)	1000	80	5500

Table 6.19: List of Selection criteria with Likert scale

Selection Criteria	High Speed Steel	Mild Steel	Carbide (Tungsten)
Wear Resistance	4	1	5
Toughness	4	4	2
Machinability	5	4	1

Table 6.20: Determination of relative importance of material selection criteria for Cutter Blade using Digital Logic Method

Selection Criteria	Numbers of positive decision, n(n-1)/2=6(6-1)/2=15															Positive Decisions	Relative Emphasis Coefficient, α
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15		
Strength	1	1	1	0	0											3	0.2
Hardness	0					1	1	0	0							2	0.133
Cost		0				0				0	0	1				1	0.067
Wear Resistance			0				0			1			0	1		2	0.133
Toughness				1				1			1		1		0	4	0.267
Machinability					1				1			0		0	1	3	0.2
Total numbers of positive decisions																15	Σα=1

Table 6.21: Calculation of the performance index for material selection of Frame and Table Legs

Selection Criteria	Relative Emphasis Coefficient α	High Speed Steel		Mild Steel		Carbide (Tungsten)	
		Scaled property β	Weighted score αβ	Scaled property β	Weighted score αβ	Scaled property β	Weighted score αβ
Strength	0.2	46.15	9.23	21.15	4.23	100	20
Hardness	0.133	84.21	11.20	42.15	5.61	100	13.3
Cost	0.067	8	0.536	100	6.7	1.45	0.097
Wear Resistance	0.133	80	10.64	20	2.66	100	13.3
Toughness	0.267	100	26.7	100	26.7	50	13.35
Machinability	0.2	100	16	80	16	20	4
Material Performance Index			**72.63**		61.90		64.05

Table 6.21 shows the weighted score for different selection criteria for three different materials and the performance index for each of the materials. Among the three materials, High Speed Steel has the maximum performance index. Therefore, for Supporting Feet, **High Speed Steel** should be selected as material which we also selected in the previous chapter.

6.5.6 Cutter Holder and Pulley

Table 6.22: List of selection criteria with numerical values

Selection Criteria	Cast Iron	Mild Steel	Stainless Steel
Strength (MPa)	500	550	579
Hardness (Mohs Scale)	5	4	5.2
Cost (BDT/kg)	105	80	430

Table 6.23: List of selection criteria with Likert Scale

Selection Criteria	Cast Iron	Mild Steel	Stainless Steel
Wear Resistance	5 [14]	2	4
Availability	5	4	3

Table 6.24: Determination of relative importance of material selection criteria for Cutter Holder and Pulley using Digital Logic Method

Selection Criteria	Numbers of positive decision, n(n-1)/2=5(5-1)/2=10										Positive Decisions	Relative Emphasis Coefficient, α
	1	2	3	4	5	6	7	8	9	10		
Strength	1	0	0	1							2	0.2
Hardness	0				0	1	1				2	0.2
Cost		1			1			1	1		4	0.4
Wear Resistance			1			0		0		0	1	0.1
Availability				0			0		0	1	1	0.1
Total numbers of positive decisions											10	Σα=1

Table 6.25: Calculation of the performance index for material selection of Cutter Holder and Pulley

Selection Criteria	Relative Emphasis Coefficient α	Cast Iron		Mild Steel		Stainless Steel	
		Scaled property β	Weighted score αβ	Scaled property β	Weighted score αβ	Scaled property β	Weighted score αβ
Strength	0.2	86.36	17.27	95	19	100	20
Hardness	0.2	96.15	19.23	76.92	15.38	100	20
Cost	0.4	76.19	30.48	100	40	18.60	7.44
Wear Resistance	0.1	100	10	40	4	80	8
Availability	0.1	100	10	80	8	60	6
Material Performance Index			**86.98**		86.38		61.44

Table 6.25 shows the weighted score for different selection criteria for three different materials and the performance index for each of the materials. Among the three materials, Cast Iron has the maximum performance index. Therefore, for Cutter Holder and Pulley, **Cast Iron** should be selected as material which we also selected in the previous chapter.

6.5.7 Rope

Table 6.26: List of selection criteria with numerical values

Selection Criteria	**Polypropylene**	**Nylon 6**	**Jute**
Cost (BDT/m)	20	12	5
Strength (MPa)	323	614	60 [15]

Table 6.27: List of selection criteria with Likert scale [16]

Selection Criteria	**Polypropylene**	**Nylon**	**Jute**
Durability	5	4	2
Stretch Resistance	3	5	1
Resistance to Shrinking	3	5	1
Wear Resistance	5	4	2

Table 6.28: Determination of relative importance of material selection criteria for Rope using Digital Logic Method

Selection Criteria	Numbers of positive decision, n(n-1)/2=6(6-1)/2=15															Positive Decisions	Relative Emphasis Coefficient, α
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15		
Cost	0	1	0	0	0											1	0.067
Strength	1					1	1	0	1							4	0.267
Durability		0				0				0	0	1				1	0.067
Stretch Resistance			1				0			1			0	1		3	0.2
Resistance to Shrinking				1				1			1		1		0	4	0.267
Wear Resistance					1				0			0		0	1	2	0.133
Total numbers of positive decisions																15	Σα=1

Table 6.29: Calculation of the performance index for material selection of Rope

Selection Criteria	Relative Emphasis Coefficient α	Polypropylene		Nylon		Jute	
		Scaled property β	Weighted score αβ	Scaled property β	Weighted score αβ	Scaled property β	Weighted score αβ
Cost	0.067	25	1.675	41.67	2.79	100	6.7
Strength	0.267	52.61	14.05	100	26.7	9.8	2.62
Durability	0.067	100	6.7	80	5.36	40	2.68
Stretch Resistance	0.2	60	12	100	20	20	4
Resistance to Shrinking	0.267	60	16.02	100	26.7	20	5.34
Wear Resistance	0.133	100	13.3	80	10.64	40	5.32
Material Performance Index			63.745		**92.19**		26.66

Table 6.29 shows the weighted score for different selection criteria for three different materials and the performance index for each of the materials. Among the three materials, Nylon has the maximum performance index. Therefore, for Rope, **Nylon** should be selected as material which we also selected in the previous chapter.

6.5.8 Rammer

Table 6.30: List of selection criteria with numerical values

Selection Criteria	Mild Steel	Cast Iron	Wood
Cost (BDT/kg)	80	105	50
Strength (MPa)	550	500	110
Charpy Impact Energy (J)	35	13	9

Table 6.31: List of selection criteria using Likert scale

Selection Criteria	Mild Steel	Cast Iron	Wood
Durability	5	4	3
Availability	3	4	5

Table 6.32: Determination of relative importance of material selection criteria for Rammer using Digital Logic Method

Selection Criteria	Numbers of positive decision, n(n-1)/2=5(5-1)/2=10										Positive Decisions	Relative Emphasis Coefficient, α
	1	2	3	4	5	6	7	8	9	10		
Cost	0	0	1	0							1	0.1
Strength	1				1	1	1				4	0.4
Impact Energy		1			0			1	1		3	0.3
Durability			0			0		0		1	1	0.1
Availability				1			0		0	0	1	0.1
Total numbers of positive decisions											10	Σα=1

Table 6.33: Calculation of the performance index for material selection of Rammer

Selection Criteria	Relative Emphasis Coefficient α	Mild Steel		Cast Iron		Wood	
		Scaled property β	Weighted score αβ	Scaled property β	Weighted score αβ	Scaled property β	Weighted score αβ
Cost	0.1	62.5	6.25	47.62	4.76	100	10
Strength	0.4	100	40	90.91	36.36	20	8
Impact Energy	0.3	100	30	37.14	11.14	25.71	7.71
Durability	0.1	100	10	80	8	60	6
Availability	0.1	60	6	80	8	100	10
Material Performance Index			**92.25**		67.86		41.71

Table 6.33 shows the weighted score for different selection criteria for three different materials and the performance index for each of the materials. Among the three materials, Steel has the maximum performance index. Therefore, for Rammer, **Mild Steel** should be selected as material which we also selected in the previous chapter.

6.5.9 Bearings

Table 6.34: List of selection criteria with numerical values

Selection Criteria	**Mild Steel**	**Stainless Steels**	**Chrome Steel**
Strength (MPa)	550	579	700 [17]
Sp. Heat Capacity (J/g-°C)	0.466	0.5	0.477 [17]

Table 6.35: List of selection criteria with Likert scale

Selection Criteria	**Mild Steel**	**Stainless Steel**	**Chrome Steel**
Coeff. Of Friction, μ_k	1	5	4
Availability	3	4	5
Cost	5	4	2

Table 6.36: Determination of relative importance of material selection criteria for Bearings using Digital Logic Method

Selection Criteria	Numbers of positive decision, n(n-1)/2=5(5-1)/2=10										Positive Decisions	Relative Emphasis Coefficient, α
	1	2	3	4	5	6	7	8	9	10		
Strength	1	0	1	1							3	0.3
Sp. Heat Capacity	0				0	1	1				2	0.2
Coeff. of Friction		1			1			0	0		2	0.2
Availability			0			0		1		0	1	0.1
Cost				0			0		1	1	2	0.2
Total numbers of positive decisions											10	Σα=1

Table 6.37: Calculation of the performance index for material selection of Bearings

Selection Criteria	Relative Emphasis Coefficient α	Mild Steel		Stainless Steel		Chrome Steel	
		Scaled property β	Weighted score αβ	Scaled property β	Weighted score αβ	Scaled property β	Weighted score αβ
Strength	0.3	78.57	23.57	82.71	24.81	100	30
Sp. Heat Capacity	0.2	93.2	18.64	100	20	95.4	19.08
Coeff. of Friction	0.2	100	20	20	4	25	5
Availability	0.1	60	6	80	8	100	10
Cost	0.2	40	8	80	16	100	20
Material Performance Index			76.21		72.81		**84.08**

Table 6.37 shows the weighted score for different selection criteria for three different materials and the performance index for each of the materials. Among the three materials, Chrome Steel has the maximum performance index. Therefore, for Bearings, **Chrome Steel** should be selected as material which we also selected in the previous chapter.

6.6 List of Manufacturing Processes and Joining Processes

6.6.1 Manufacturing Processes Used:

1. Forging
2. Welding
3. Machining
4. Grinding
5. Casting

6.6.2 Joining Methods Used:

1. Permanent Joint
2. Temporary Joint

6.7 Quantitative Analysis of Manufacturing Process Selection

6.7.1 Manufacturing Process Selection for Frame and Table Legs

Table 6.38: List of selection criteria with value from Likert scale for manufacturing of Frame and Table Legs

Selection Criteria	Forging	Welding and Machining
Cost	5	4
Strength	3	5
Operational Complexity	3	4
Dimensional Accuracy	4	3
Surface Finish	3	4

From these criteria, we want to maximize Strength, Dimensional Accuracy and Surface Finish; and minimize Cost and Operational complexity.

Table 6.39: Determination of relative importance of manufacturing process selection criteria for Frame and Table Legs using Digital Logic Method

Selection Criteria	Numbers of Positive Decision, N = n(n-1)/2 = 5(5-1)/2 = 10										Positive Decision	Relative Emphasis Coefficient, α
	1	2	3	4	5	6	7	8	9	10		
Cost	1	1	1	1							4	0.4
Strength	0				1	0	1				2	0.2
Operational Complexity		0			0			1	0		1	0.1
Dimensional Accuracy			0			1		0		1	2	0.2
Surface Finish				0			0		1	0	1	0.1
Total Number of Positive Decisions											10	∑α=1

Table 6.39 shows the Digital Logic Method for manufacturing processes selection criteria for Frame and Table Legs. Among these criteria, Cost has the highest relative emphasis coefficient.

Table 6.40: Calculation of the performance index of manufacturing process selection criteria for Frame and Table Legs

Selection Criteria	Relative Emphasis Coefficient, α	Forging		Welding and Machining	
		Scaled Property, β	Weighted Score, αβ	Scaled Property, β	Weighted Score, αβ
Cost	0.4	80	32	100	40
Strength	0.2	60	12	100	20
Operational Complexity	0.1	100	10	75	7.5
Dimensional Accuracy	0.2	100	20	75	15
Surface Finish	0.1	75	7.5	100	10
Process Performance Index			81.5		**92.5**

Table 6.40 shows the weighted score of selection criteria for two different manufacturing processes and the performance index for each of the processes. Among the two processes, Welding and Machining has the maximum performance index. Therefore, for Frame and Table Legs, **Welding and Machining** should be selected as manufacturing processes which we also selected in the previous chapter.

6.7.2 Manufacturing Process Selection for Cutter Holder and Rammer

Table 6.41: List of selection criteria with value from Likert scale for manufacturing of Cutter Holder and Rammer

Selection Criteria	Forging	Casting and Machining
Cost	5	3
Strength	4	4
Operational Complexity	5	3
Dimensional Accuracy	4	3
Surface Finish	3	3

From these criteria, we want to maximize Strength, Dimensional Accuracy and Surface Finish; and minimize Cost and Operational complexity.

Table 6.42: Determination of relative importance of manufacturing process selection criteria for Cutter Holder and Rammer using Digital Logic Method

Selection Criteria	Numbers of Positive Decision, N = n(n-1)/2 = 5(5-1)/2 = 10										Positive Decision	Relative Emphasis Coefficient, α
	1	2	3	4	5	6	7	8	9	10		
Cost	1	1	1	1							4	0.4
Strength	0				1	0	1				2	0.2
Operational Complexity		0			0			1	0		1	0.1
Dimensional Accuracy			0			1		0		1	2	0.2
Surface Finish				0			0		1	0	1	0.1
Total Number of Positive Decisions											10	∑α=1

Table 6.42 shows the Digital Logic Method for manufacturing processes selection criteria for Cutter Holder and Rammer. Among these criteria, Cost has the highest relative emphasis coefficient.

Table 6.43: Calculation of the performance index of manufacturing process selection criteria for Cutter Holder and Rammer

Selection Criteria	Relative Emphasis Coefficient, α	Forging		Casting and Machining	
		Scaled Property, β	Weighted Score, αβ	Scaled Property, β	Weighted Score, αβ
Cost	0.4	60	24	100	40
Strength	0.2	100	20	100	20
Operational Complexity	0.1	60	6	100	10
Dimensional Accuracy	0.2	100	20	75	15
Surface Finish	0.1	100	10	100	10
Process Performance Index			80		**95**

Table 6.43 shows the weighted score of selection criteria for two different manufacturing processes and the performance index for each of the processes. Among the two processes, Casting and Machining has the maximum performance index. Therefore, for Cutter Holder and Rammer, **Casting and Machining** should be selected as manufacturing processes which we also selected in the previous chapter.

6.7.3 Manufacturing Process Selection for Cutter Blade

Table 6.44: List of selection criteria with value from Likert scale for manufacturing of Cutter Blade

Selection Criteria	EDM and Machining	Forging and Machining
Cost	5	3
Strength	5	4
Operational Complexity	5	3
Dimensional Accuracy	4	3
Surface Finish	3	4

From these criteria, we want to maximize Strength, Dimensional Accuracy and Surface Finish; and minimize Cost and Operational complexity.

Table 6.45: Determination of relative importance of manufacturing process selection criteria for Cutter Blade using Digital Logic Method

Selection Criteria	Numbers of Positive Decision, N = n(n-1)/2 = 5(5-1)/2 = 10										Positive Decision	Relative Emphasis Coefficient, α
	1	2	3	4	5	6	7	8	9	10		
Cost	1	1	1	1							4	0.4
Strength	0				1	0	1				2	0.2
Operational Complexity		0			0			1	0		1	0.1
Dimensional Accuracy			0			1		0		1	2	0.2
Surface Finish				0			0		1	0	1	0.1
Total Number of Positive Decisions											10	∑α=1

Table 6.45 shows the Digital Logic Method for manufacturing processes selection criteria for Cutter Blade. Among these criteria, Cost has the highest relative emphasis coefficient.

Table 6.46: Calculation of the performance index of manufacturing process selection criteria for Cutter Blade

Selection Criteria	Relative Emphasis Coefficient, α	EDM and Machining		Forging and Machining	
		Scaled Property, β	Weighted Score, αβ	Scaled Property, β	Weighted Score, αβ
Cost	0.4	60	24	100	40
Strength	0.2	100	20	80	16
Operational Complexity	0.1	60	6	100	10
Dimensional Accuracy	0.2	100	20	75	15
Surface Finish	0.1	75	7.5	100	10
Process Performance Index			77.5		**91**

Table 6.46 shows the weighted score of selection criteria for two different manufacturing processes and the performance index for each of the processes. Among the two processes, Forging and Machining has the maximum performance index. Therefore, for Cutter Blade, **Forging and Machining** should be selected as manufacturing processes which we also selected in the previous chapter.

6.8 Quantitative Analysis of Permanent Joining Process Selection

6.8.1 Joining Process Selection of Frame parts and Table Legs with frame:

Table 6.47: List of selection criteria with numerical values or values from Likert Scale for Permanent Joints:

Selection Criteria	Arc Welding	MIG Welding	TIG Welding
Cost (BDT/Product)	100	150	180
Surface Finish	4	5	5
Strength	4	5	5
Operation Time (Sec)	20	30	25
Defect Rate (%)	3	2	2

From these criteria, we want to maximize strength and surface finish; and minimize cost, operation time and defect rate.

Table 6.48: Calculation of the relative emphasis coefficient for joining process selection of Frame parts and Table Legs with Frame using Digital Logic Method:

Selection Criteria	Number of Positive Decisions, N = n(n-1)/2=5(5-1)/2=10										Positive Decisions	Relative Emphasis Coefficient α
	1	2	3	4	5	6	7	8	9	10		
Cost	1	1	1	1							4	0.4
Surface Finish	0				0	1	0				1	0.1
Strength		0			1			1	0		2	0.2
Operation Time			0			0		0		1	1	0.1
Defect Rate				0			1		1	0	2	0.2
Total Number of Positive Decisions											10	Σα=1

Table 6.49: Calculation of performance index for joining process selection of Frame parts and Table Legs with Frame.

Selection Criteria	Relative Emphasis Coefficient α	Arc Welding		MIG Welding		TIG welding	
		Scaled Property, β	Weighted Property, αβ	Scaled Property, β	Weighted Property, αβ	Scaled Property, β	Weighted Property, αβ
Cost	0.4	100	40	66.67	26.66	55	22
Surface Finish	0.1	80	8	100	10	100	10
Strength	0.2	80	16	100	20	100	20
Operation Time	0.1	100	10	66.67	10	80	8
Defect rate	0.2	66.67	13.33	100	20	100	20
Process Performance Index			**87.33**		86.66		80

Since the performance index for Arc welding is highest among all so, Arc welding will be selected as our joining process of Frame parts and Table Legs with Frame**.**

6.8.2 Joining Process Selection of Bamboo Storage with Frame, Cutter Blades, Pulley with Rammer.

Table 6.50: List of selection criteria with numerical values or values from Likert Scale for Permanent Joints:

Selection Criteria	Arc Welding	MIG Welding	TIG Welding
Cost (BDT/Product)	100	150	180
Surface Finish	4	5	5
Strength	4	5	5
Precision	2	3	4
Defect Rate (%)	3	2	2

From these criteria, we want to maximize strength and surface finish; and minimize cost, precision and defect rate.

Table 6.51: Calculation of the relative emphasis coefficient for joining process selection of Bamboo storage with frame, cutter blades and Pulley with Rammer using Digital Logic Method:

Selection Criteria	Number of Positive Decisions, N =n(n-1)/2=5(5-1)/2=10										Positive Decisions	Relative Emphasis Coefficient α
	1	2	3	4	5	6	7	8	9	10		
Cost	1	0	0	1							2	0.2
Surface Finish	0				1	0	0				1	0.1
Strength		1			0			1	1		3	0.3
Precision			1			1		0		1	3	0.3
Defect Rate				0			1		0	0	1	0.1
Total Number of Positive Decisions											10	Σα=1

Table 6.52: Calculation of performance index for joining process selection of Bamboo storage with frame, cutter blades and Pulley with Rammer.

Selection Criteria	Relative Emphasis Coefficient α	Arc Welding		MIG Welding		TIG welding	
		Scaled Property, β	Weighted Property, αβ	Scaled Property, β	Weighted Property, αβ	Scaled Property, β	Weighted Property, αβ
Cost	0.2	100	20	66.67	13.33	55	11
Surface Finish	0.1	80	8	100	10	100	10
Strength	0.3	80	24	100	30	100	30
Precision	0.3	50	15	75	22.5	100	30
Defect rate	0.1	66.67	6.67	100	10	100	10
Process Performance Index			73.67		85.83		**91**

Since the performance index for TIG welding is highest among all so, TIG welding will be selected as our joining process of Bamboo storage with frame, cutter blades and Pulley with Rammer.

6.9 Quantitative Analysis of Temporary Joining Process Selection

6.9.1 Joining Process Selection of Bamboo Cutter with Cutter holder and Cutter holder with Frame:

Table 6.53: List of selection criteria with numerical values or values from Likert Scale for Temporary Joints:

Selection Criteria	Nut-Bolt	Bolted
Cost	5	5
Strength	5	4
Availability	5	3
Surface Finish	4	5
Design Flexibility	5	3

From these criteria, we want to maximize strength, surface finish, availability and design flexibility and minimize cost.

Table 6.54: Calculation of the relative emphasis coefficient for joining process selection of bamboo cutter with cutter holder and cutter holder with frame using Digital Logic Method:

Selection Criteria	Number of Positive Decisions, N=n(n-1)/2=5(5-1)/2=10										Positive Decisions	Relative Emphasis Coefficient α
	1	2	3	4	5	6	7	8	9	10		
Cost	1	0	1	0							2	0.2
Strength	0				0	1	0				1	0.1
Availability		1			1			1	0		3	0.3
Surface Finish			0			0		0		1	1	0.1
Design Flexibility				1			1		1	0	3	0.3
Total Number of Positive Decisions											10	Σα=1

Table 8. shows the digital logic method for joining process selection of bamboo cutter with cutter holder and cutter holder with frame. Among these criteria, design flexibility and availability have the highest relative emphasis coefficient.

Table 6.55: Calculation of performance index for joining process selection of bamboo cutter with cutter holder and cutter holder with frame:

Selection Criteria	Relative Emphasis Coefficient α	Nut-Bolt		Bolted	
		Scaled Property, β	Weighted Property, αβ	Scaled Property, β	Weighted Property, αβ
Cost	0.2	100	20	100	20
Strength	0.1	100	10	80	8
Availability	0.3	100	30	60	18
Surface Finish	0.1	80	8	100	10
Design Flexibility	0.3	100	30	60	18
Process Performance Index			**98**		74

Since the performance index for Nut-bolt is larger than bolted, so Nut-bolt will be used for the temporary joint of bamboo cutter with cutter holder and cutter holder with frame.

6.9.2 Joining Process Selection of Rammer with Bearings:

Table 6.56: List of selection criteria with numerical values or values from Likert Scale for Temporary Joints:

Selection Criteria	Nut-Bolt	Bolted	Rivet
Cost	5	4	2
Strength	4	3	5
Availability	5	3	4
Surface Finish	3	4	5
Design Flexibility	5	3	4

From these criteria, we want to maximize strength, surface finish, availability and design flexibility and minimize cost.

Table 6.57: Calculation of the relative emphasis coefficient for joining process selection of Rammer with Bearings using Digital Logic Method.

Selection Criteria	Number of Positive Decisions, N=n(n-1)/2=5(5-1)/2=10										Positive Decisions	Relative Emphasis Coefficient α
	1	2	3	4	5	6	7	8	9	10		
Cost	0	1	1	1							3	0.3
Strength	1				1	1	0				3	0.3
Availability		0			0			1	0		1	0.1
Surface Finish			0			0		0		1	1	0.1
Design Flexibility				0			1		1	0	2	0.2
Total Number of Positive Decisions											10	Σα=1

Table 6.58: Calculation of performance index for joining process selection of Rammer with bearings.

Selection Criteria	Relative Emphasis Coefficient α	Nut-Bolt		Bolted		Rivet	
		Scaled Property, β	Weighted Property, αβ	Scaled Property, β	Weighted Property, αβ	Scaled Property, β	Weighted Property, αβ
Cost	0.3	40	12	50	15	100	30
Strength	0.3	80	24	60	18	100	30
Availability	0.1	100	10	60	6	80	8
Surface Finish	0.1	60	6	80	8	100	10
Design Flexibility	0.2	100	20	60	12	8	16
Process Performance Index			72		59		**94**

Since the performance index Riveting is larger than others, so Riveting will be used for the temporary joint of Rammer with bearings.

6.9.3 Joining Process Selection of Supporting Feet with Table Legs:

Table 6.59: List of selection criteria with numerical values or values from Likert Scale for Temporary Joints:

Selection Criteria	Fastening	Hot Glue	Epoxy Adhesive
Cost	4	5	3
Strength	4	4	5
Availability	5	5	5
Surface Finish	5	5	4
Design Flexibility	5	3	4

From these criteria, we want to maximize strength, surface finish, availability and design flexibility and minimize cost.

Table 6.60: Calculation of the relative emphasis coefficient for joining process selection of supporting feet with table legs using Digital Logic Method:

Selection Criteria	Number of Positive Decisions, N=n(n-)/2=5(5-1)/2=10										Positive Decisions	Relative Emphasis Coefficient α
	1	2	3	4	5	6	7	8	9	10		
Cost	0	1	1	1							3	0.3
Strength	1				1	0	1				3	0.3
Availability		0			0			1	0		1	0.1
Surface Finish			0			1		0		1	2	0.2
Design Flexibility				0			0		1	0	1	0.1
Total Number of Positive Decisions											10	Σα=1

Table 6.61: Calculation of performance index for joining process selection supporting feet with table legs.

Selection Criteria	Relative Emphasis Coefficient α	Fastening		Hot Glue		Epoxy Adhesive	
		Scaled Property, β	Weighted Property, αβ	Scaled Property, β	Weighted Property, αβ	Scaled Property, β	Weighted Property, αβ
Cost	0.3	75	22.5	60	18	100	30
Strength	0.3	80	24	80	24	100	30
Availability	0.1	100	10	100	10	100	10
Surface Finish	0.2	100	20	100	20	80	16
Design Flexibility	0.1	100	10	60	10	80	8
Process Performance Index			88.5		82		**94**

Since the performance index of epoxy adhesive is larger than others, so epoxy adhesive will be used for the temporary joint of table feet with table legs.

Chapter-7

Design Analysis

7.1 Introduction

Design analysis is the systematic process of evaluating and scrutinizing a design to ensure it meets specific criteria and performs optimally within the intended context. It serves as a critical bridge between creative ideation and practical implementation in various fields, from product design and architecture to engineering and software development.

At its core, design analysis involves a comprehensive examination of a design's structural, functional, and performance aspects. Engineers, architects, and designers employ a variety of tools and techniques, such as computer-aided design (CAD), simulation software, and mathematical modelling, to simulate and assess how a design will behave under real-world conditions. This process enables the identification of potential flaws, weaknesses, and areas for improvement while ensuring that the design aligns with safety standards and user expectations.

Design analysis plays a pivotal role in enhancing innovation and minimizing risks in the development process. It enables designers to make informed decisions, refine concepts, and optimize designs, ultimately leading to more reliable and efficient end products. This introduction to design analysis sets the stage for exploring its diverse applications and demonstrating how it empowers creative minds to transform imaginative ideas into tangible, high-performing solutions.

7.2 Static Structural Analysis (ANSYS)

7.2.1 Static Structural Analys of Cutter:

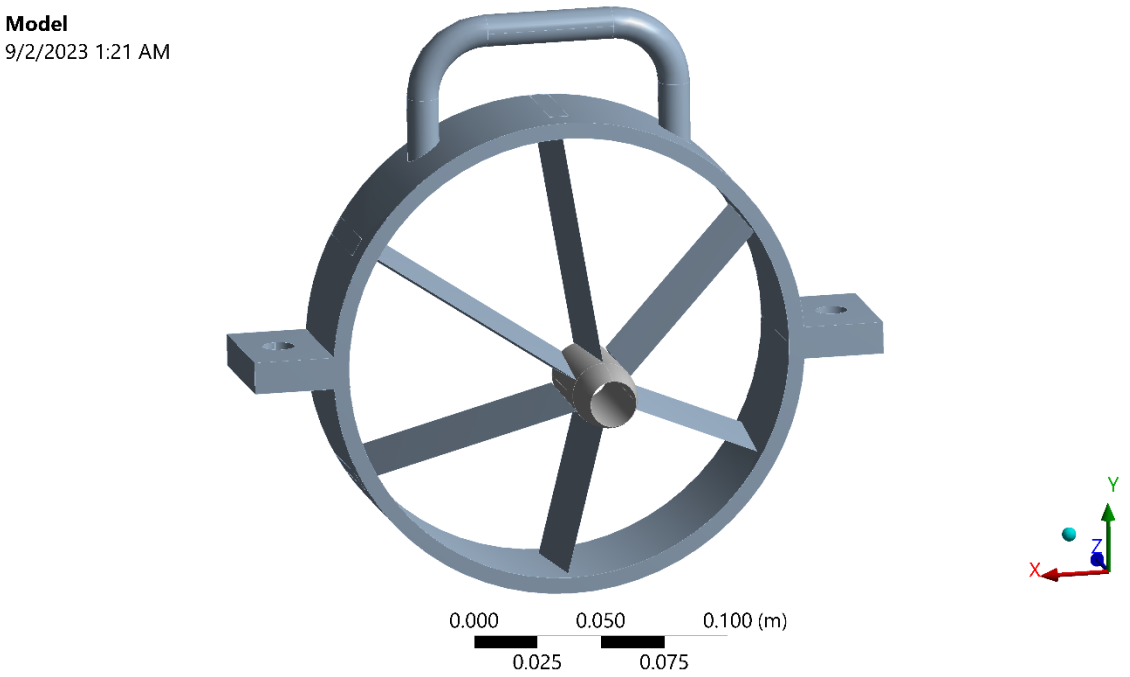


Figure 7.1 Cutter

Load:

Types	Force
X-Component	0 N (ramped)
Y-Component	0 N (ramped)
Z-Component	500 N (ramped)

1.Total Deformation:

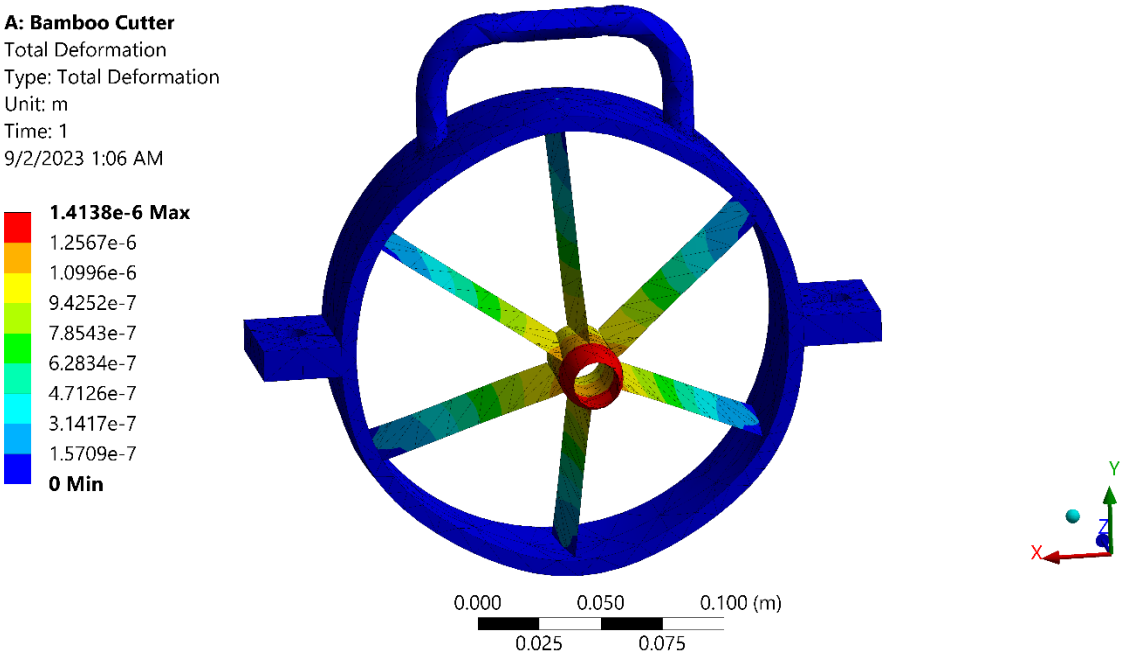


Figure 7.2 Total Deformation

Result:

Range	Total Deformation
Minimum	0 (mm)
Maximum	0.0014138 (mm)

2.Equivalent Stress:

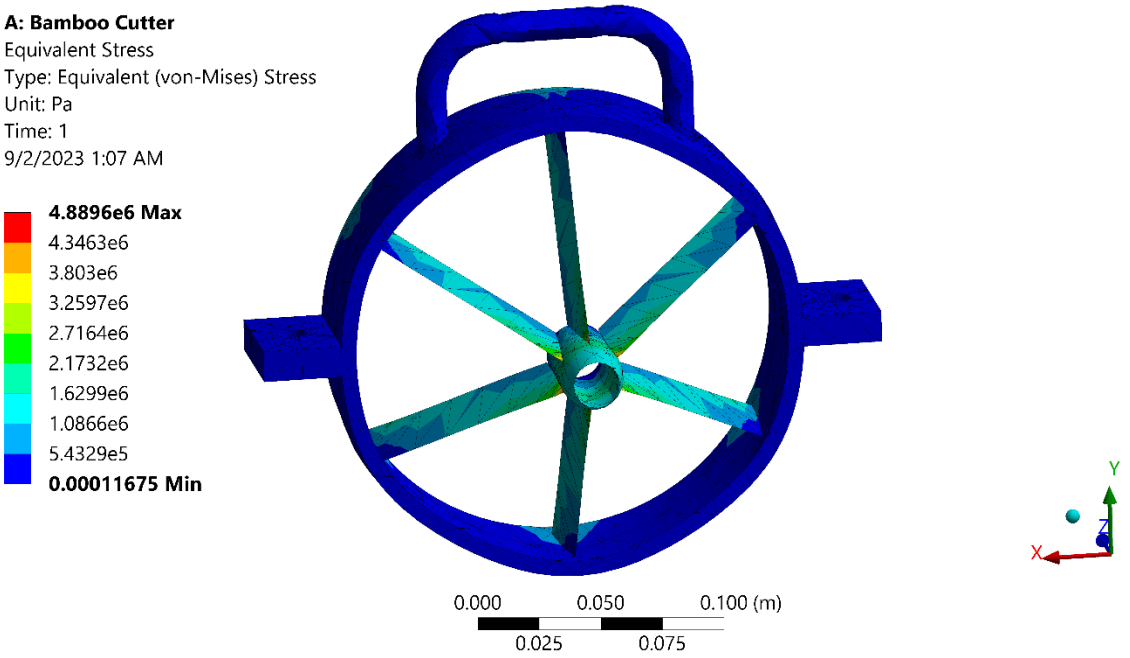


Figure 7.3 Equivalent Stress

Result:

Range	Equivalent (von-Mises) Stress
Minimum	1.1675e-10 (MPa)
Maximum	4.8896 (MPa)

3.Fatigue Life:

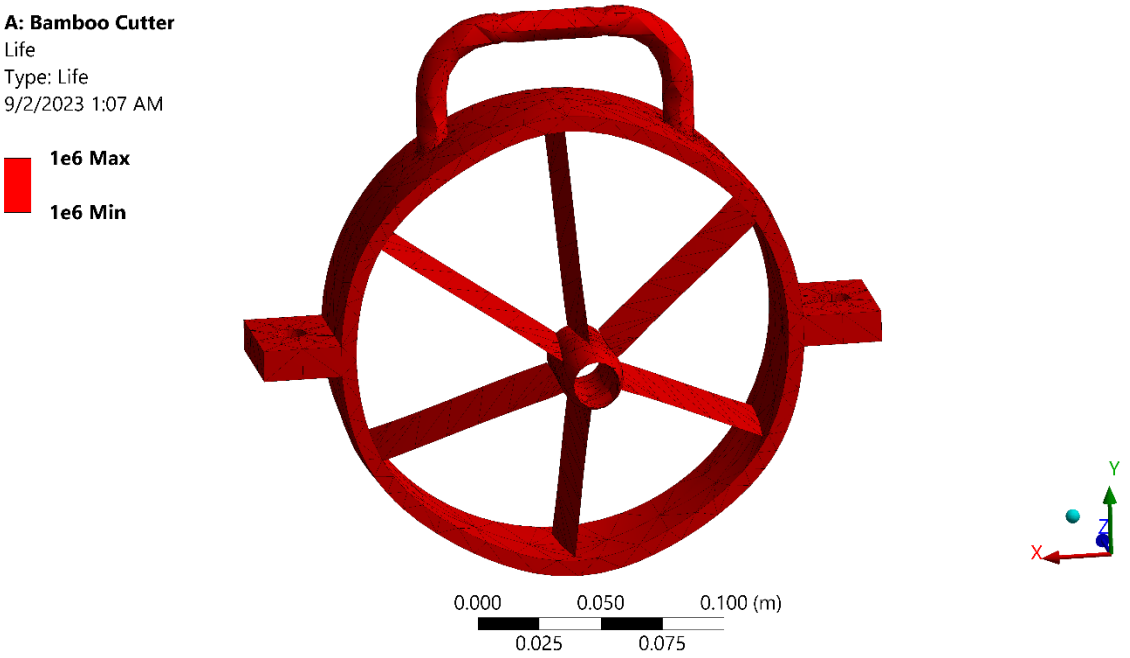


Figure 7.4 Fatigue Life

Result:

Range	Life (Cycles)
Minimum	1e6
Maximum	1e6

4.Safety Factor:

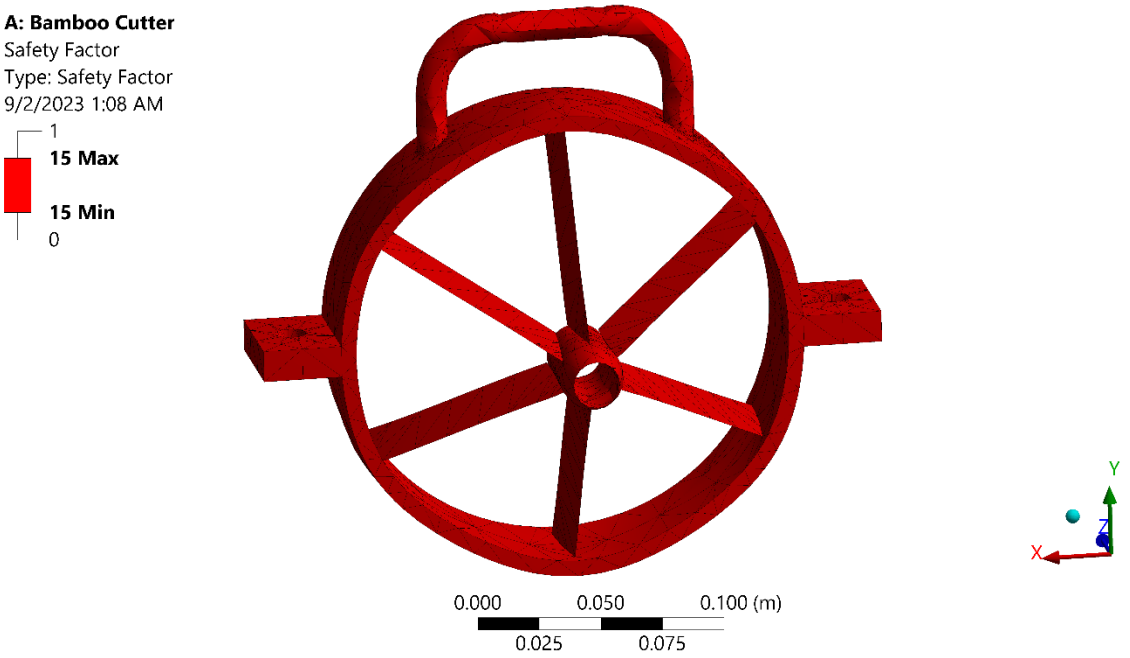


Figure 7.5 Safety Factor

Result:

Range	Safety Factor
Minimum	15
Maximum	15

5.Equivalent Alternating Stress:

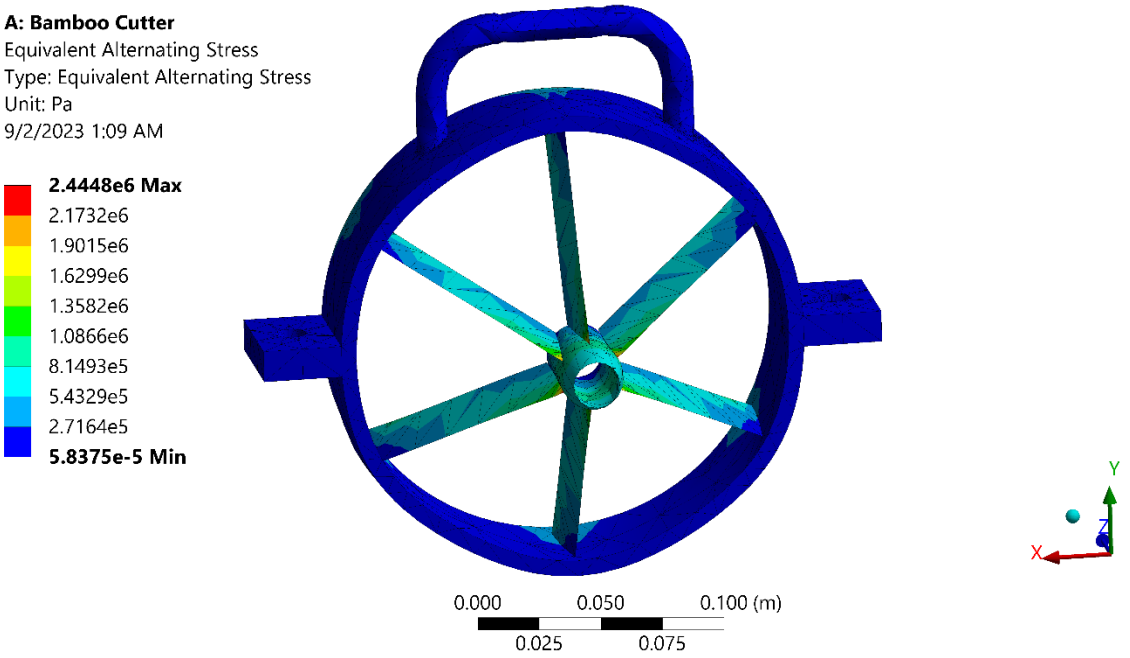


Figure 7.6 Equivalent Alternating Stress

Result:

Range	Equivalent Alternating Stress
Minimum	5.8375e-11 (MPa)
Maximum	2.4448 (MPa)

7.2.2 Static Structural Analysis of Cutter Holder:

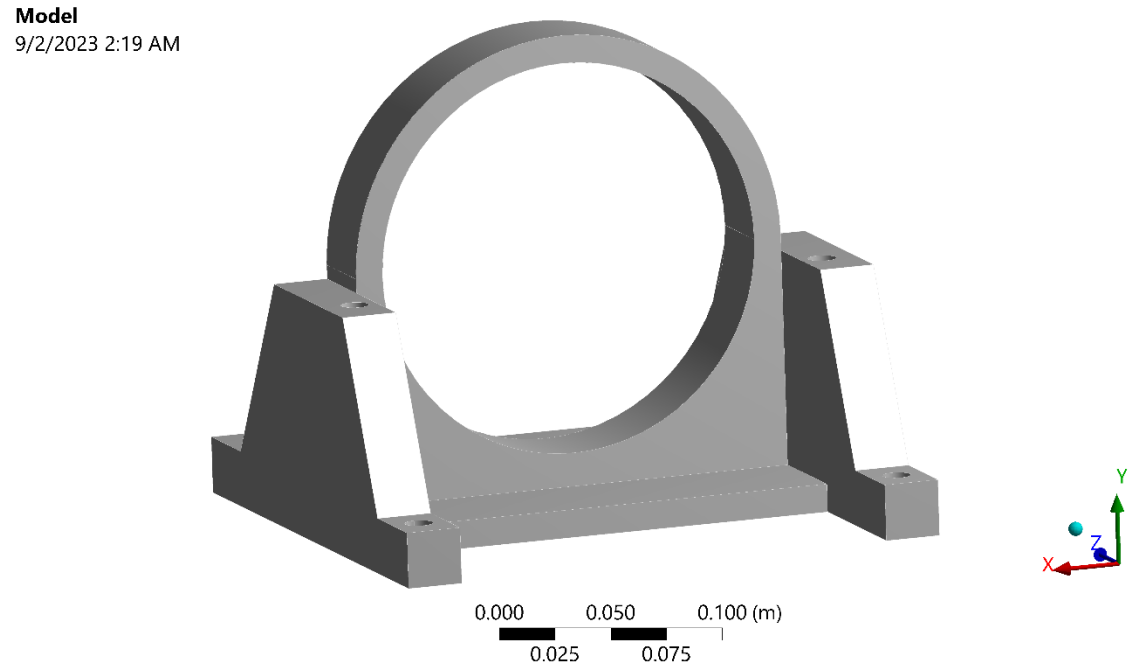


Figure 7.7 Cutter Holder

Load:

Types	Force
X-Component	0 N (ramped)
Y-Component	0 N (ramped)
Z-Component	500 N (ramped)

1.Total Deformation:

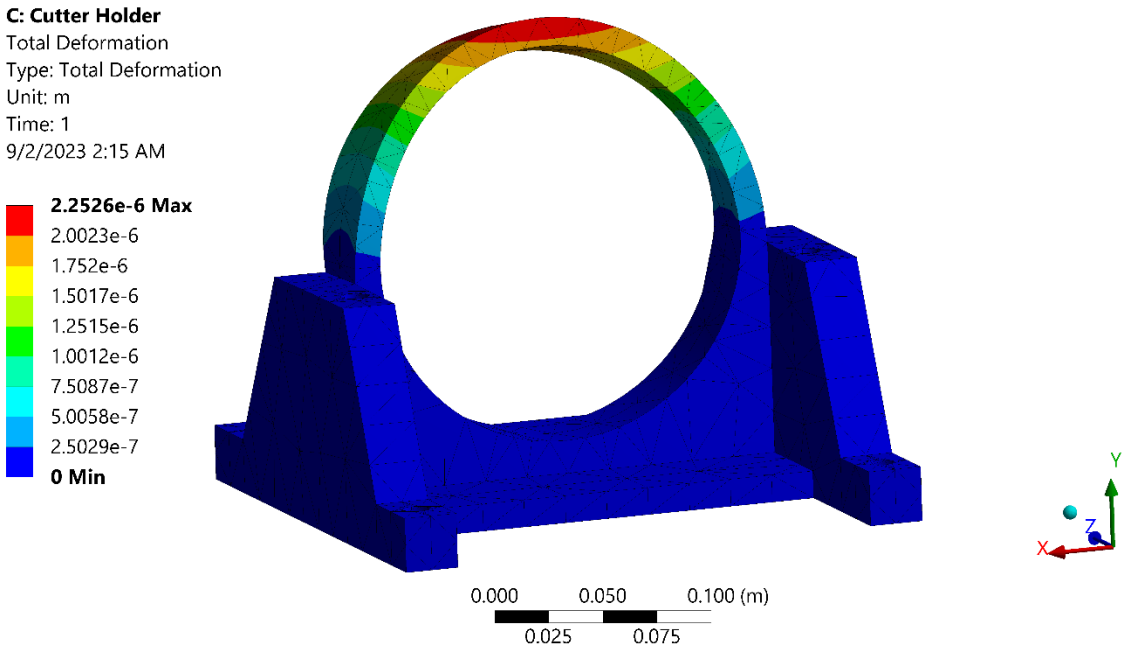


Figure 7.8 Total Deformation

Result:

Range	Total Deformation
Minimum	0 (mm)
Maximum	0.0014138 (mm)

2.Equivalent Stress:

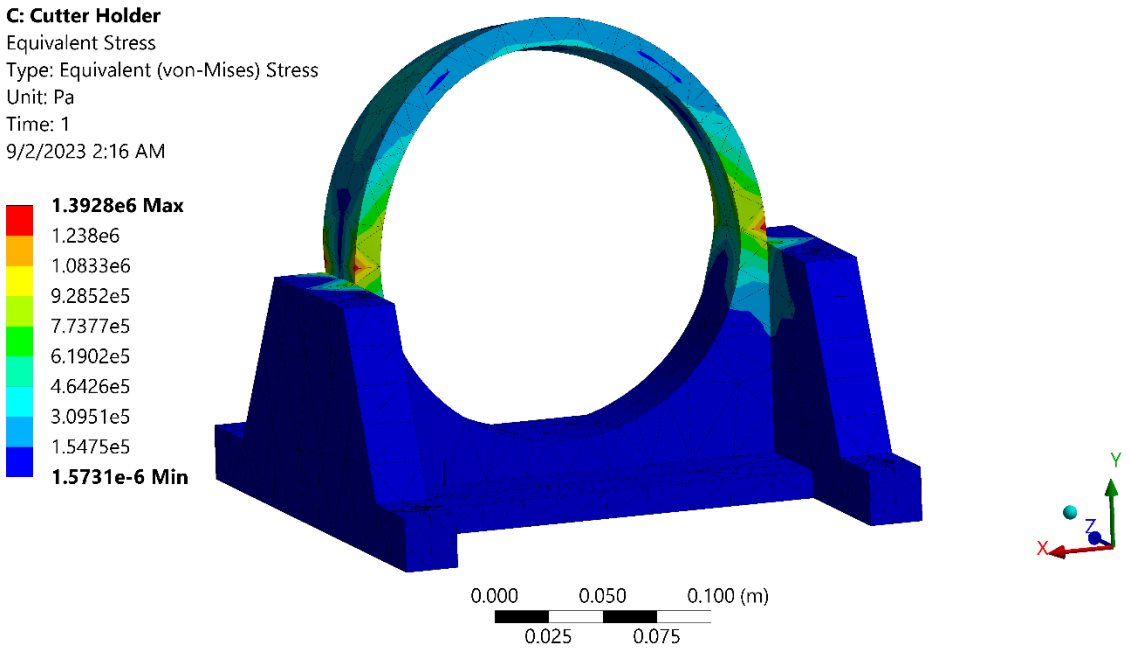


Figure 7.9 Equivalent Stress

Result:

Range	Equivalent (von-Mises) Stress
Minimum	1.5731e-12 (MPa)
Maximum	1.3928 (MPa)

3.Fatigue Life:

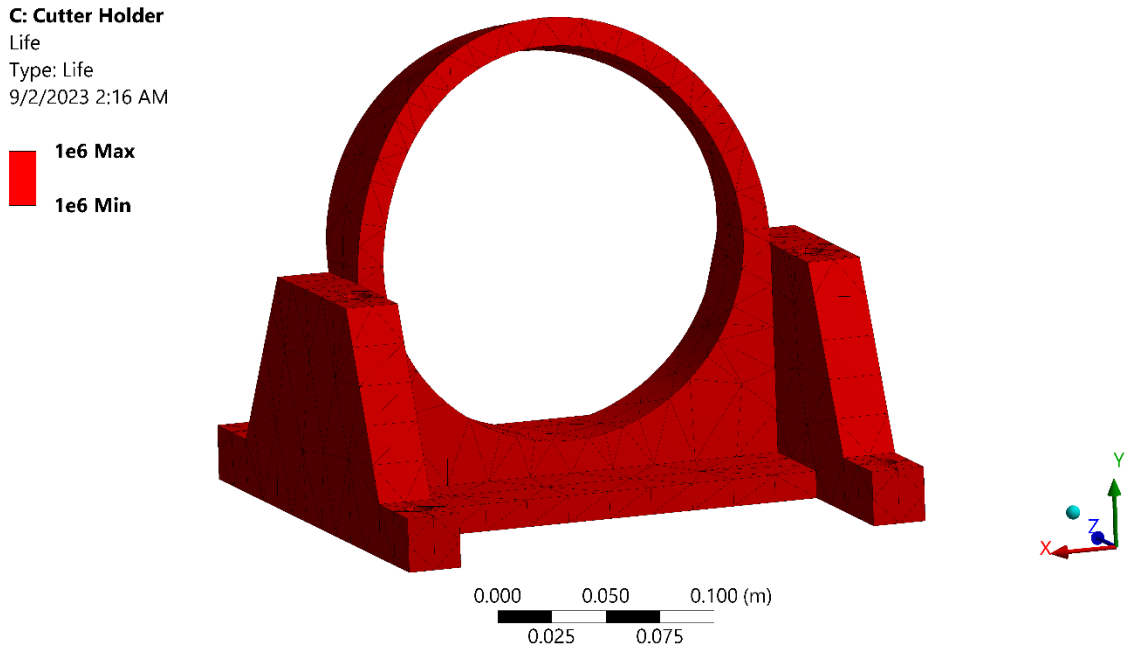


Figure 7.10 Fatigue Life

Result:

Range	Life (Cycles)
Minimum	1e6
Maximum	1e6

4.Safety Factor:

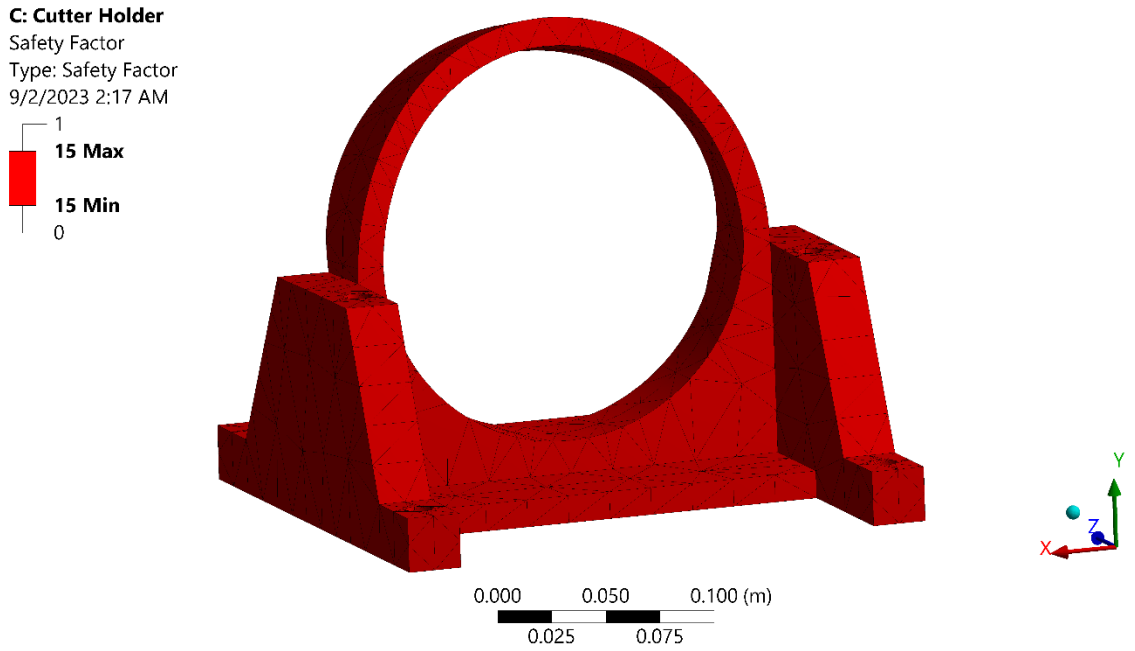


Figure 7.11 Safety Factor

Result:

Range	Safety Factor
Minimum	15
Maximum	15

5.Equivalent Alternating Stress:

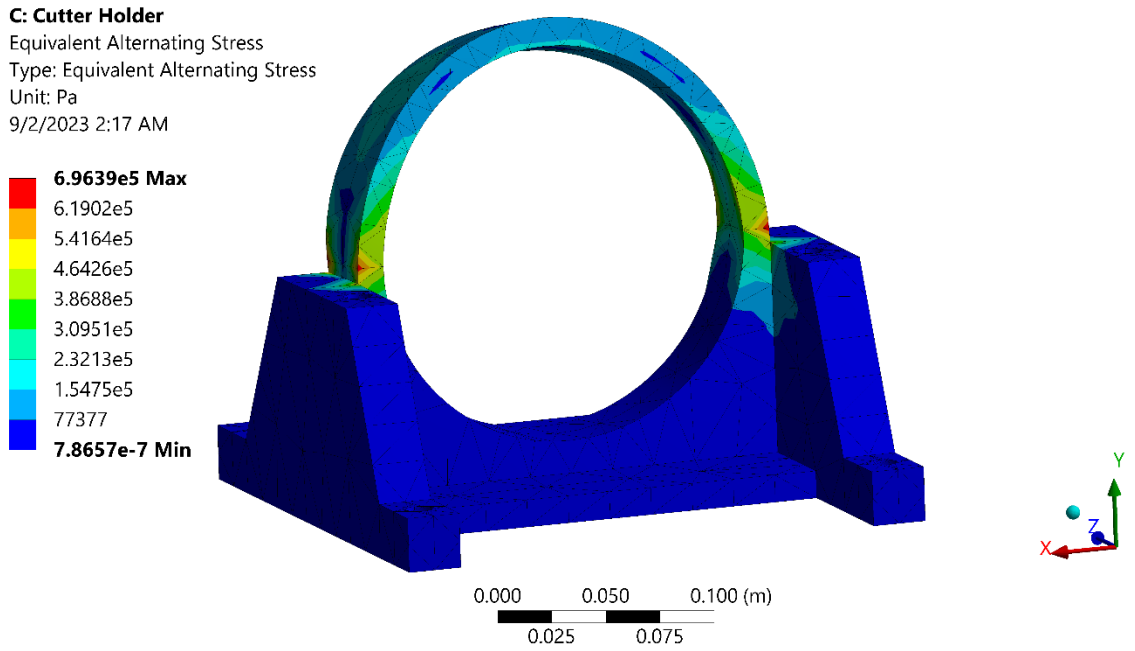


Figure 7.12 Equivalent Alternating Stress

Result:

Range	Equivalent Alternating Stress
Minimum	7.8657e-13 (MPa)
Maximum	0.69639 (MPa)

7.2.3 Static Structural Analysis of Bearings:

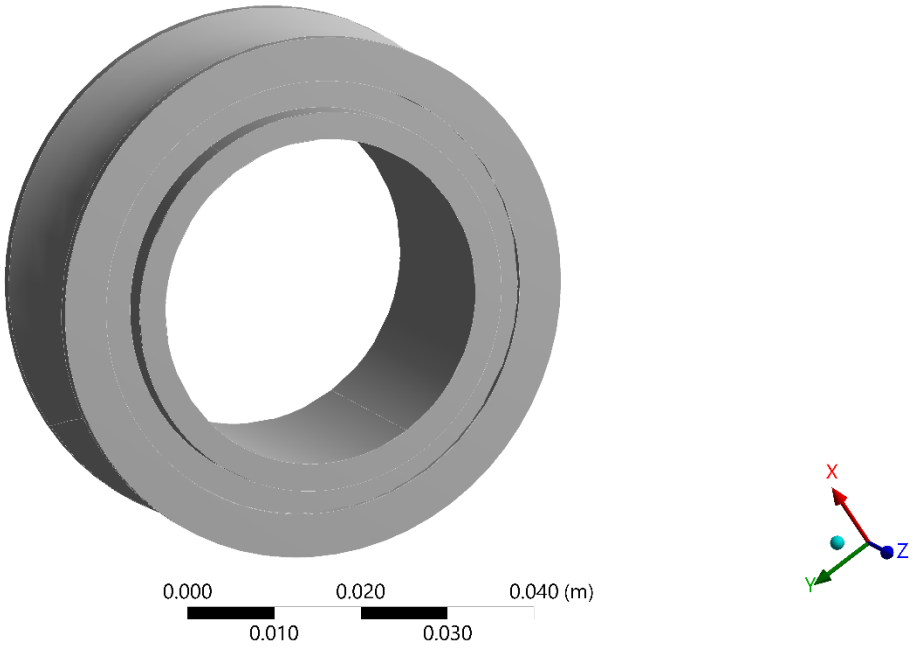


Figure 7.13 Bearings

Load:

Types	Force
X-Component	0 N (ramped)
Y-Component	200 N (ramped)
Z-Component	0 N (ramped)

1.Total Deformation:

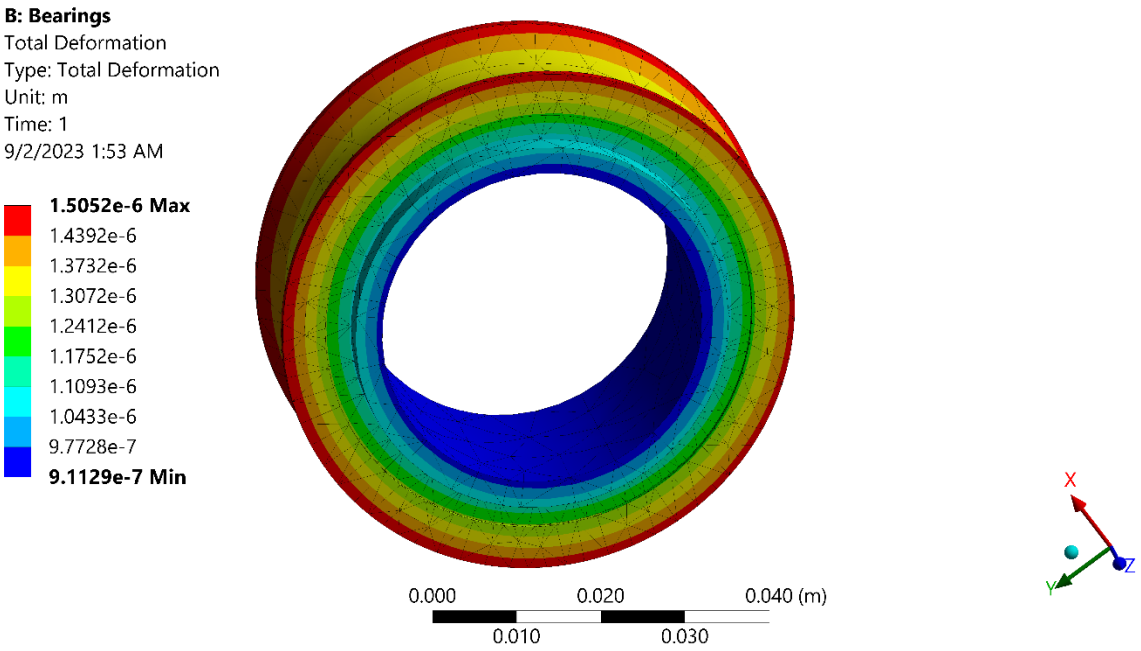


Figure 7.14 Total Deformation

Result:

Range	Total Deformation
Minimum	0.00091129 (mm)
Maximum	0.0015052 (mm)

2.Equivalent Stress:

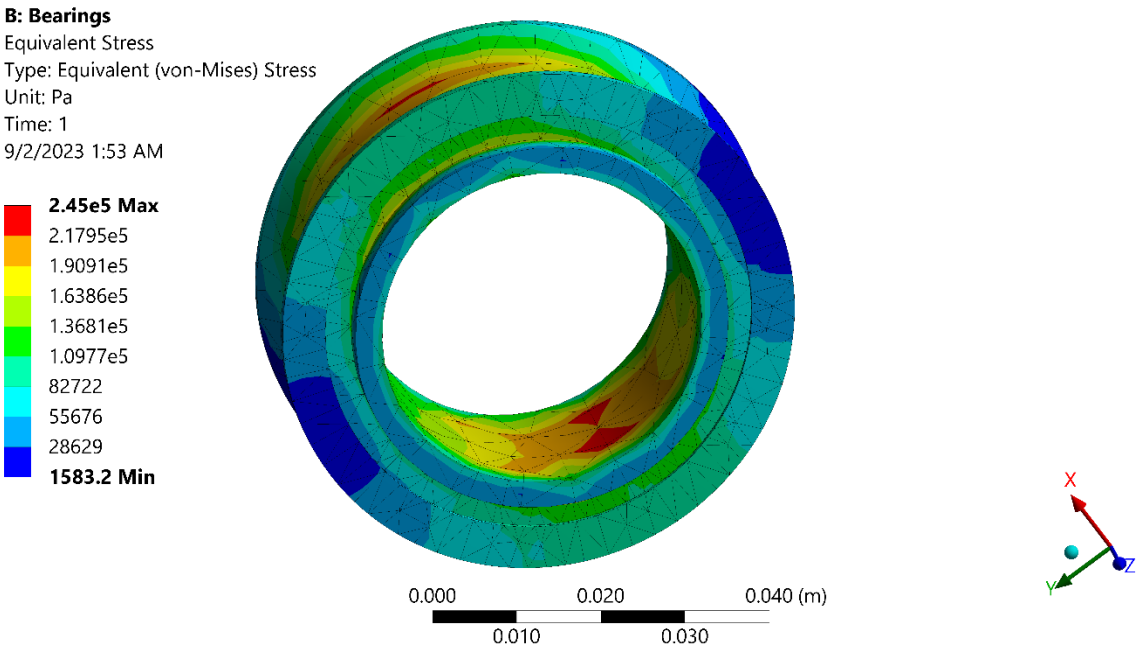


Figure 7.15 Equivalent Stress

Result:

Range	Equivalent (von-Mises) Stress
Minimum	0.0015832 (MPa)
Maximum	0.245 (MPa)

3.Fatigue Life:

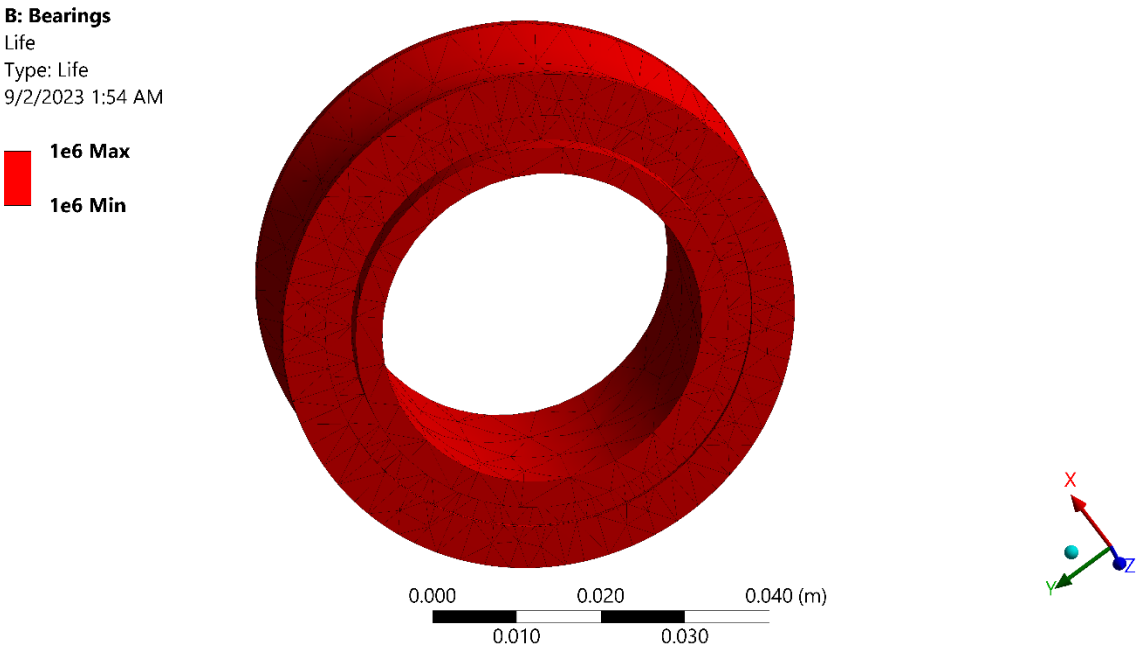


Figure 7.16 Fatigue Life

Result:

Range	Life (Cycles)
Minimum	1e6
Maximum	1e6

4.Safety Factor:

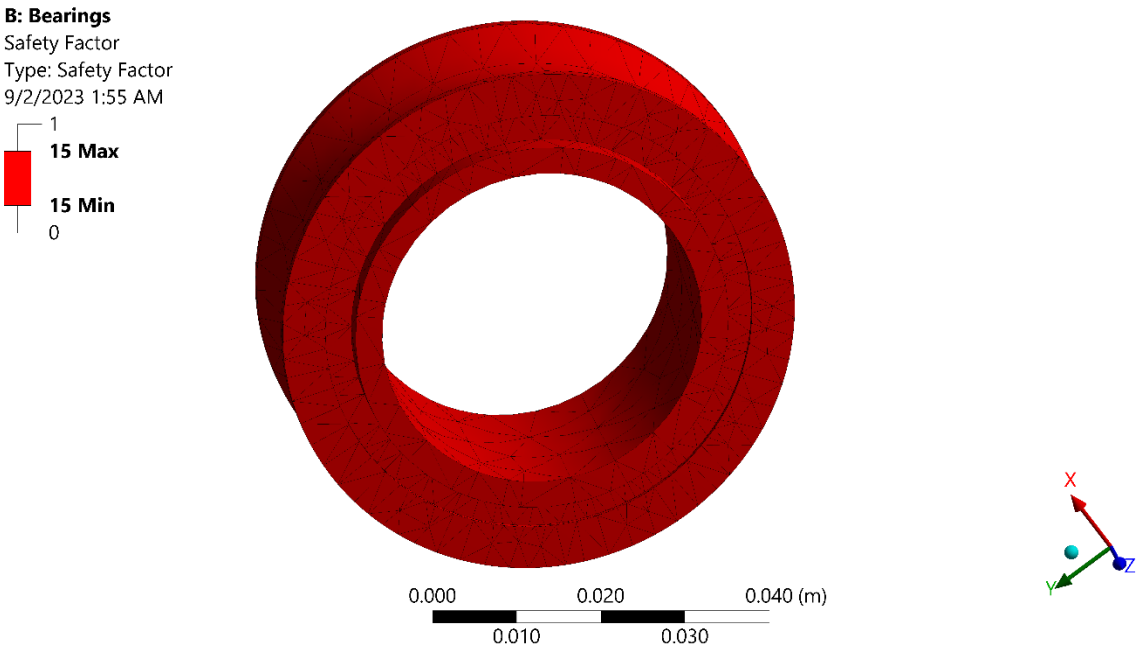


Figure 7.17 Safety Factor

Result:

Range	Safety Factor
Minimum	15
Maximum	15

5.Equivalent Alternative Stress:

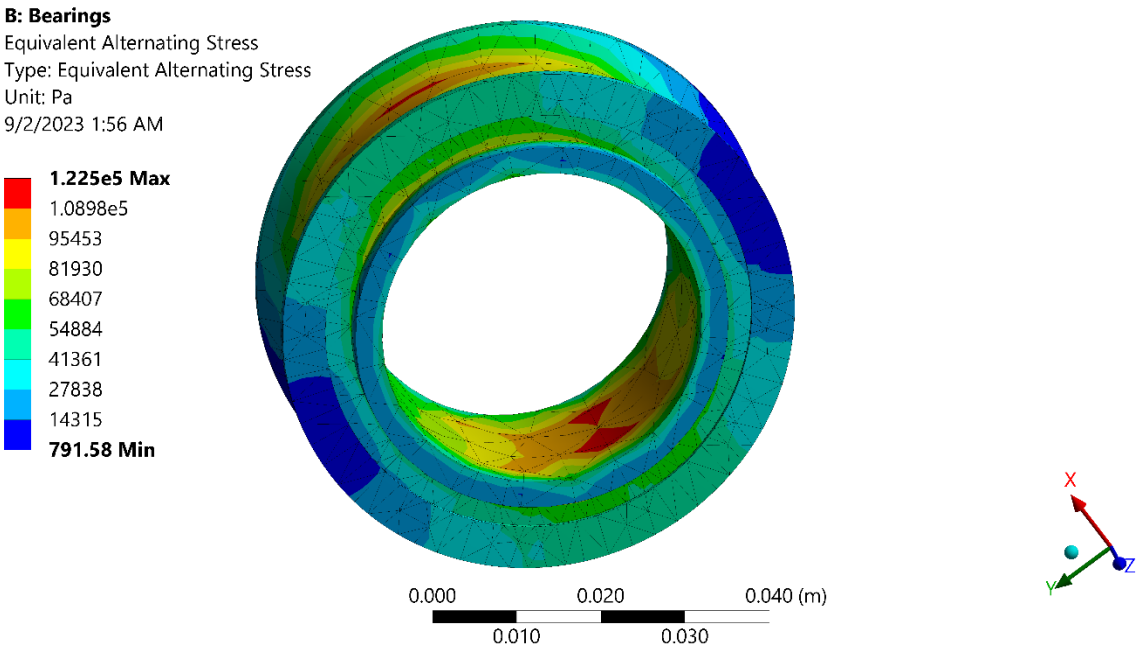


Figure 7.18 Equivalent Alternative Stress

Result:

Range	Equivalent Alternating Stress
Minimum	0.00079158 (MPa)
Maximum	0.1225 (MPa)

7.2.4 Static Structural Analysis of Pulley Holder:

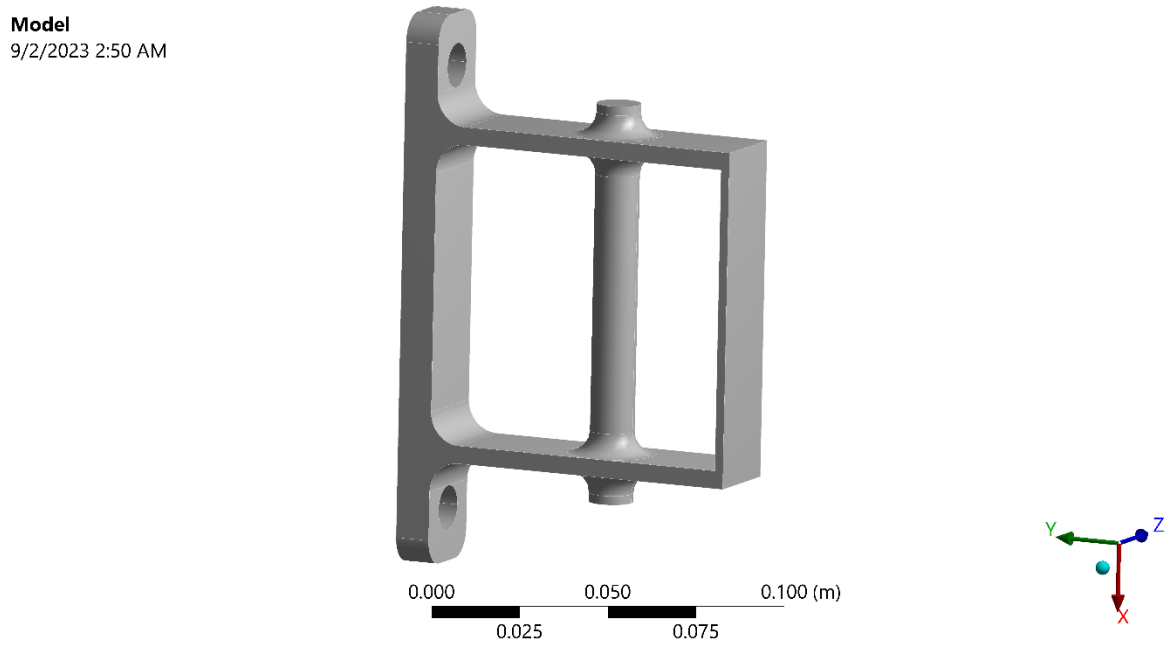


Figure 7.19 Pulley Holder

Load:

Types	Force
X-Component	0 N (ramped)
Y-Component	-1000 N (ramped)
Z-Component	0 N (ramped)

1.Total Deformation:

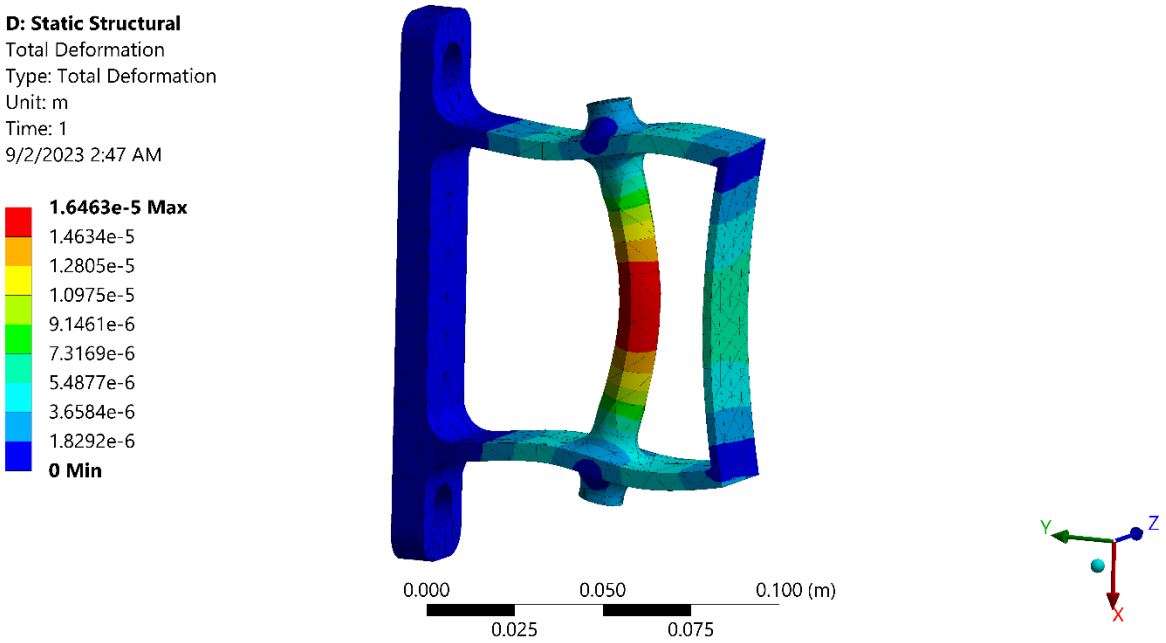


Figure 7.20 Total Deformation

Result:

Range	Total Deformation
Minimum	0 (mm)
Maximum	0.0014138 (mm)

2.Equivalent Stress:

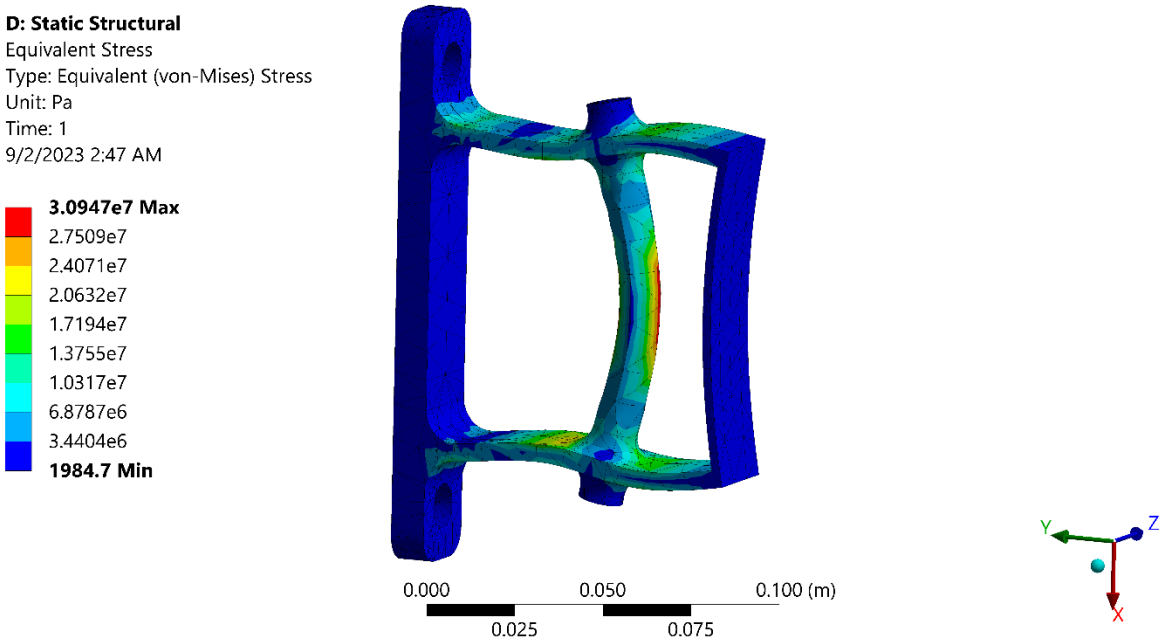


Figure 7.21 Equivalent Stress

Result:

Range	Equivalent (von-Mises) Stress
Minimum	0 (mm)
Maximum	0.016463 (mm)

3.Fatigue Life:

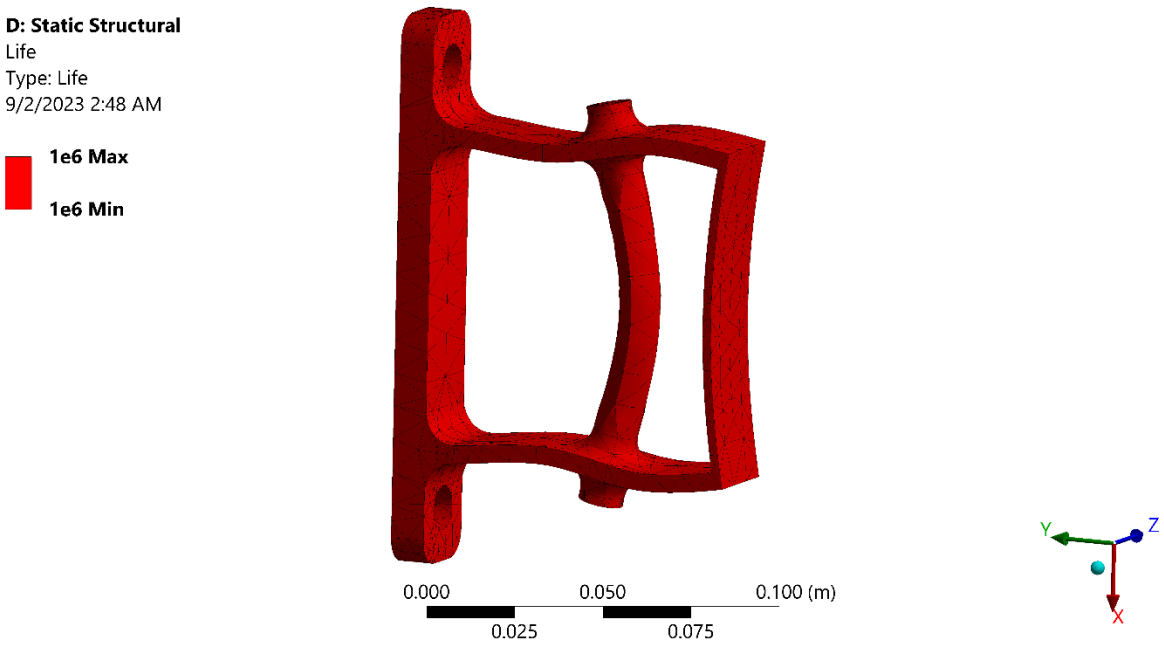


Figure 7.22 Fatigue Life

Result:

Range	Life (Cycles)
Minimum	1e6
Maximum	1e6

4.Safety Factor:

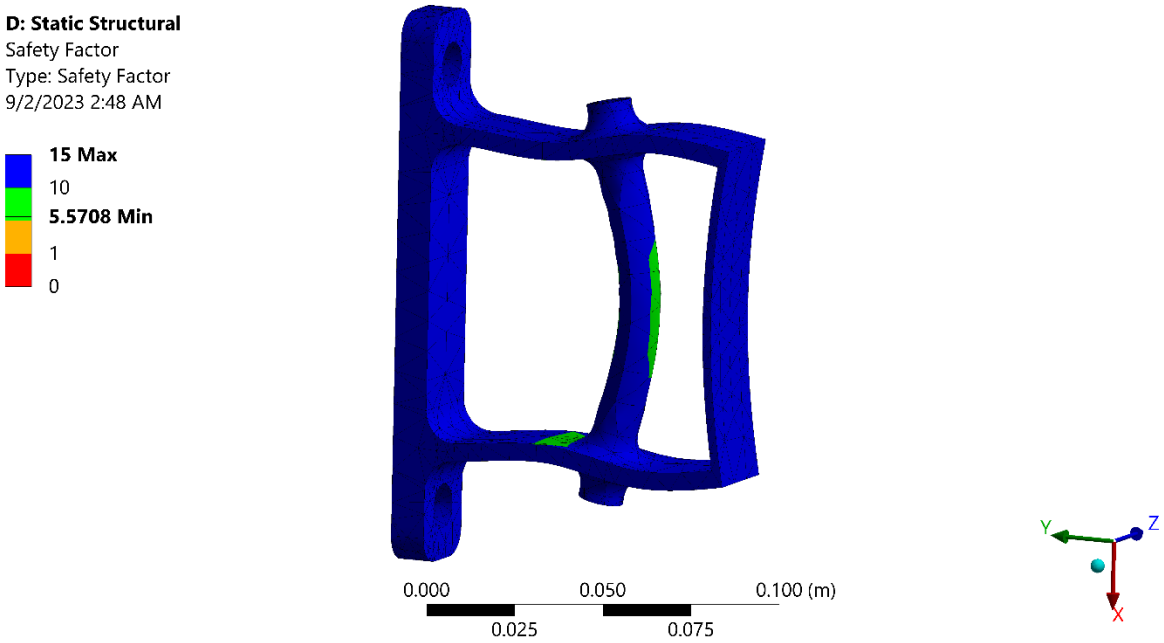


Figure 7.23 Safety Factor

Result:

Range	Safety Factor
Minimum	5.5708
Maximum	15

5.Equivalent Alternating Stress:

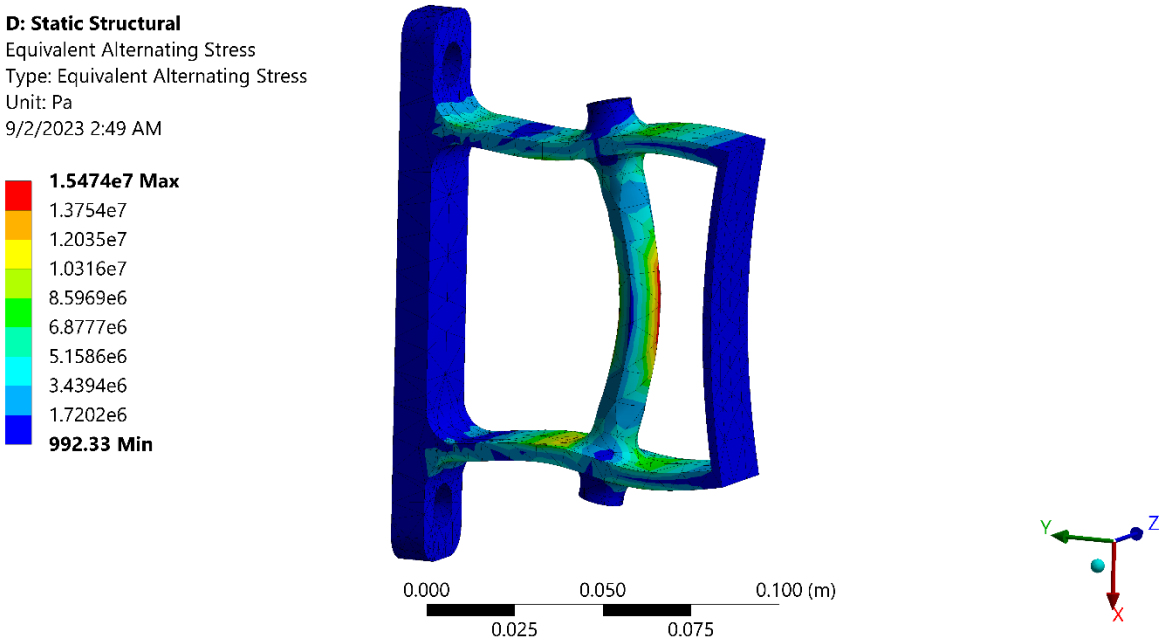


Figure 7.24 Equivalent Alternating Stress

Result:

Range	Equivalent Alternating Stress
Minimum	0.00099233 (MPa)
Maximum	15.474 (MPa)

7.2.5 Static Structural Analysis of Rammer:

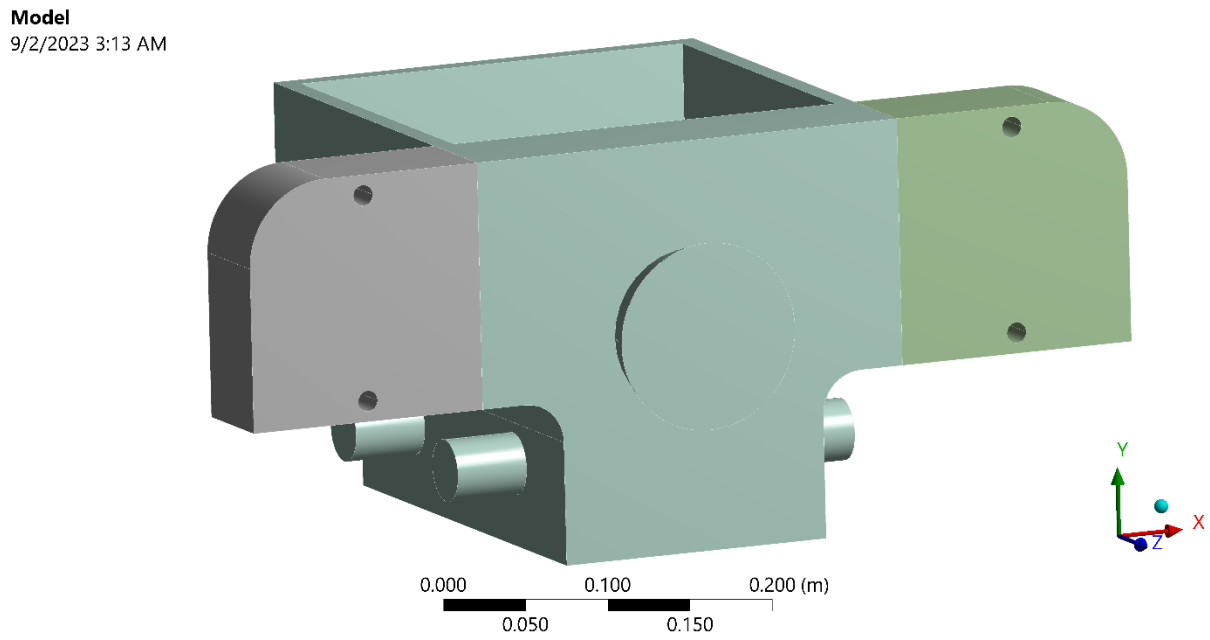


Figure 7.25 Rammer

Load:

Types	Force
X-Component	0 N (ramped)
Y-Component	0 N (ramped)
Z-Component	1000 N (ramped)

1.Total Deformation:

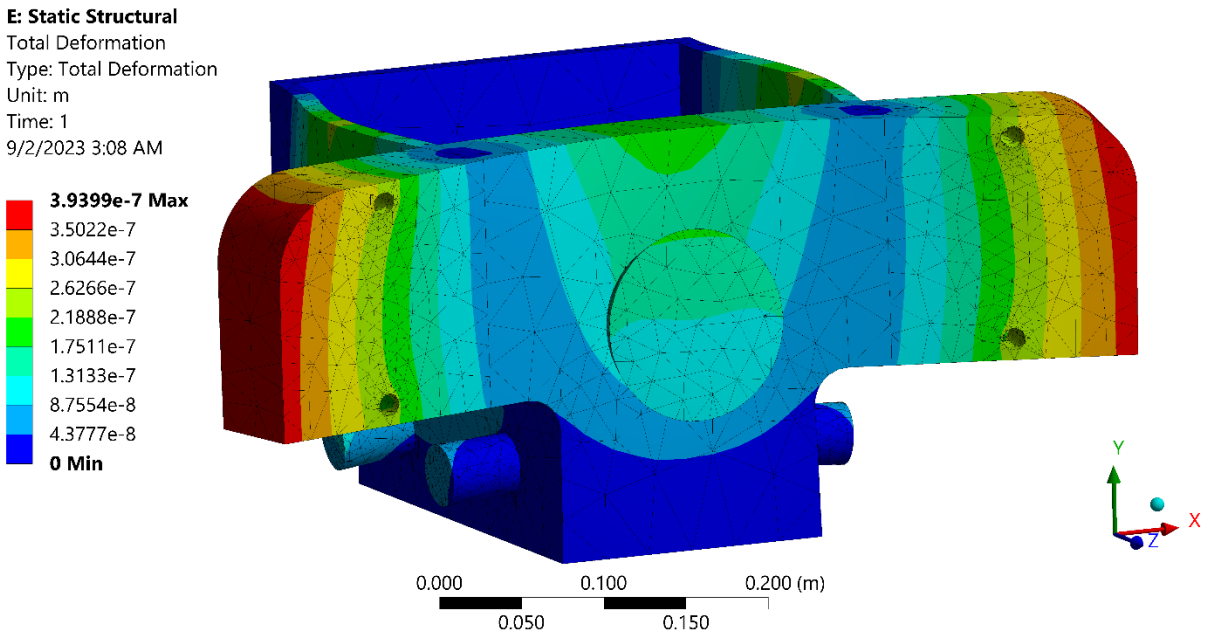


Figure 7.26 Total Deformation

Result:

Range	Total Deformation
Minimum	0 (mm)
Maximum	0.00039399 (mm)

2.Equivalent Stress:

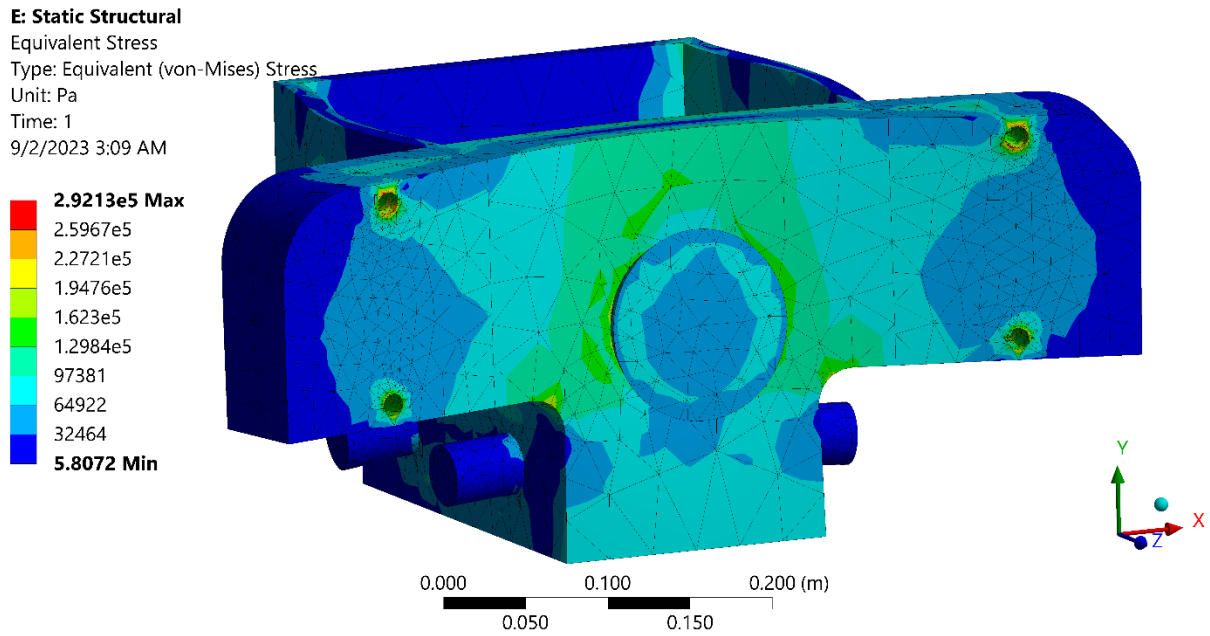


Figure 7.27 Equivalent Stress

Result:

Range	Equivalent (von-Mises) Stress
Minimum	5.8072e-6 (MPa)
Maximum	0.29213 (MPa)

3.Fatigue Life:

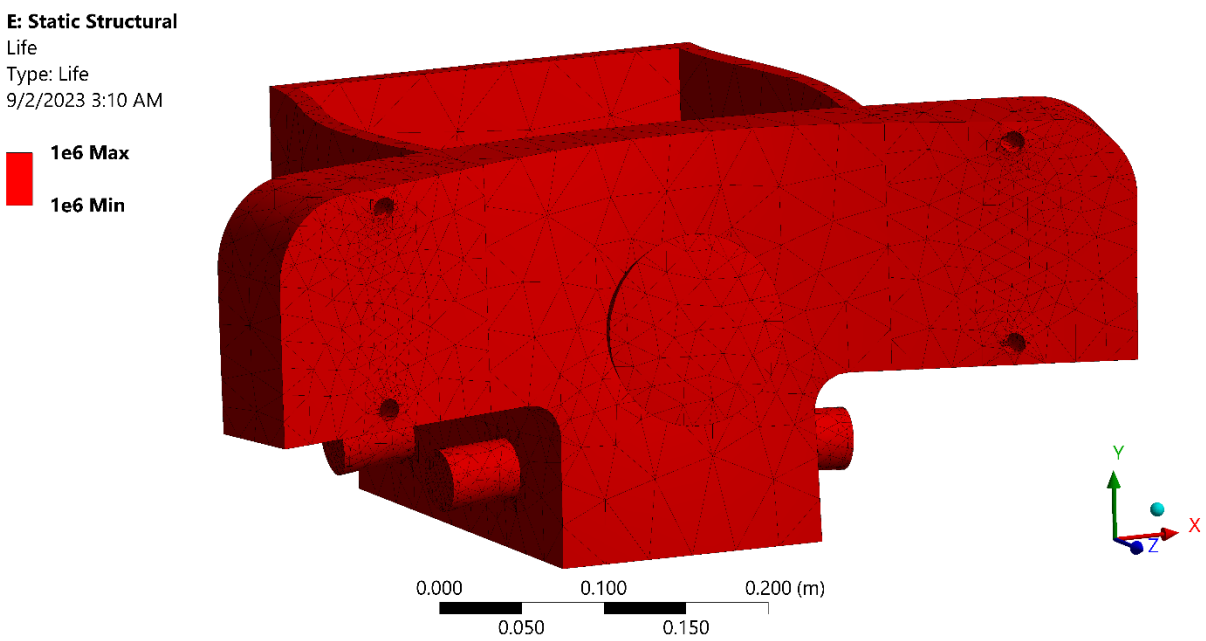


Figure 7.28 Fatigue Life

Result:

Range	Life (Cycles)
Minimum	1e6
Maximum	1e6

4.Safety Factor:

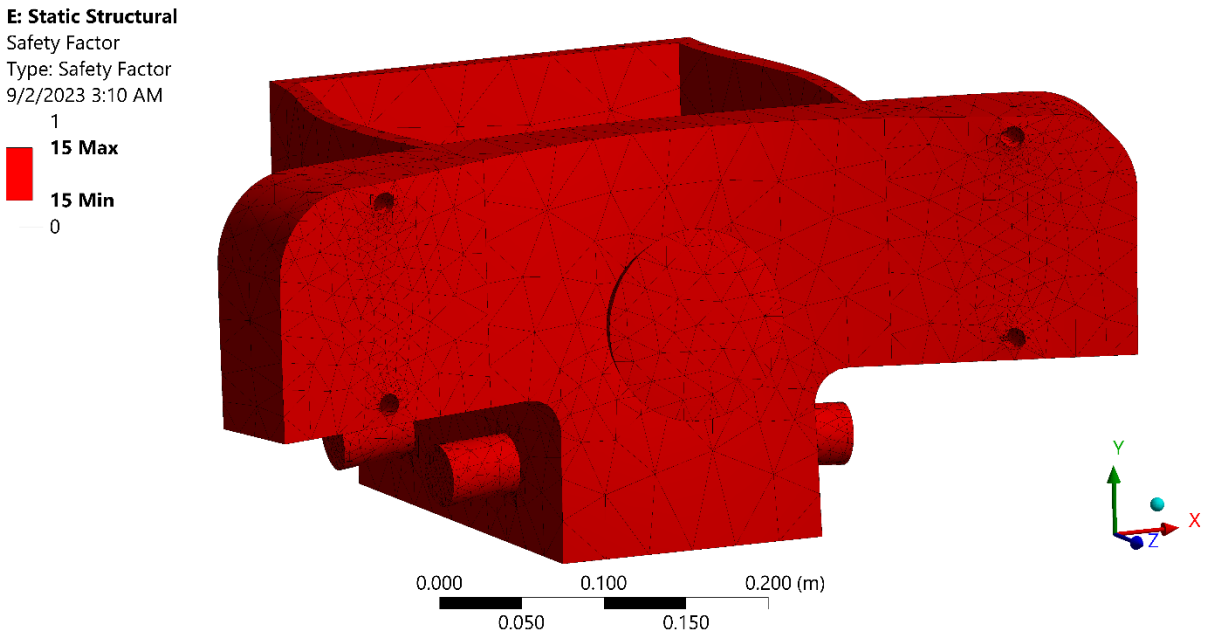


Figure 7.29 Safety Factor

Result:

Range	Safety Factor
Minimum	15
Maximum	15

5.Equivalent Alternating Stress:

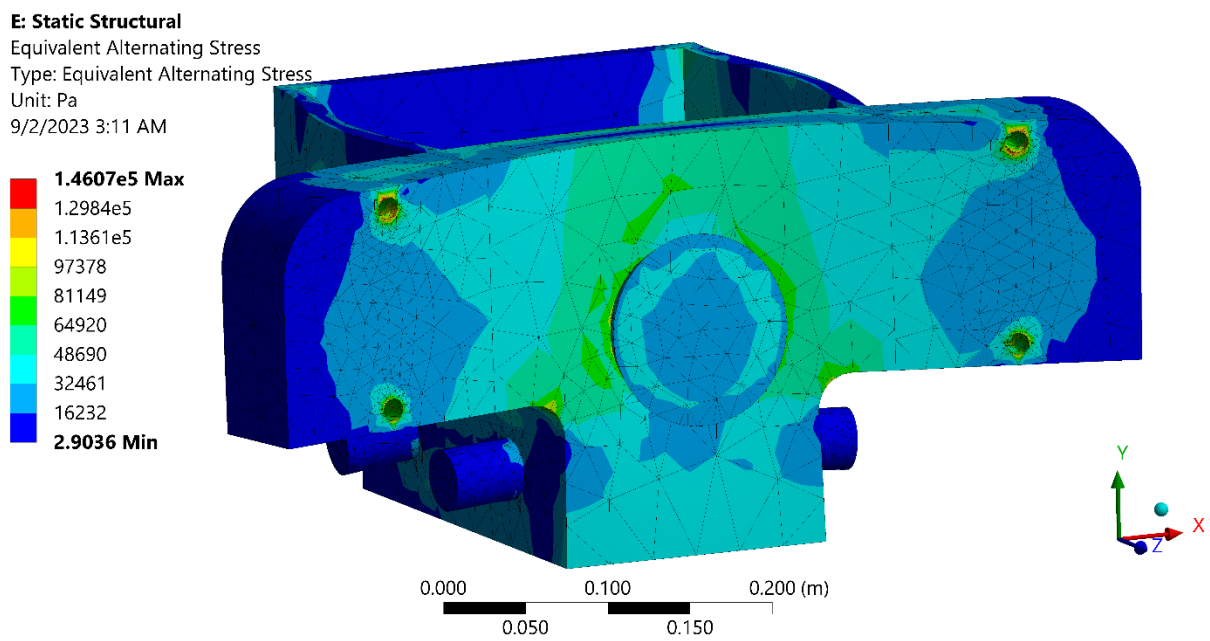


Figure 7.30 Equivalent Alternating Stress

Result:

Range	Equivalent Alternating Stress
Minimum	2.9036e-6 (MPa)
Maximum	0.14607 (MPa)

Chapter-8

Cost and Sensitivity Analysis

8.1 Introduction

Cost is intrinsically linked to the design of a product, as the decisions made during the design phase significantly influence the resulting cost. In the current competitive landscape, alongside technological advancements and aesthetics, cost emerges as the primary driving force behind consumer choices. Customers often have financial constraints that direct their purchasing decisions toward affordability as a crucial consideration. A successful product supplier must prioritize cost management. "Design to cost" emerges as a crucial management strategy. An effective approach to ensure affordability for customers involves establishing a target cost as an independent design parameter, a commitment maintained throughout the product development stages.

Cost analysis comprises several fundamental elements:

- Identifying the drivers of product cost and integrating these considerations into product specifications, with a focus on cost reduction.
- Striking a balance between meeting customer requirements and maintaining affordability.
- Actively factoring costs into the development process as a critical design parameter, alongside other decision-making criteria.
- Ensuring consistency in accounting methods between cost systems and product cost models, periodically validating the accuracy of these models.
- Establishing and allocating target costs, even down to component levels, to enable effective cost management and exploring various concepts and design alternatives to develop cost-effective design approaches.
- Pursuing continuous improvement through process value engineering to enhance the long-term value of the product.

In the development of our "Convenient Bamboo Cutting Machine," we conducted a comprehensive cost analysis, particularly from the perspective of a mass production system. Additionally, we performed break-even and sensitivity analyses to further enhance our understanding of the financial aspects of this endeavor.

8.2 Manufacturing Cost

8.2.1 Cost of Machinery

1. **Name: Forging Furnace** (Model No: SX3-5-12; Brand: TOPLINE, Korea)
   Buying Cost: 2, 00,500 Tk. [18]
   Effective working life: 15
   Salvage value: 30,000 Tk.
   Quantity: 1

2. **Name: Stand Bench Drill** (Model No: SBD1601; Brand: Semprox, China)
   Buying Cost: 27,500 Tk. [19]
   Effective working life: 20 years
   Salvage value: 1,500Tk.
   Quantity: 1

3. **Name: Grinder Tool** (Model No: E9961-3; Brand: KOGEEK, China)
   Buying Cost: 14,220 Tk. [20]
   Effective working life: 18 years
   Salvage value: 2,000 Tk.
   Quantity: 1

4. **Name: Welding Machine** (Model No: ST - MMA 300; Brand: Smartec, China)
   Buying Cost: 8,190 Tk. [21]
   Effective working life: 20 years
   Salvage value: 1,000 Tk.
   Quantity: 1

5. **Name: Casting setup** (Local market)
   Buying Cost: 15,000 Tk.
   Effective working life: 5 years
   Salvage value: 2,000 Tk
   Quantity: 1

6. **Name: Spray Painting Machine** (Model No: TT5006; Brand: Total, China)
   Buying Cost: 4,150 Tk [22]
   Effective working life: 5 years
   Quantity: 1

Total Cost of Machineries: 2, 69,560 Tk

8.2.2 Cost of Raw Material

1. Mild Steel (ASTM A36; GRADE: A36)
   Required Quantity: 65 kg.
   Market price: 90 Tk /kg
   Total Cost Per Product: 5,850 Tk

2. High Speed Steel (ASTM M2; Grade: M2)
   Required Quantity: 2 kg.
   Market price: 120Tk/ kg
   Total Cost Per Product: 240 Tk

3. Cast Iron (ASTM A48; GRADE: 25A)
   Required Quantity: 10 kg.
   Market price: 65 Tk / KG
   Total Cost Per Product: 650 Tk

4. Welding Rod
   Required Quantity: 1 packet.
   Market price: 450 Tk/ packet
   Total Cost Per Product: 450 Tk

5. Wood
   Total Cost Per Product: 400 Tk

6. Paint
   Total Cost Per Product: 200 Tk

Total Cost of Raw Materials Per Product: 7,790 Tk
No of Production per Year: 300
Total Cost of Raw Materials Per Year: 23,37,000 Tk

8.2.3 Cost of Labor

1. Name of job: Machining
   Number of workers: 1
   Wage rate: 25,000 Tk.
   Total labor cost: 25,000 Tk.

2. Name of job: Blade Forging
   Number of workers: 1
   Wage rate: 20,000 Tk.
   Total labor cost: 20,000 Tk.

3. Name of job: Welding
   Number of workers: 2
   Wage rate: 20,000 Tk.
   Total labor cost: 40,000 Tk.

4. Name of job: Casting
   Number of workers: 2
   Wage rate: 15,000 Tk.
   Total labor cost: 30,000 Tk.

5. Name of job: Painting
   Number of workers: 1
   Wage rate: 8000 Tk.
   Total labor cost: 8000 Tk.

Total Cost of Labor Per Month: 1, 23,000 Tk.

Total Cost of Labor Per Year: 14, 76,000 Tk

8.2.4 Purchasing Cost Per Unit of Product

Table 8.1: Purchasing Cost Per Unit of Product

Parts	**Price per unit (Tk.)**	**Quantity**	**Total Cost**
Pulley	350	2	700
Bearings	150	4	600
Rope	5	45 feet	225
Supporting feet	50	6	300
Total			**1,825**

Total Purchasing cost Per Year: 5, 47,500 Tk

8.2.5 Manufacturing Overhead Cost

Table 8.2: Manufacturing Overhead Cost

Cost Item	No. of Post	Salary per Person (Tk)	Total Cost (Tk)
Production Manager	1	40,000	40,000
Manufacturing Engineer	1	30,000	30,000
Design Engineer	1	30,000	30,000
Supply Chain & Quality Control Manager	1	35,000	35,000
Power Consumption			15,000
Factory Rent			20,000
Monthly Manufacturing Overhead			**1,70,000**

Yearly Manufacturing Overhead = 20, 40,000 Tk

8.2.6 Total Manufacturing Cost

Total Cost of Raw Materials Per Year: 23, 37,000 Tk

Total Cost of Purchasing Per Year: 5, 47,500 Tk

Total Cost of Labor Per Year: 14, 76,000 Tk

Total Manufacturing Overhead Cost Per Year: 20, 40,000 Tk

Total Manufacturing Cost Per Year: 64, 00,500 Tk

8.3 Non-Manufacturing Costs

Non-manufacturing costs are the costs that are not directly related to manufacturing, but these are needed to maintain the supply chain properly. Generally, these include selling and administrative costs. Non-manufacturing costs are generally incurred at the later portion of the supply chain.

8.3.1 Administrative Cost (per month):

Table 8.3: Administrative Cost

Post	No. of Post	Salary/Person (Tk)	Total Cost (Tk)
Chief Executive Officer	1	45,000	45,000
HR Manager	1	35,000	35,000
Accountant	1	25,000	25,000
Secretary	1	20,000	20,000
Clerk	1	10,000	10,000
Guard	1	10,000	10,000
Office Rent			10,000
Power Consumption			3,000
Water Bill			500
Total			**1,58,500**

Yearly Administrative cost: 19, 02,000 Tk

8.3.2 Sales and Marketing Expenses

Table 8.4 sales and Marketing expenses

Cost Items	No. of post	Salary/person	Total (TK)
Marketing Executive	1	25,000	25,000
Advertising			10,000
Total			**35,000**

Yearly Sales and Marketing costs: 4, 20,000 Tk

Total Non-Manufacturing costs: 23, 22,000 Tk

8.4 Bank Loan and Interest

Total bank loan= 1, 00, 00,000 Tk

Interest rate, r = 15%

Repayment periods, n= 20 years.

Amount to be paid to the bank annually, A= P (A/P, 15, 20)

= 15, 97,615 Tk

8.5 Total Cost

Total Manufacturing Cost Per Year: 64, 00,500 Tk

Total Non-manufacturing Cost Per Year: 23, 22,000 Tk

Total Bank Loan Repayment Per Year: 15, 97,615 Tk

Total Cost Per Year: 1,03,20,115 Tk

8.6 Break-even Analysis

Fixed Cost for First Year

Bank payment: 15, 97,615 Tk

Machine Costs: 2, 69,560 Tk

Manufacturing Overhead: 20,40,000 Tk

Selling and Administrative Costs: 23,22,000 Tk

Total amount of fixed cost: 62, 29,175 Tk

No of Production in First Year: 300

Fixed costs per unit production: 20,764 Tk

Variable Cost for First Year

Raw material costs: 23,37,000 Tk

Purchasing Costs: 5,47,500 Tk

Labor costs: 14,76,000 Tk

Total variable costs per year: 43, 60,500 Tk

Total variable costs per unit production: 14,535 Tk

Selling price: 20,000 Tk

Profit: 5465 Tk per product

At Beak-Even Point,

Selling price × Break-even unit, (x) = Fixed costs + Variable costs

20,000x = 62,29,175 + 14,535x

This yields **x = 1140 units**

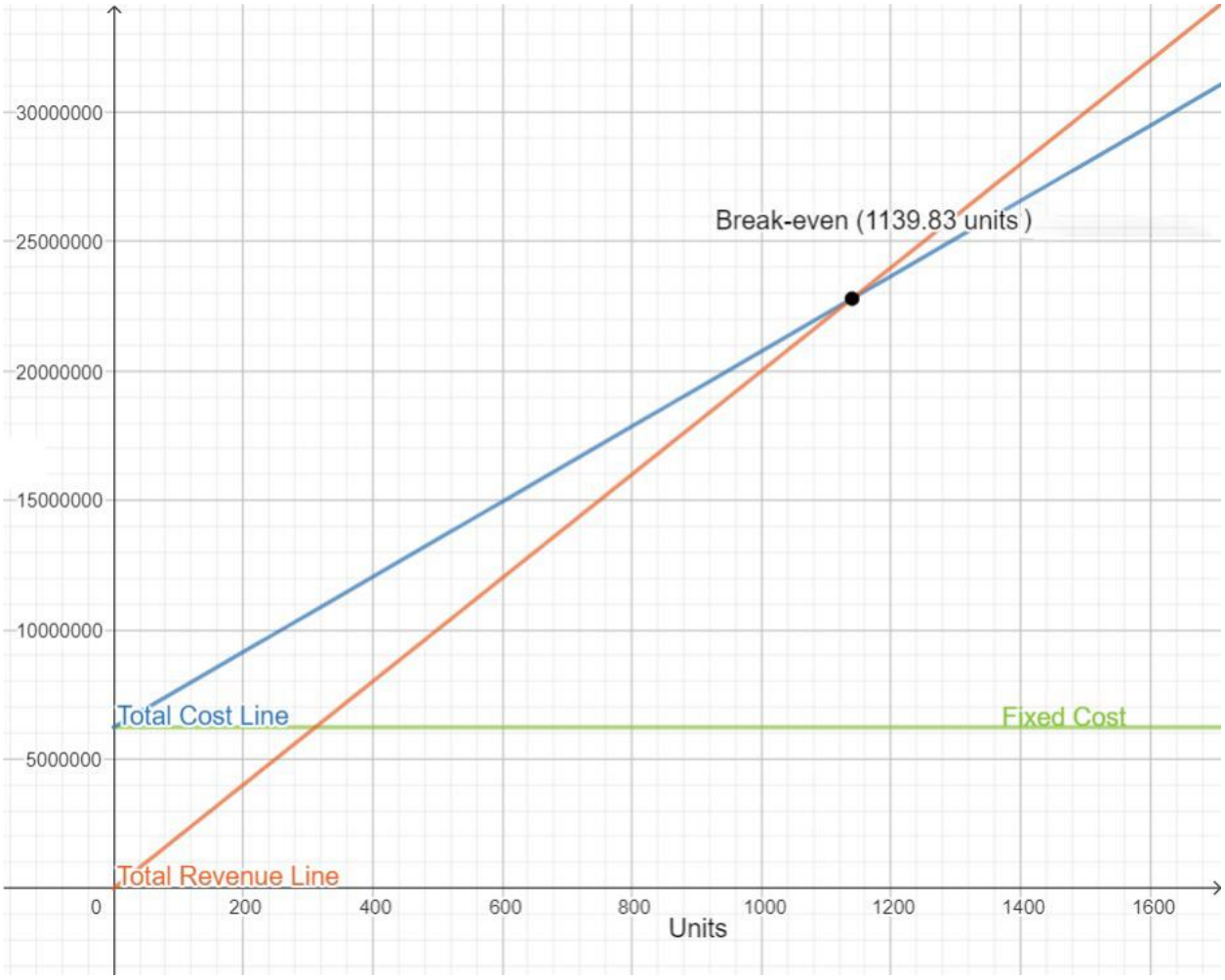


Figure 8.1 Graphical Representation of the Break-Even Analysis

8.7 Sensitivity Analysis

Sensitivity analysis gives an insight about how sensitive the break-even point is with respect to a particular cost or revenue parameter. In this sensitivity analysis, a change of 10% is made for each cost parameter separately and then a break-even point is computed. This analysis will provide the list of sensitive parameters.

8.7.1 Sensitivity of Demand:

Break Even Point,

Selling price × Q_{BEP} = Total Fixed costs + (Variable costs × Q_{BEP})

Or, 20,000 × Q_{BEP} × 1.1 = 62,29,175 + (14,535 × Q_{BEP} × 1.1)

$\therefore Q_{BEP}$ =1,037

ΔQ_{BEP} = ((1,140 -1,037)/ 1,140) ×100%

= 9.035%

So, Demand is very sensitive. The break-even point changes almost proportionally with the change in demand. So, any change in demand will have a direct impact on the break-even quantity.

8.7.2 Sensitivity of Direct Material:

At Break Even Point,

Selling price × Q_{BEP} = Total Fixed costs+ (Variable costs × Q_{BEP})

Or, 20,000 × Q_{BEP} = 62, 29,175 + ((4,920+ ((28, 84,500×1.1)/300)) × Q_{BEP})

$\therefore Q_{BEP}$ = 1,384

ΔQ_{BEP} = ((1,384 -1,140)/ 1,140) ×100%

= 21.4%

So, Direct Material is also very sensitive. The cost of direct material can significantly dominate on the break-even quantity.

8.7.3 Sensitivity of Direct Labor:

At Break Even Point,

Selling price × Q_{BEP} = Total Fixed cost+ (Variable cost × Q_{BEP})

Or, 20,000 × Q_{BE} = 62, 29,175 + ((9,615+ ((14, 76,000 × 1.1)/300)) × Q_{BEP})

$\therefore Q_{BEP}$ = 1,253

ΔQ_{BEP} = ((1,253 -1,140)/ 1,140) ×100%

= 9.91%

So, Direct Labor is very sensitive. The break-even point changes about 9.91% with 10% change in direct labor. This indicates that if somehow cost of direct labor goes up and other cost parameters stay stable, then about 9.91% production is to be increased in order to attain break-even point.

8.7.4 Sensitivity of Manufacturing Overhead:

At Break Even Point,

Selling price × Q_{BEP} = Total Fixed cost+ (Variable cost × Q_{BEP})

Or, 20,000 × Q_{BEP} = ((41, 89,175+ (20, 40,000 × 1.1)) + (14,535 × Q_{BEP})

$\therefore Q_{BEP}$ =1,178

ΔQ_{BEP} = ((1,178-1,140)/ 1,140) ×100%

= 3.33%

So, Manufacturing overhead is not very sensitive. The percentage change in break-even point is less than half of the percentage change of Manufacturing Overhead.

8.7.5 Sensitivity of Administrative Cost:

At Break Even Point,

Selling price × Q_{BEP} = Total Fixed cost+ (Variable cost × Q_{BEP})

Or, 20,000 × Q_{BEP} = ((43, 27,175+ (19, 02,000 × 1.1)) + (14,535 × Q_{BEP})

$\therefore Q_{BEP}$ = 1,175

ΔQ_{BEP} = ((1,175 -1,140)/ 1,140) ×100%

= 3.07%

So, Administrative Costs are not very sensitive. The percentage change in break-even point is almost one-third of the percentage change of administrative cost.

8.7.6 Sensitivity of Sales and Marketing Expenses:

At Break Even Point,

Selling price × Q_{BEP} = Total Fixed cost+ (Variable cost × Q_{BEP})

Or, 20,000 × Q_{BEP} = ((5, 809175+ (4, 20,000 × 1.1)) + (14,535× Q_{BEP})

$\therefore Q_{BEP}$ = 1,148

ΔQ_{BEP} = ((1,148 -1,140)/ 1,140) ×100%

= 0.7%

So, Selling Costs are not sensitive at all. The break-even point changes about 0.7% with a 10% change in selling cost. This indicates that if somehow the cost of selling goes up and other cost parameters stay stable, then only 0.7% production is to be increased in order to attain the break-even point.

The cost analysis shows that the system developed for the manufacturing of Convenient Bamboo Cutter is capable of producing units just over the break-even point. This means if any major sources of cost are included in any phase of value addition, the system may lag below the break-even point. From the sensitivity analysis, it is shown that increasing any cost parameter doesn't significantly affect the break-even point. This is a measure of the process stability.

Conclusion

Convenient Bamboo Cutter is aimed to be a useful tool for reducing the hassle and energy required in the process of cutting bamboo. It is to be employed with the unique feature of enhancing the force applied by a user, which is calculated to be sufficient in order for the bamboo to be sliced off and ready for use.

This course has provided us an opportunity to learn how to bridge the gap between customers and engineers as well as understand the sophisticated process and techniques that are involved in the development and design of a product from concept to implementation.

Appendix

A1. Survey Questionnaire

Survey Form

A Survey on Convenient Bamboo Cutter

Name:…… Profession:……..

Address:……...

Age:……. Date:……

Question 1: Have you used a bamboo cutting machine before?

- Yes
- No

Question 2: How often do you work with bamboo or bamboo related products?

- Once a year
- Once every 6 months
- Depends on seasonal use
- Frequently (2-3 times a week)
- Daily

Question 3: Are you aware of the benefits of using a bamboo cutting machine over traditional manual methods?

- Yes
- No

Question 4: Are you interested in automizing this bamboo cutting process using a machine?

- Yes
- No

Question 5: Which of the following methods do you prefer?

- Traditional (Manual)
- Electrically operable automatic machine with motor
- Mechanically designed semi-automatic

Question 6: How important is the safety feature of a bamboo cutting machine to you?

- Very Much
- At Moderate Level
- Not that much

Question 7: How does using a bamboo cutting machine impact your productivity or efficiency compared to a manual cutting method?

- Very Much
- At Moderate Level
- Little Bit
- Not That Much

Question 8: Motor-driven automatic bamboo cutters are available in the market. How likely would you prefer buying one of those, given the difficulties of manual bamboo cutting?

- I would buy one.
- I would buy one if I could afford it.
- I am only interested in traditional methods.

Question 9: If you could cut bamboo more efficiently and with less effort by means of a relatively cheaper, pulley-operated bamboo cutter, would you prefer to purchase it?

- Yes
- No
- Cannot say for sure.

Question 10: How much would you like to spend over a bamboo cutting machine?

BDT……

Question 11: On a scale 1 to 10, how likely are you to recommend a bamboo cutting machine to others?

…….

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